

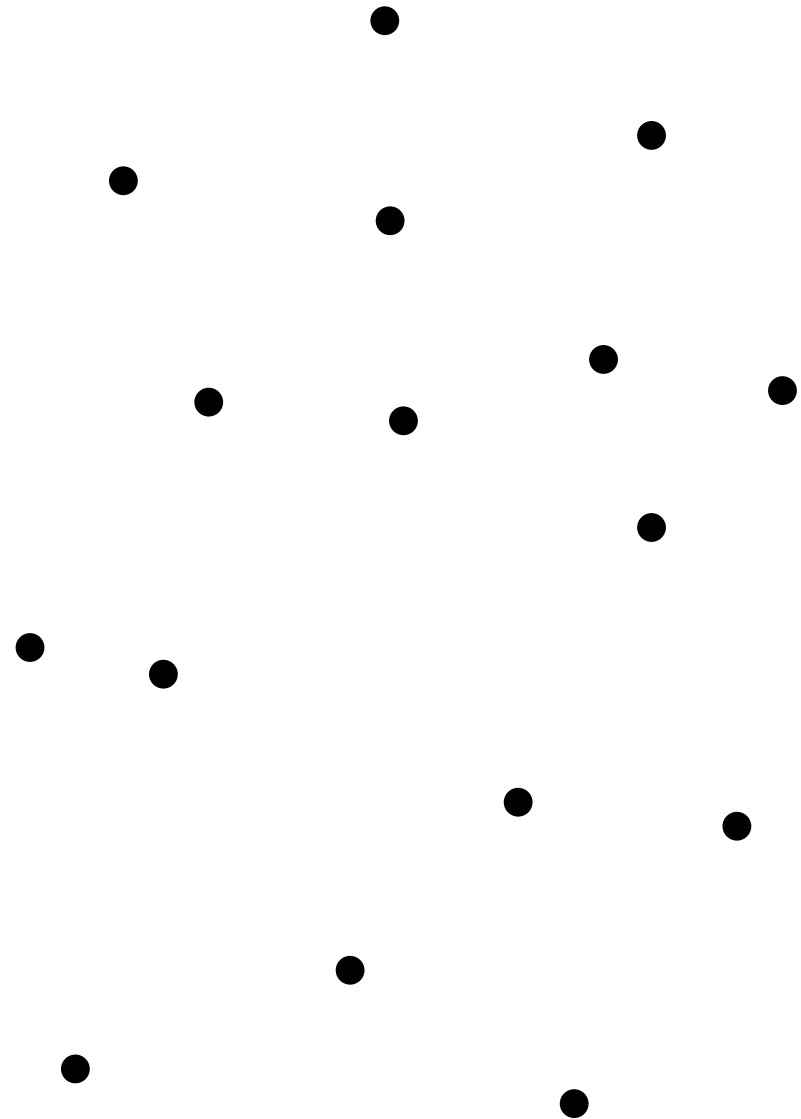
PROXIMITY

Vera Sacristán and Rodrigo Silveira

Universitat Politècnica de Catalunya

PROXIMITY

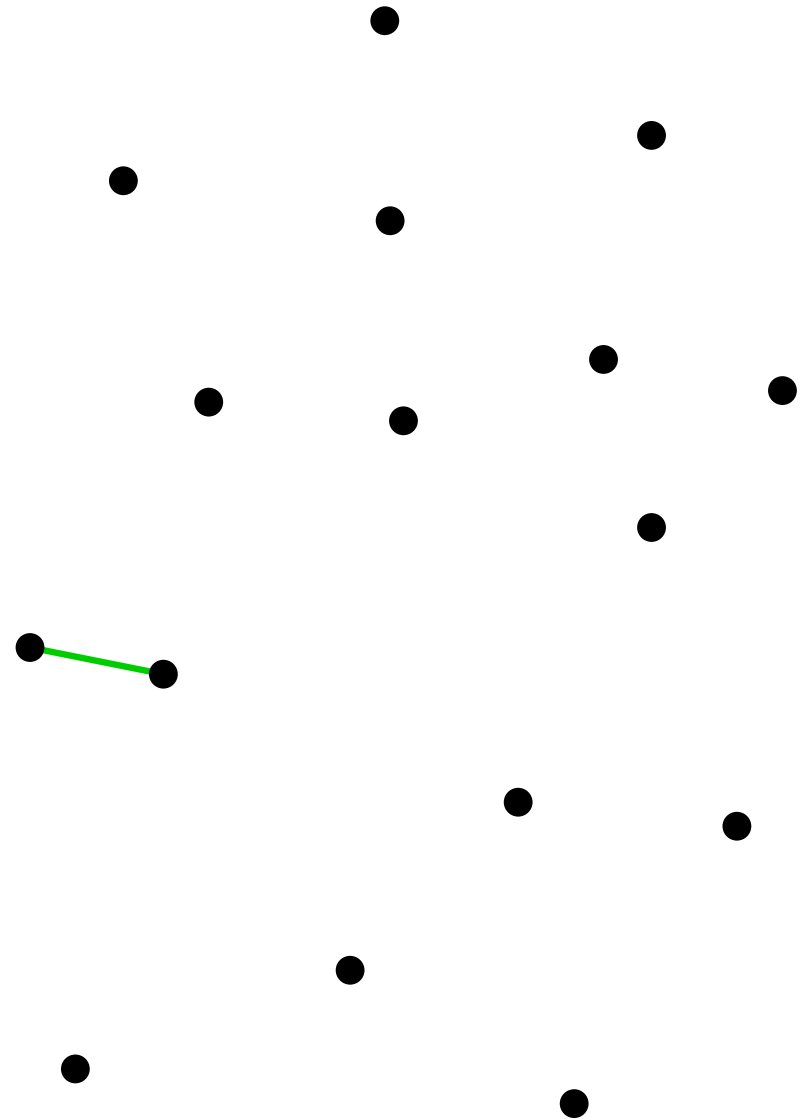
Let P be a set of n points in $\mathbb{R}^2$...



PROXIMITY

CLOSEST PAIR

Find the pair of points closest to each other



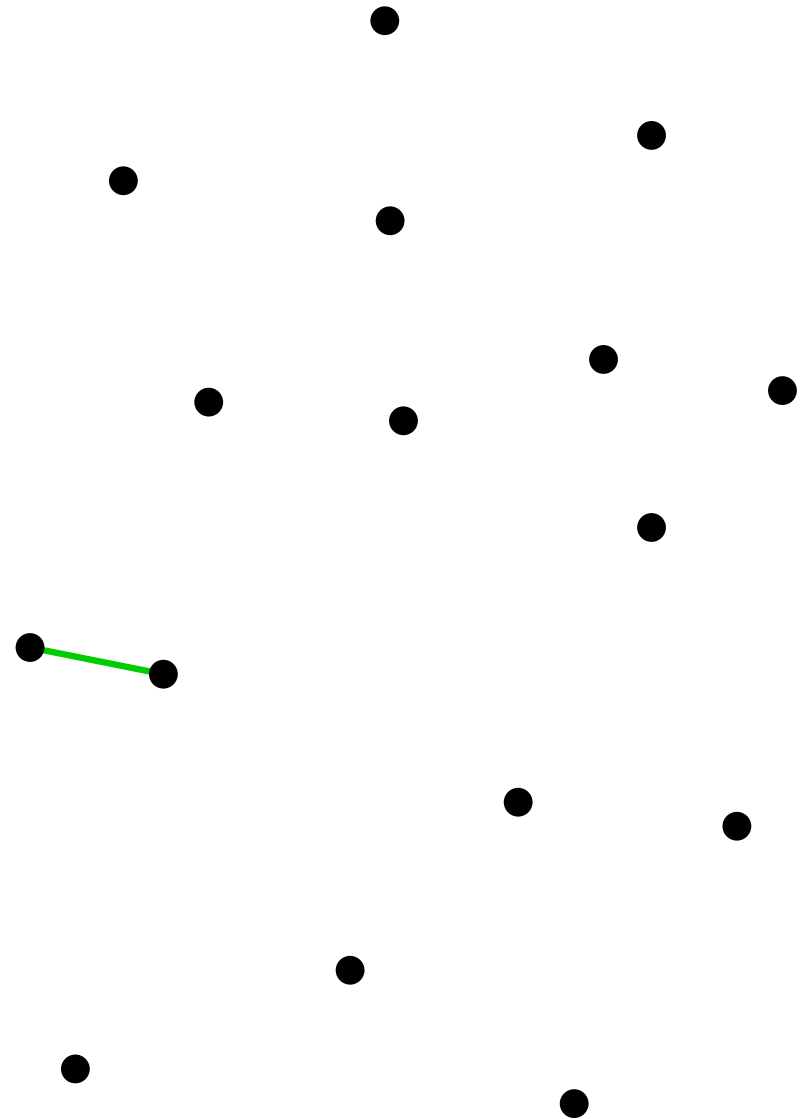
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APPLICATION

Airplanes in danger of collision



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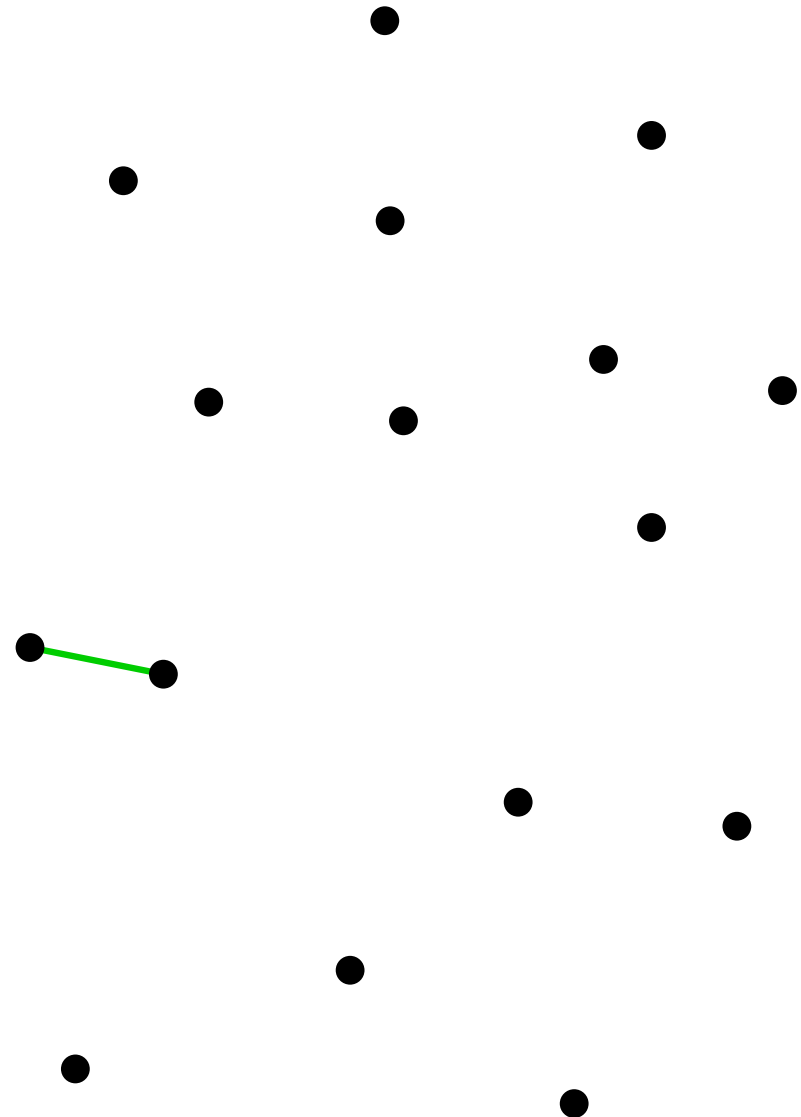
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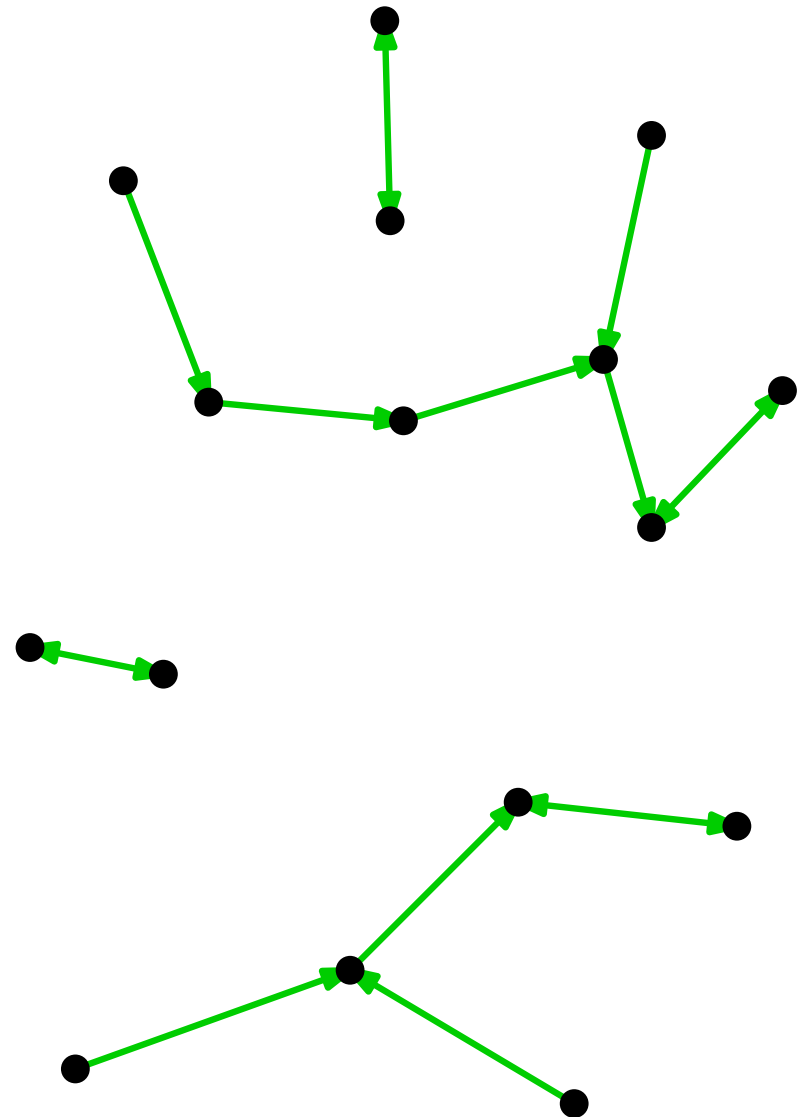
- Brute force solution: check every pair of points in $O(n^2)$ time
- This procedure does not exploit the metric properties
- In dimension 1, it can be solved in $O(n \log n)$ time by sorting the input first, and then using the fact that the solution must be a pair of consecutive points



PROXIMITY

ALL NEAREST NEIGHBORS

Build the directed graph where the existence of an (oriented) edge $\overrightarrow{p_i p_j}$ means that p_j is the point of P closest to p_i



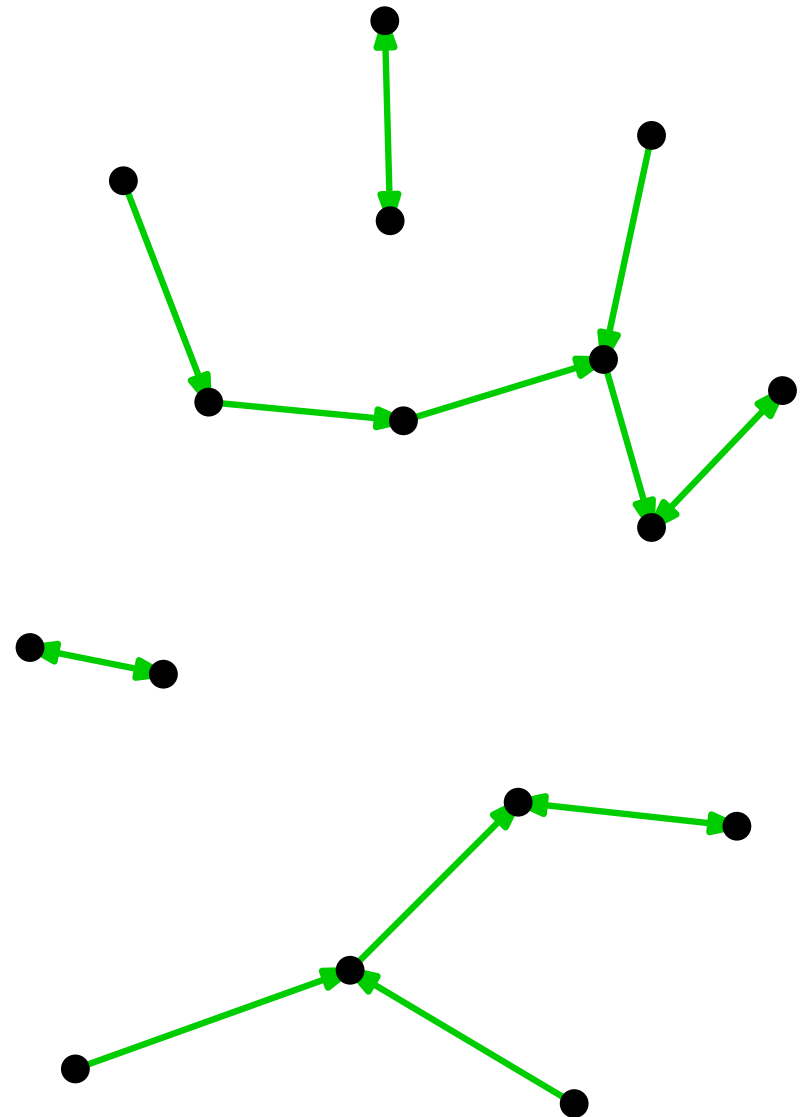
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APPLICATION

In Ecology, to study the territoriality of species



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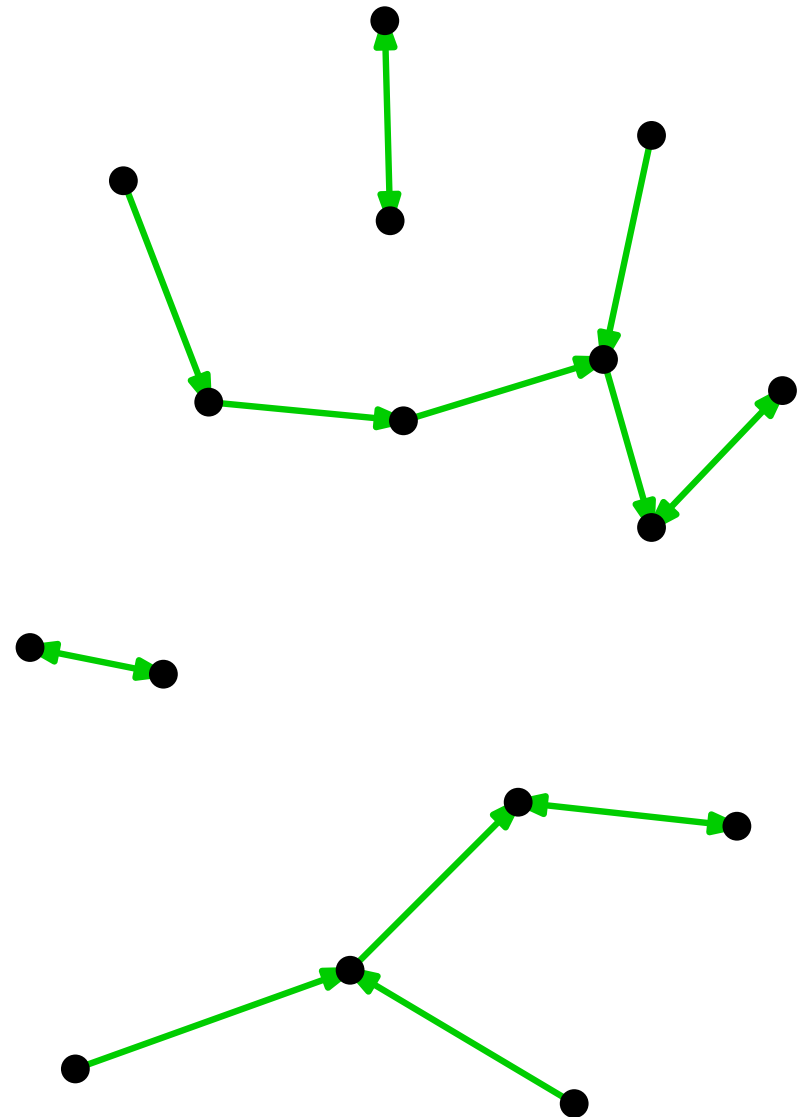
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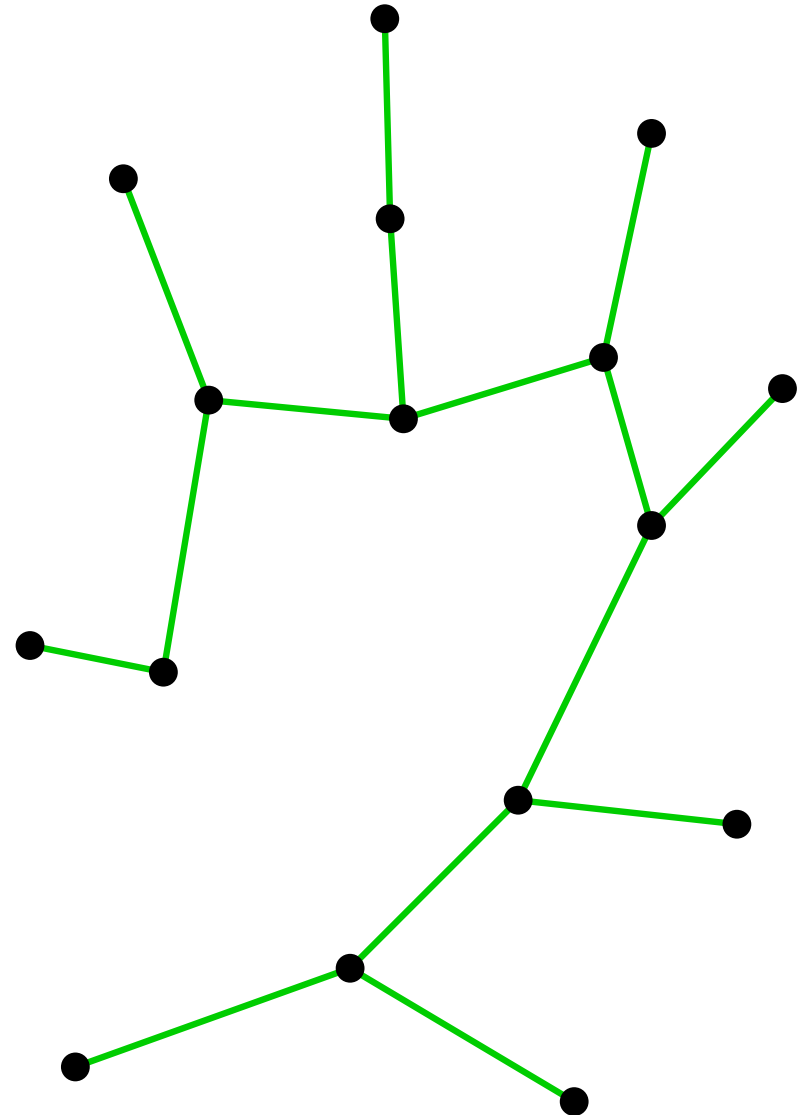
- Brute force solution: check every pair of points in $O(n^2)$ time
- This process does not exploit the metric properties
- In dimension 2, at most 6 points of P have p_i as their closest point
- In dimension 1 it can be solved in $O(n \log n)$ time (same as before)



PROXIMITY

EUCLIDEAN MINIMUM SPANNING TREE

Build the tree connecting all points of P and minimizing the sum of the lengths of its edges



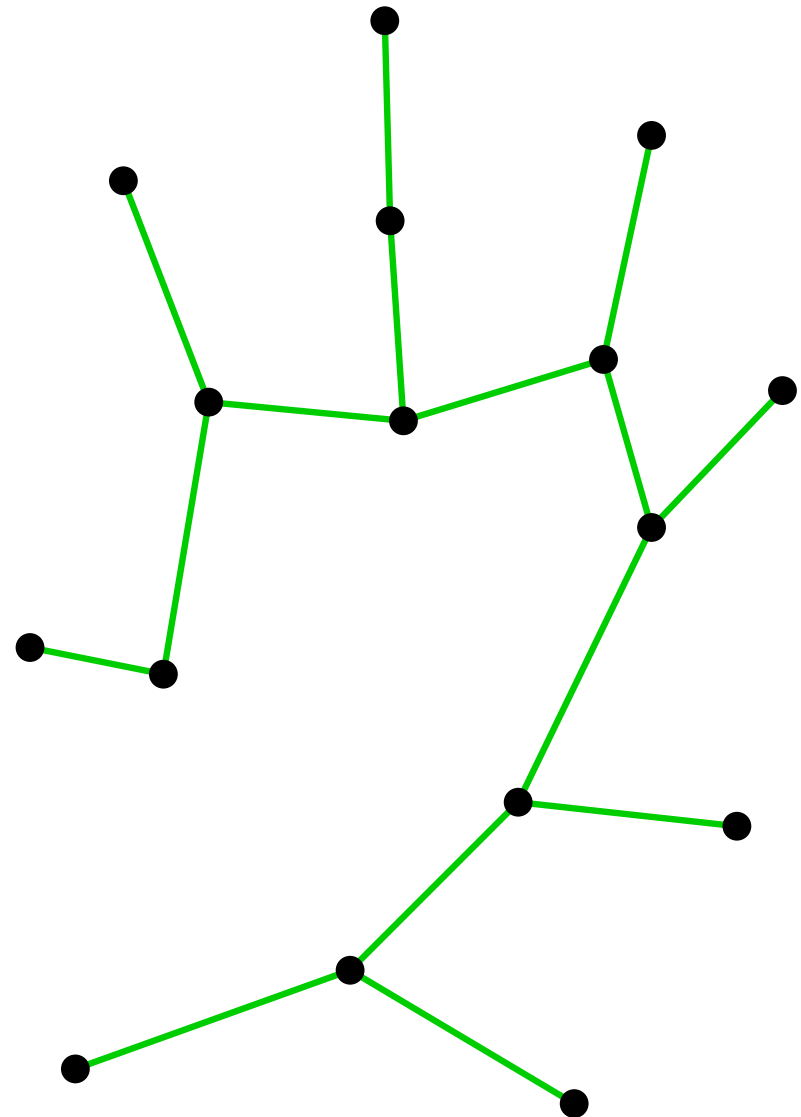
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Telecommunication networks



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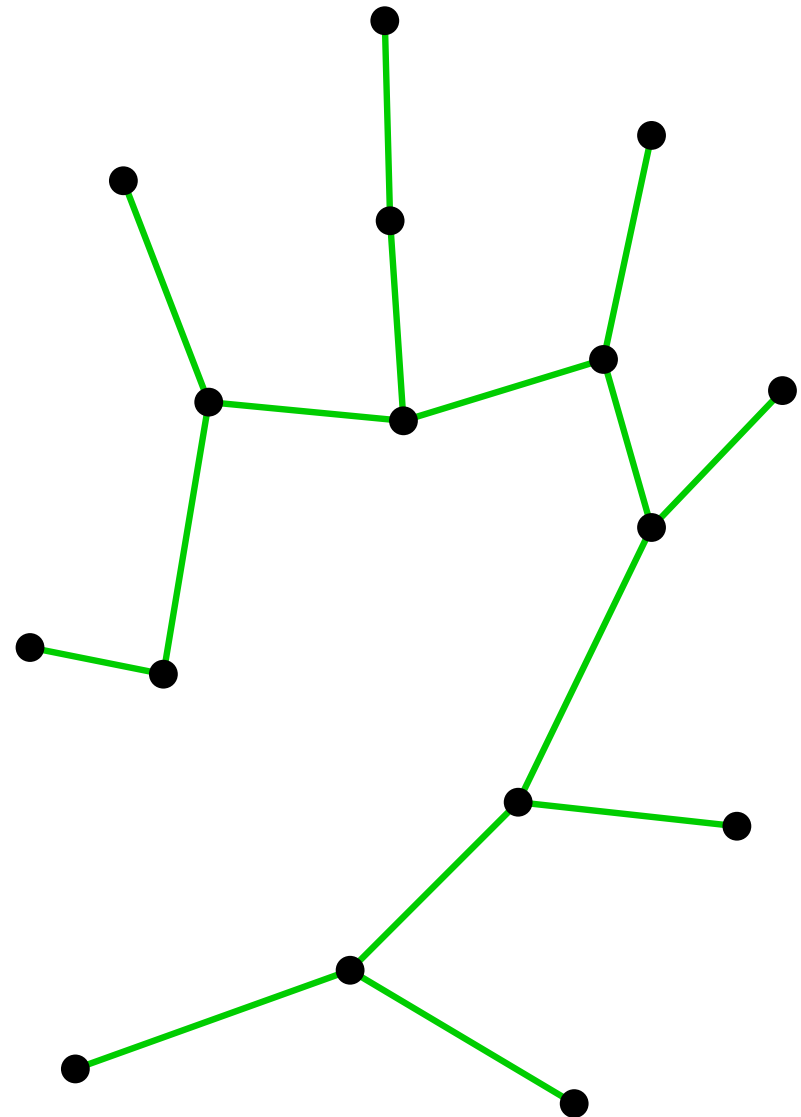
EUCLIDEAN MINIMUM SPANNING TREE

Build the tree connecting all points of P and minimizing the sum of the lengths of its edges

APPLICATION

Telecommunication networks

- Applying Prim's or Kruskal's algorithms to the complete graph of P , with Euclidean weights: $O(e \log e) = O(n^2 \log n)$ time
- Applying improved algorithms to the complete weighted graph: $O(e) = O(n^2)$ time
- These procedures do not exploit the metric properties
- In dimension 1 it can be done in $O(n \log n)$ time (same as before)



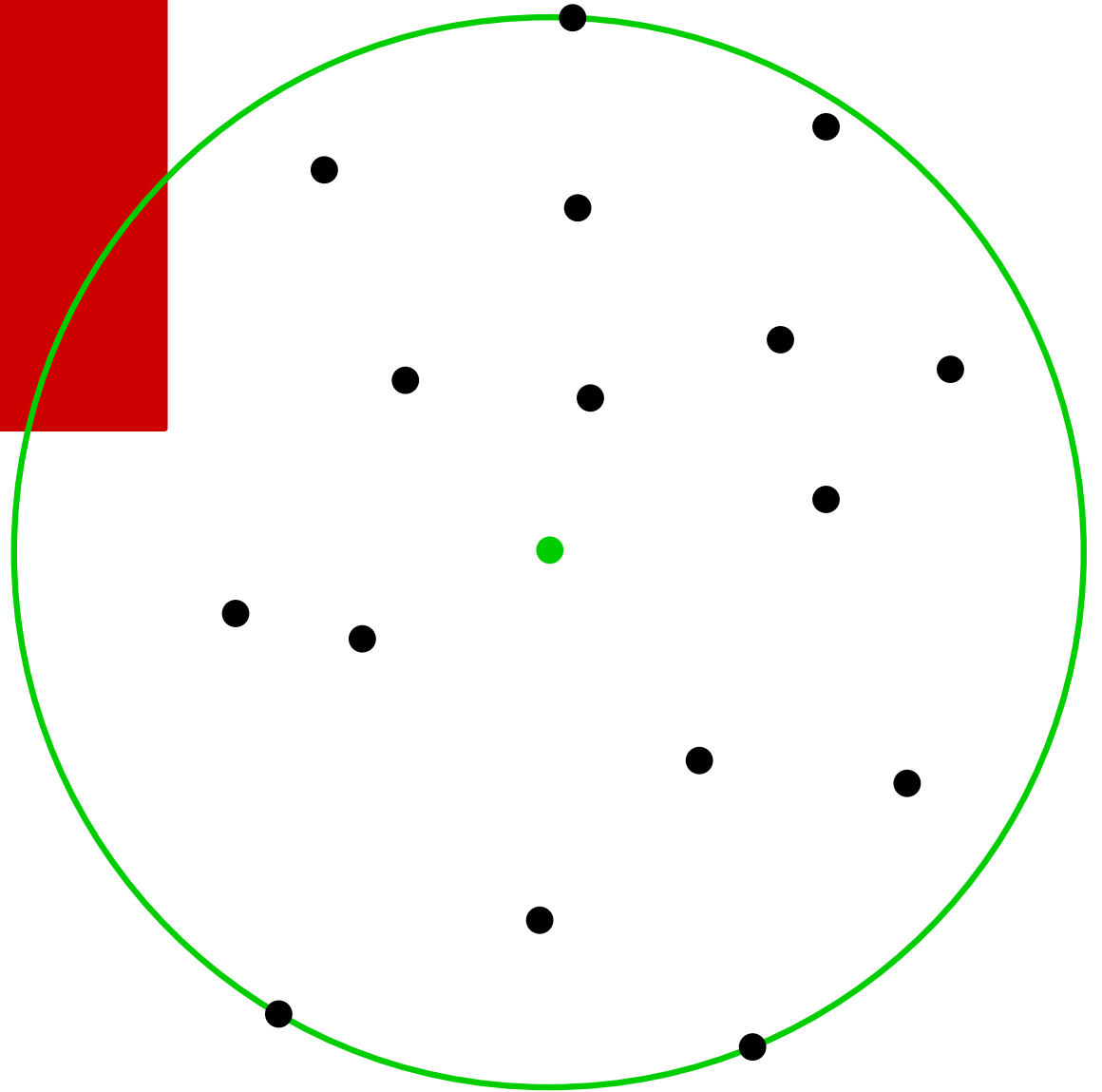
PROXIMITY

MIN-MAX FACILITY LOCATION: MINIMUM SPANNING CIRCLE

Find the point x on the plane achieving

$$\min_{x \in \mathbb{R}^2} \max_{p_i \in P} d(x, p_i)$$

Geometrically: find the center of the circle of minimum radius enclosing P



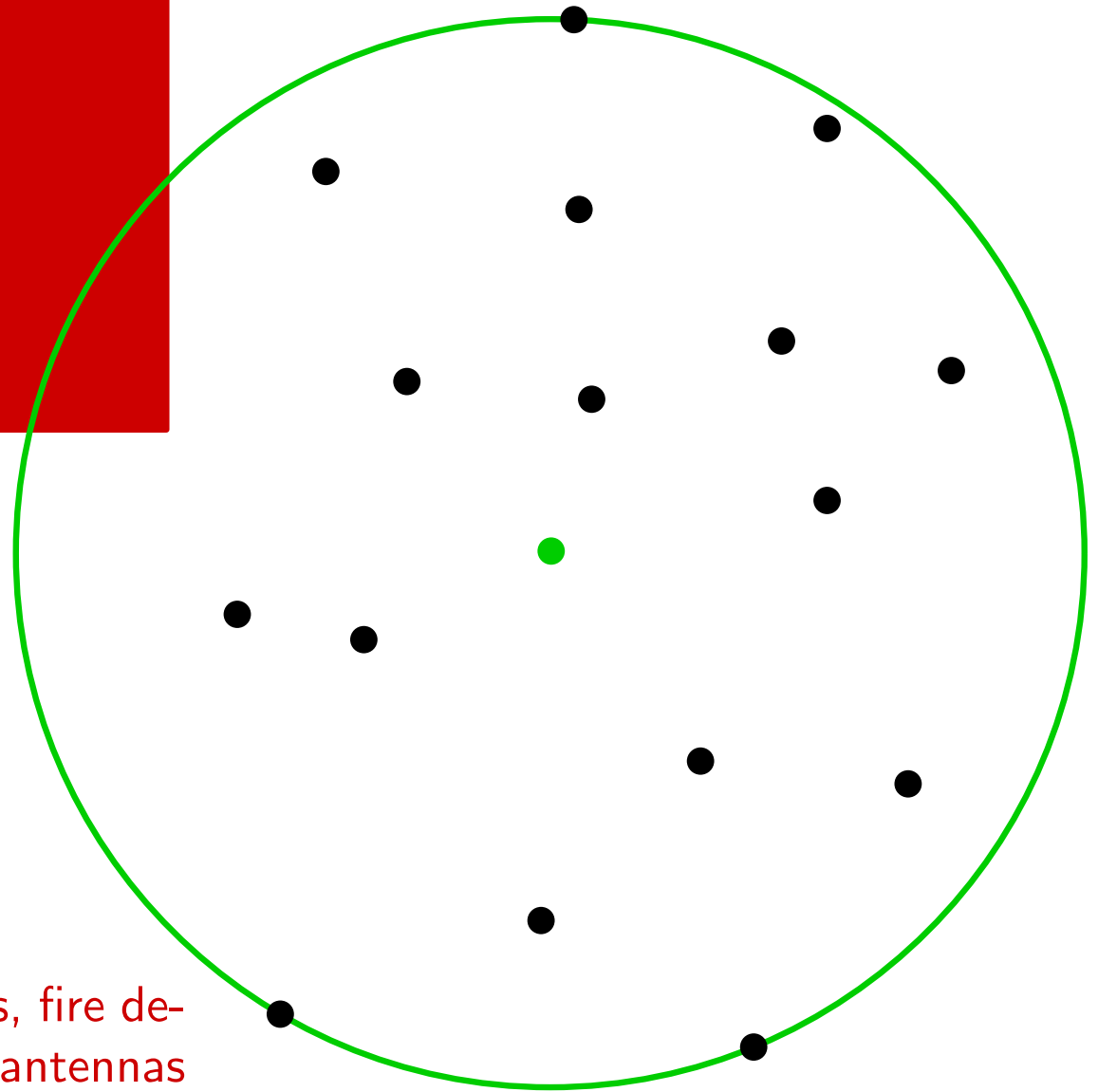
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Location of emergency services (hospitals, fire departments, ...), of radio and tv repeaters, antennas for mobile phones,...

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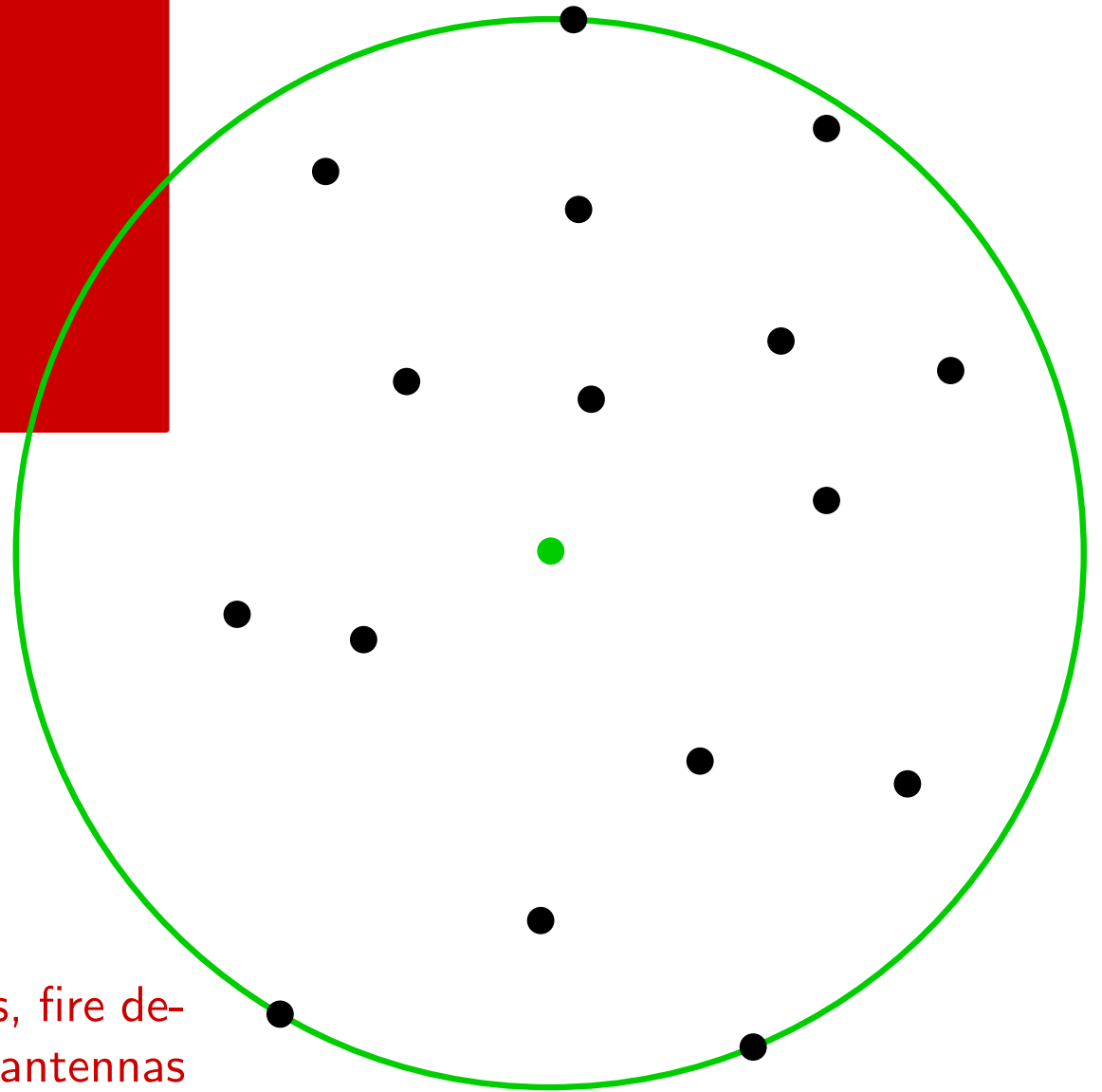
$$\min_{x \in \mathbb{R}^2} \max_{p_i \in P} d(x, p_i)$$

Geometrically: find the center of the circle of minimum radius enclosing P

- Brute force solution: consider every set of three points, in $O(n^4)$ time
- In dimension 1 it can be solved in $O(n)$ time: the solution is the midpoint of $\min(P)$ and $\max(P)$

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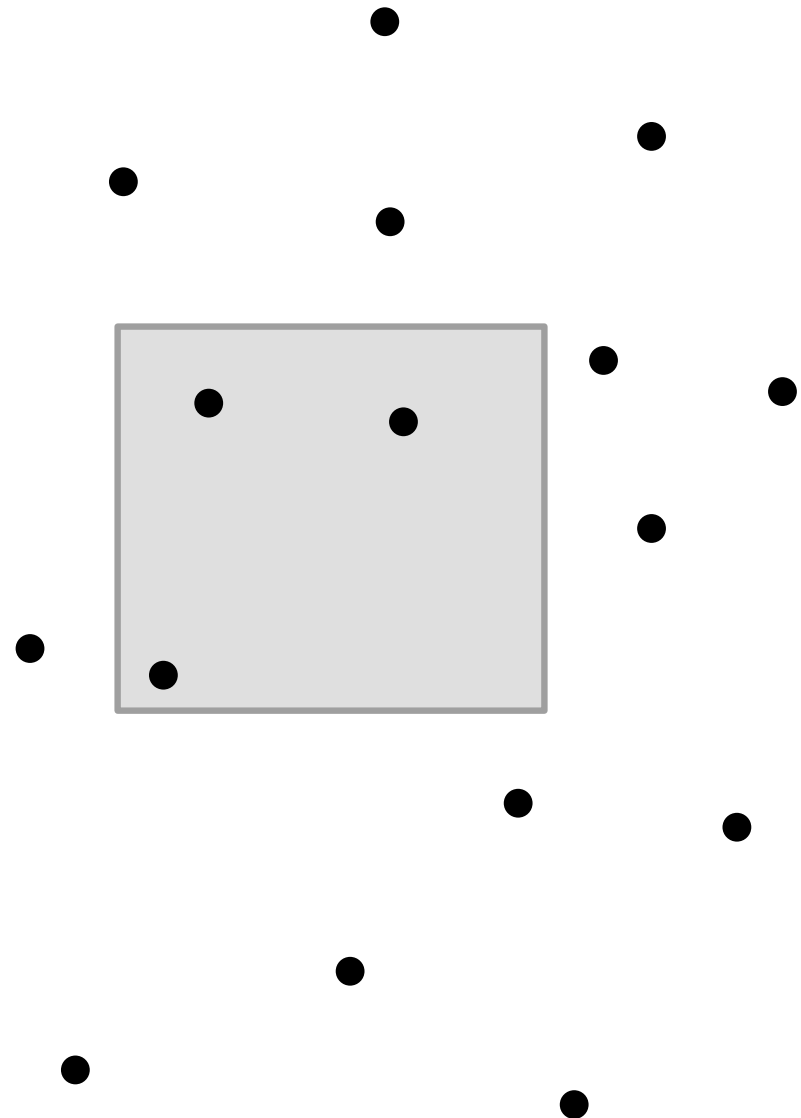
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Find the point x in the region A achieving

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Geometrically: find in A the center of circle of maximum radius not enclosing any point of P



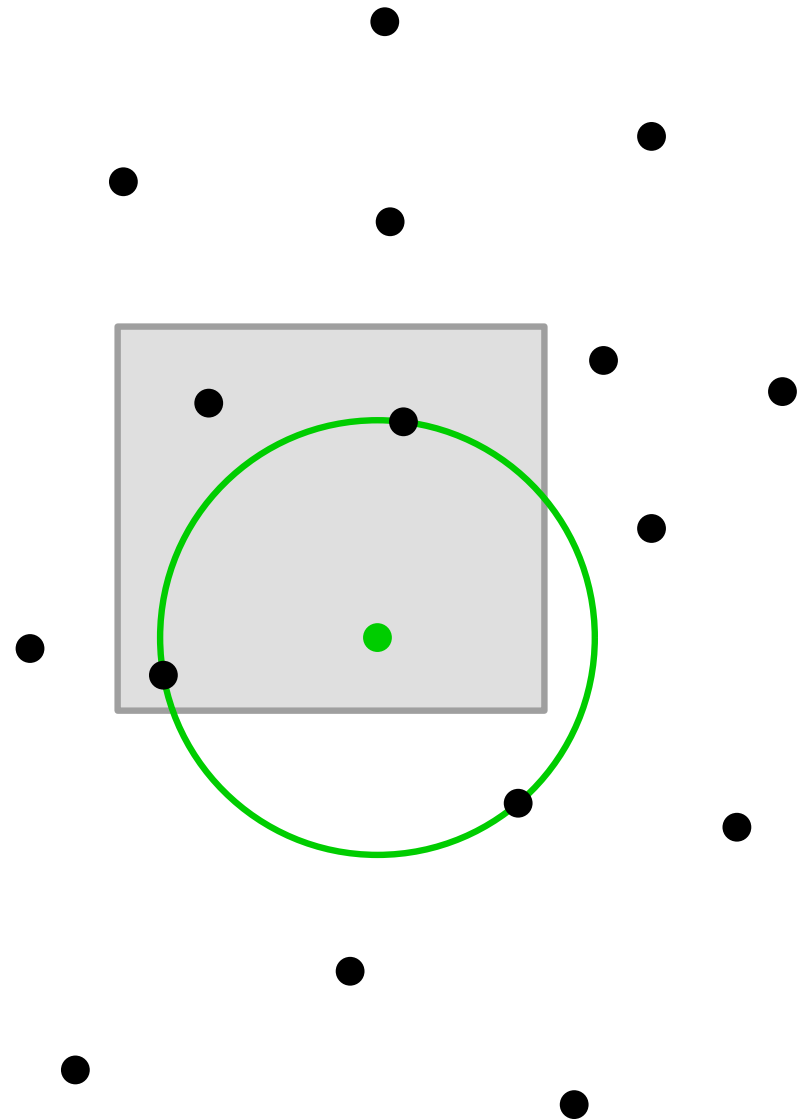
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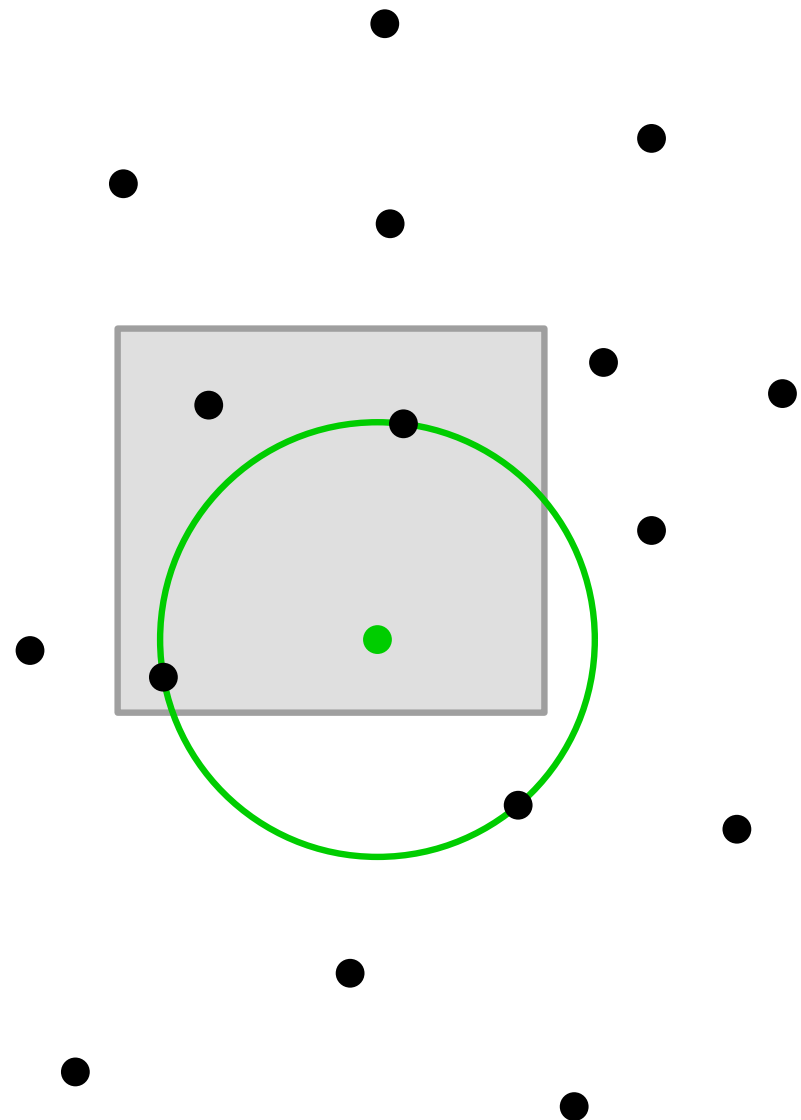
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Location of undesired services (polluting industry, landfills, ...) or new shopping malls in competence with preexistent centers

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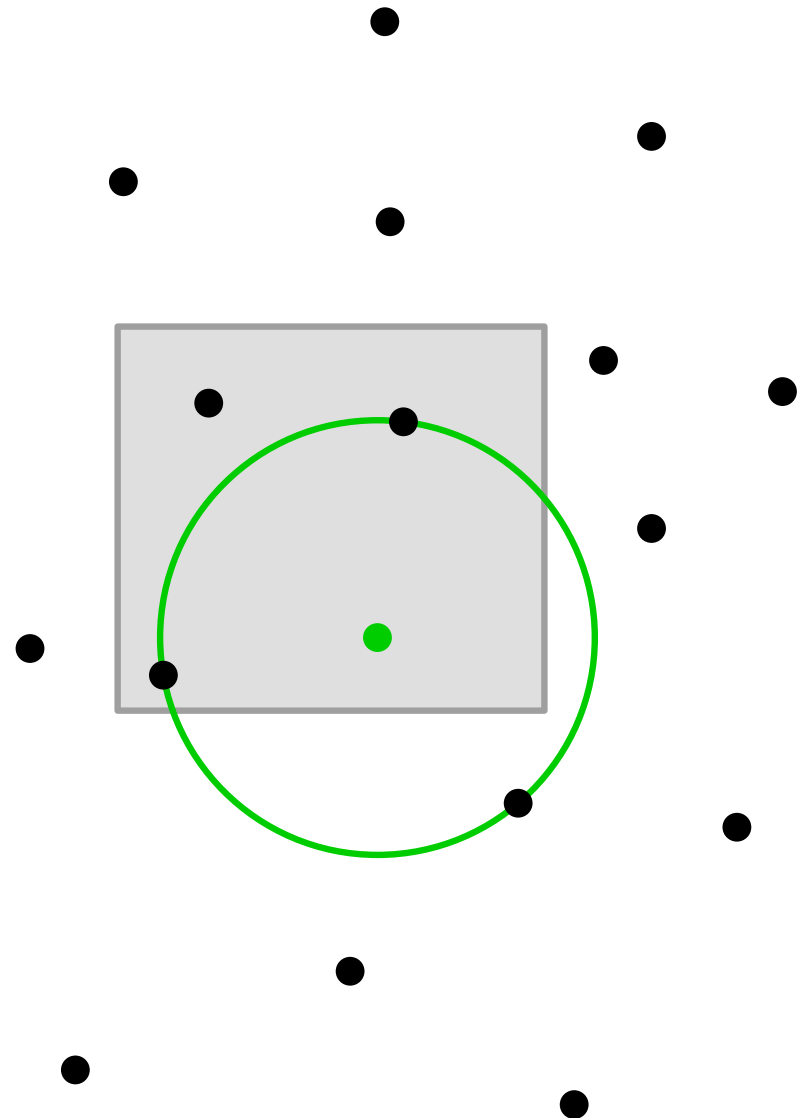
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APPLICATION

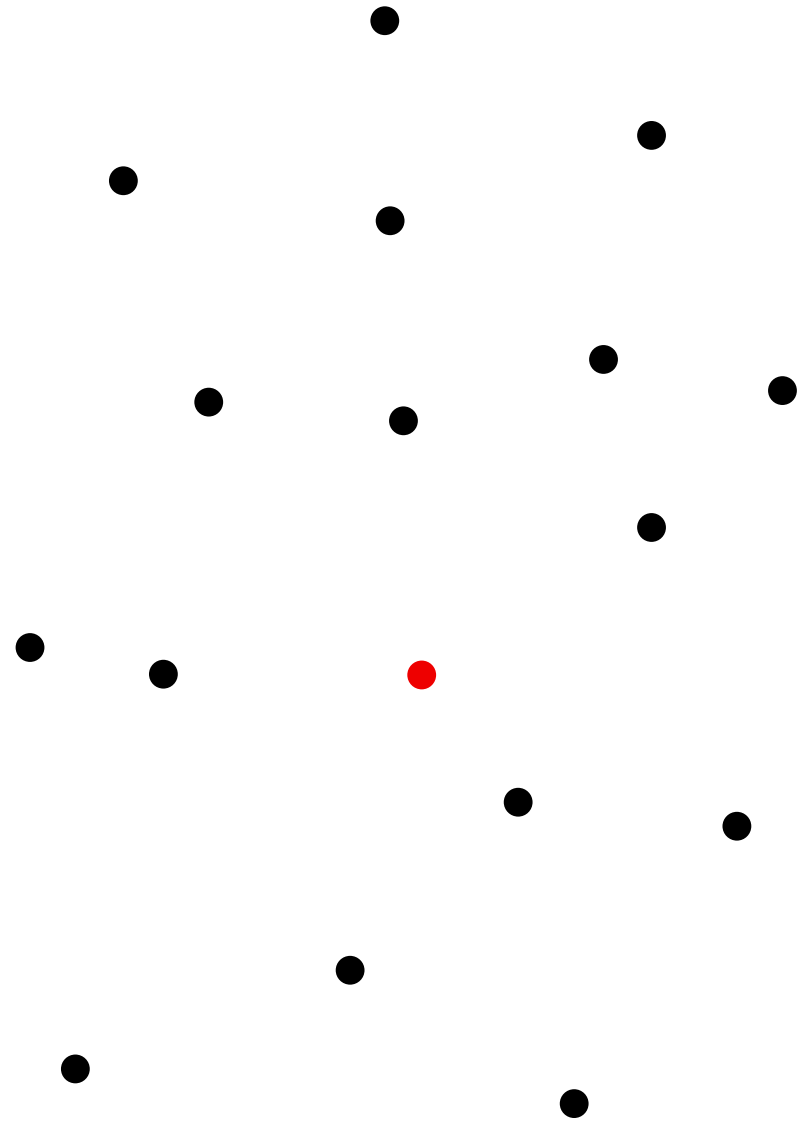
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NEAREST NEIGHBOR QUERY

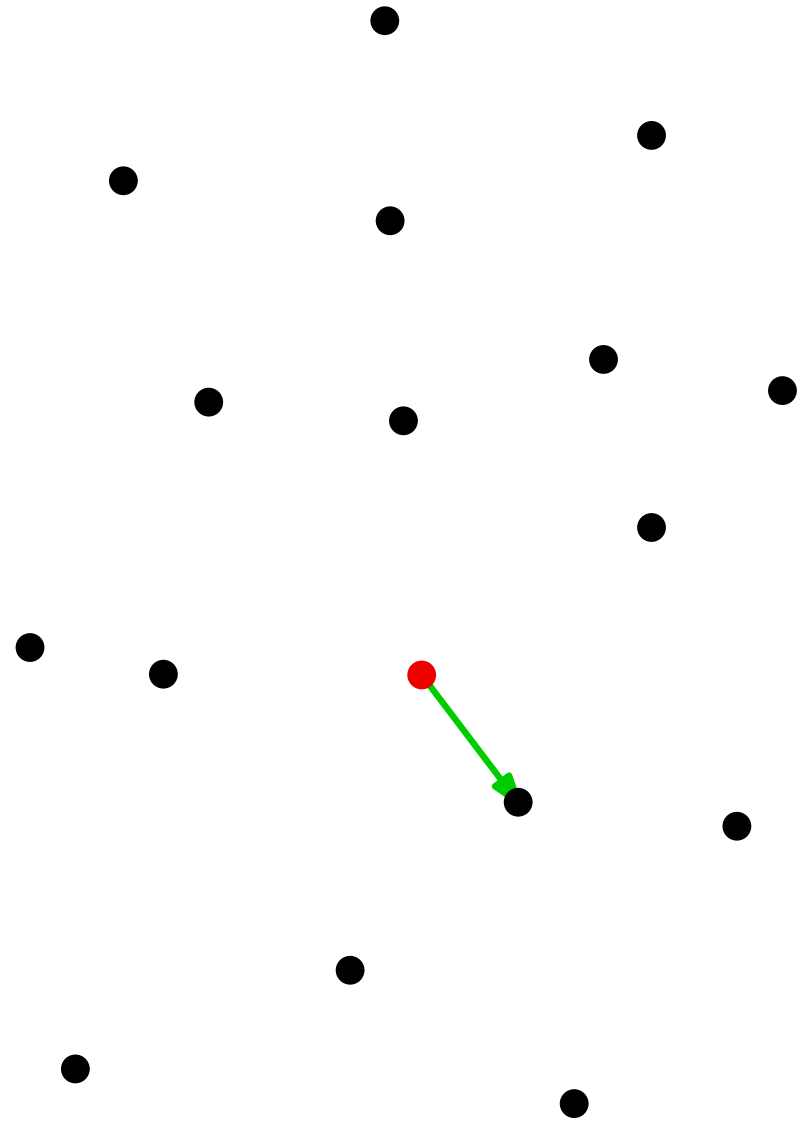
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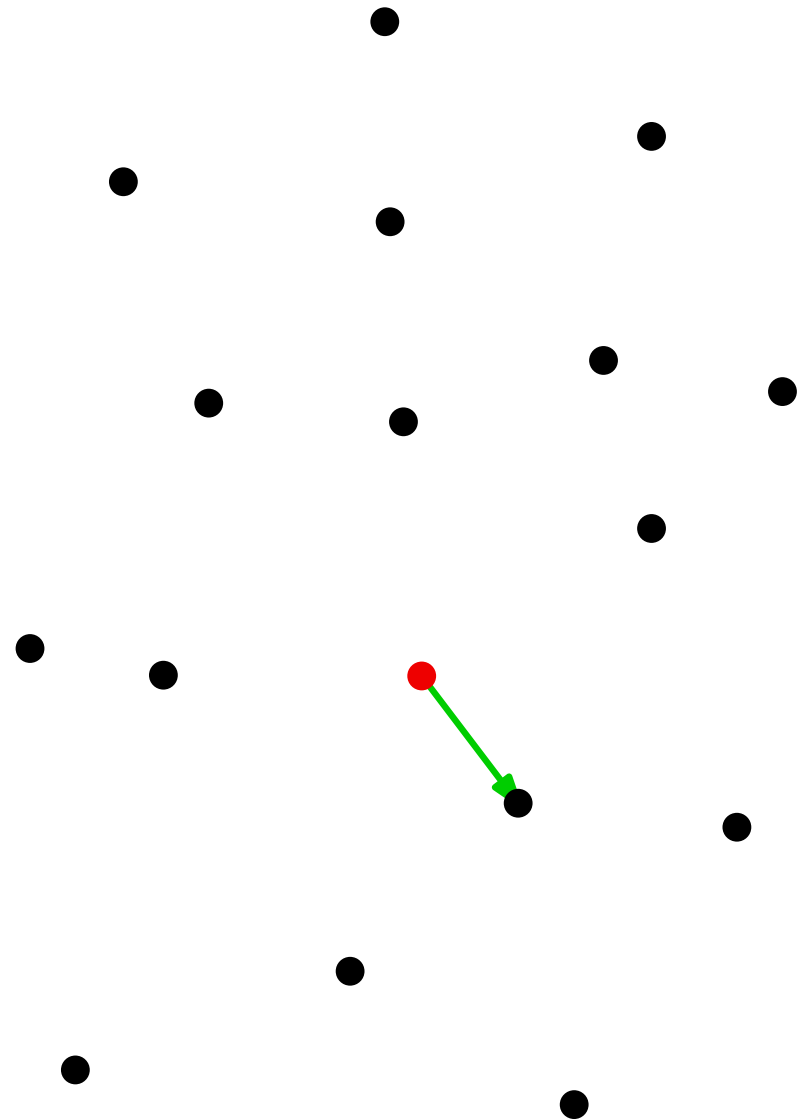
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APPLICATION

The post office problem, pattern (voice) recognition and, in general, matching problems

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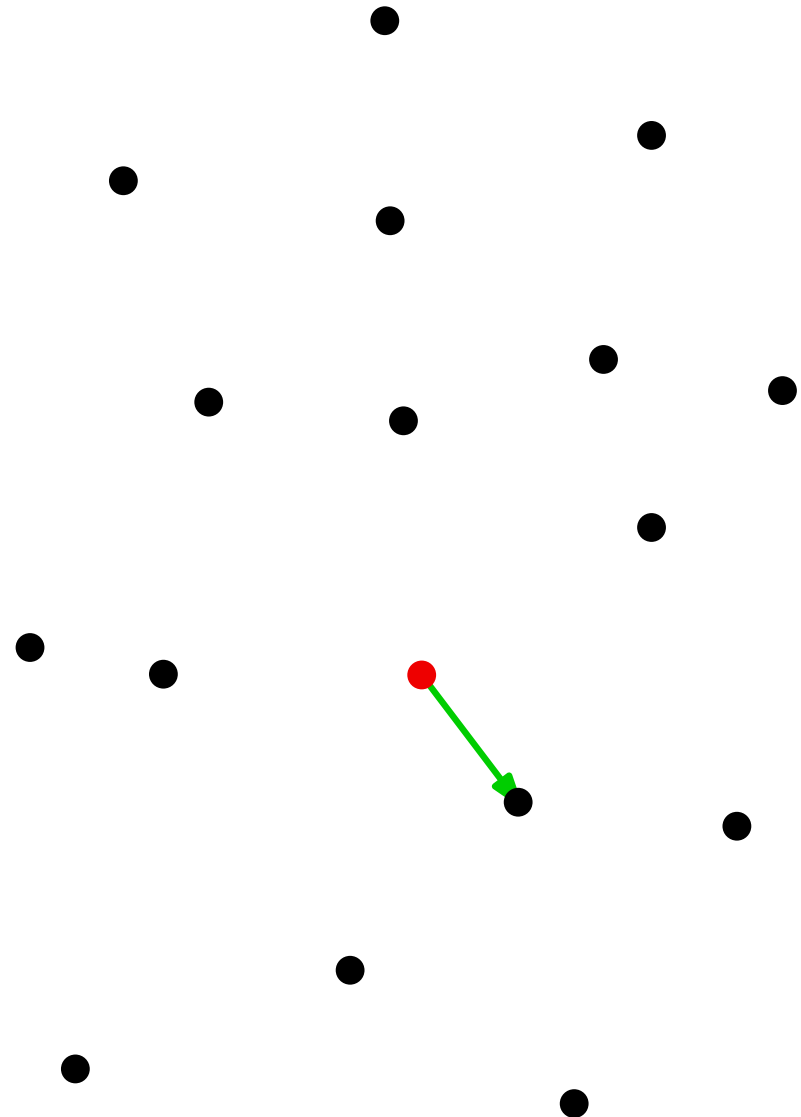
NEAREST NEIGHBOR QUERY

Given a new point q , quickly find its closest site p_i

- Brute force: compare q with every site p_i , in $O(n)$ time
- Preprocessing P in order to speed up the queries is worth the effort when the query is to be repeatedly called
- In dimension 1, first sort the points of P in $O(n \log n)$ time, and then each query will be answered in $O(\log n)$ time by binary search

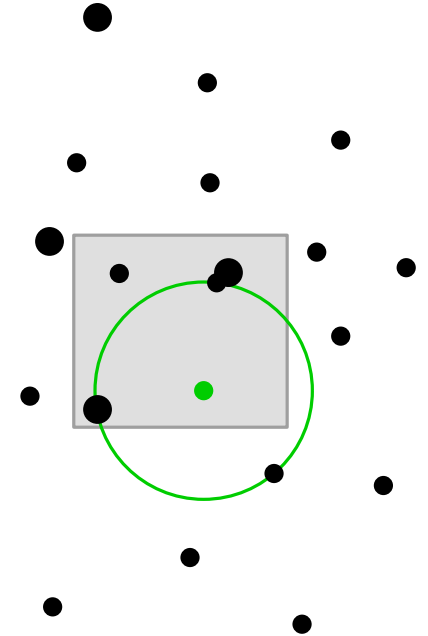
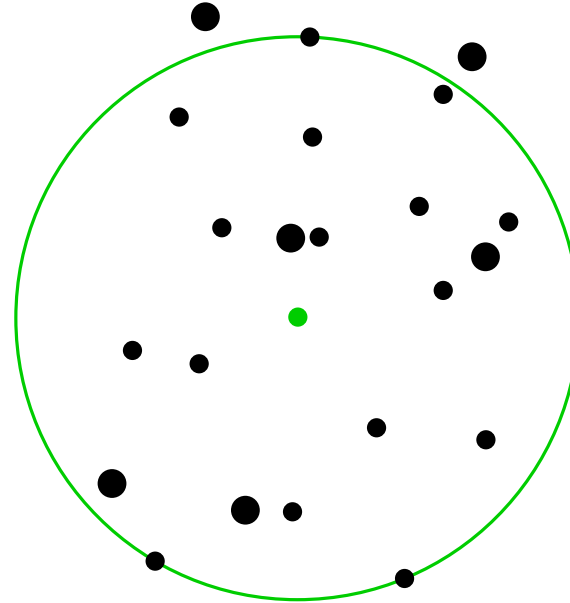
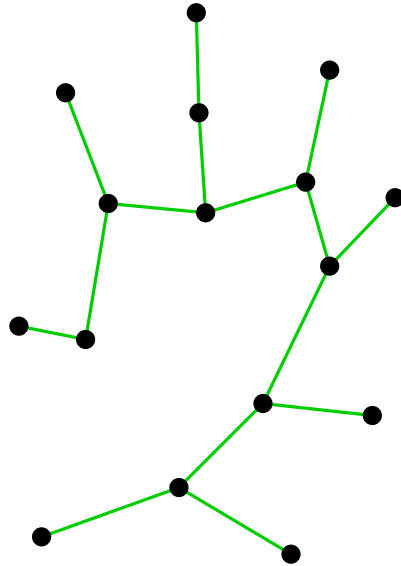
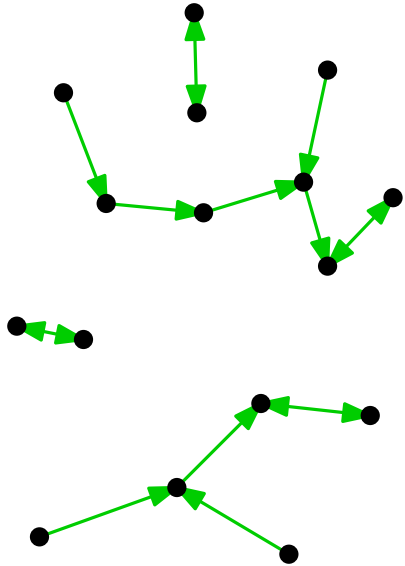
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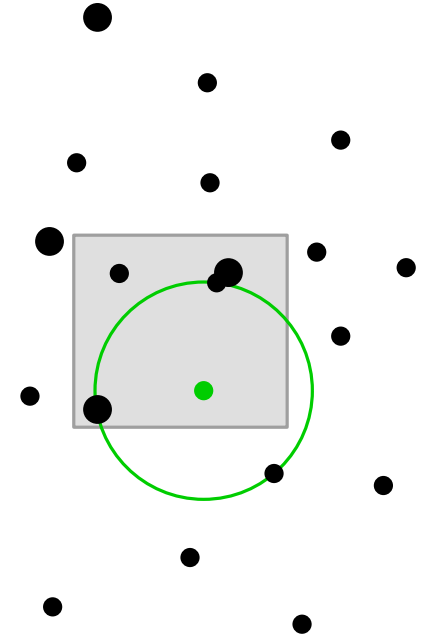
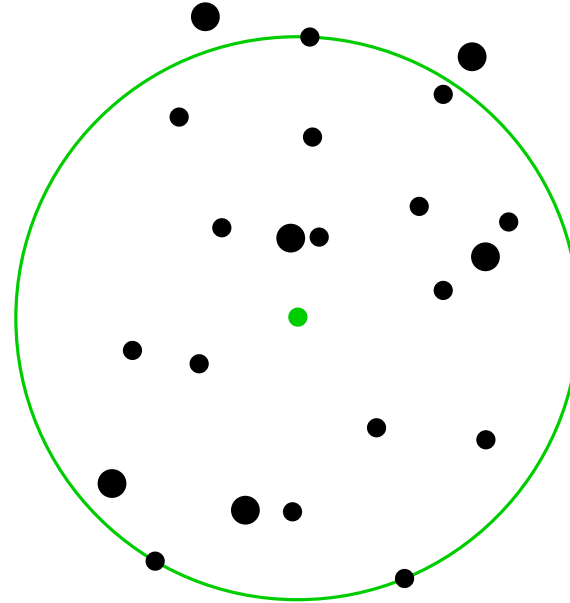
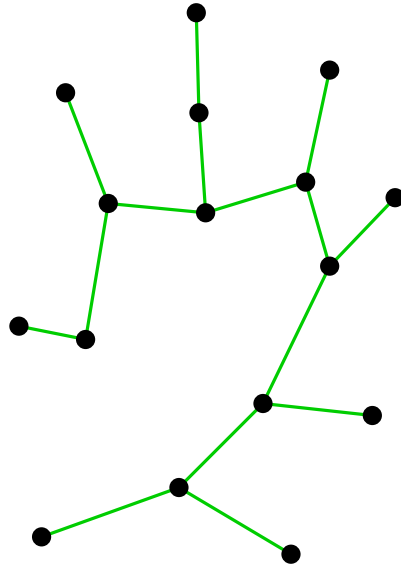
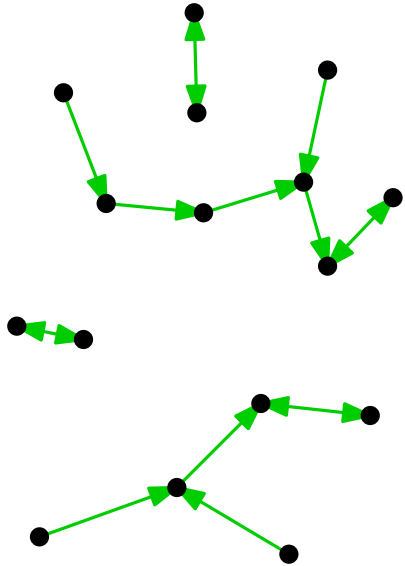
MANY DIFFERENT PROBLEMS



ALL OF THEM INVOLVE DISTANCES AND PROXIMITY

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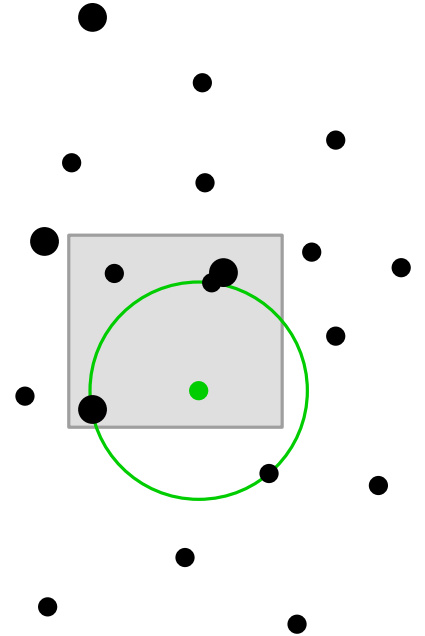
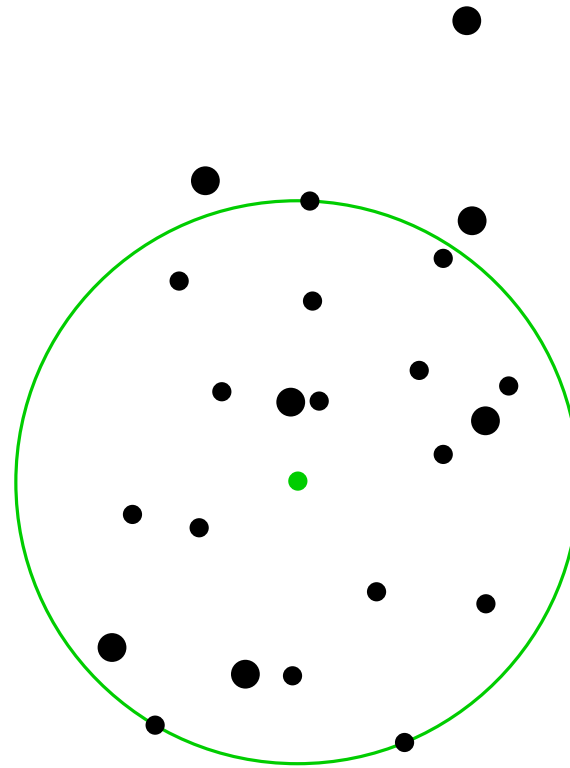
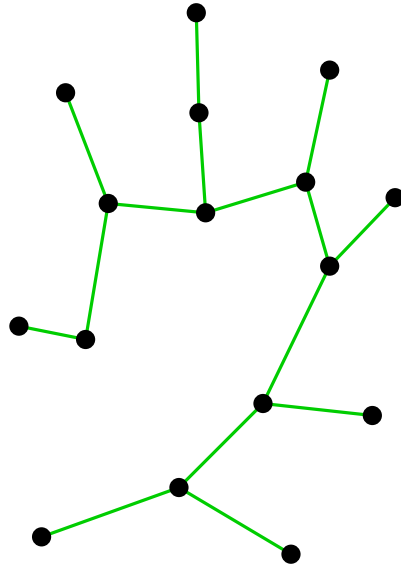
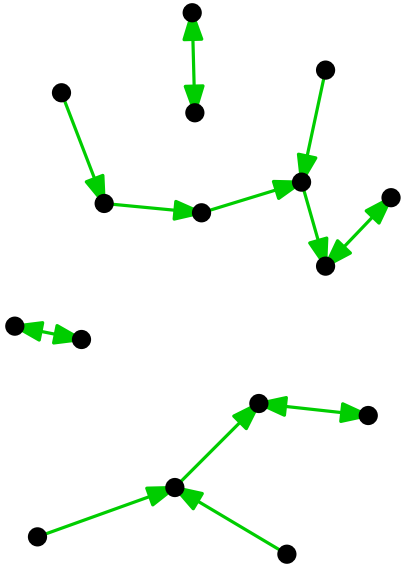


ALL OF THEM INVOLVE DISTANCES AND PROXIMITY

ONE COMMON SOLUTION:

PROXIMITY

MANY DIFFERENT PROBLEMS



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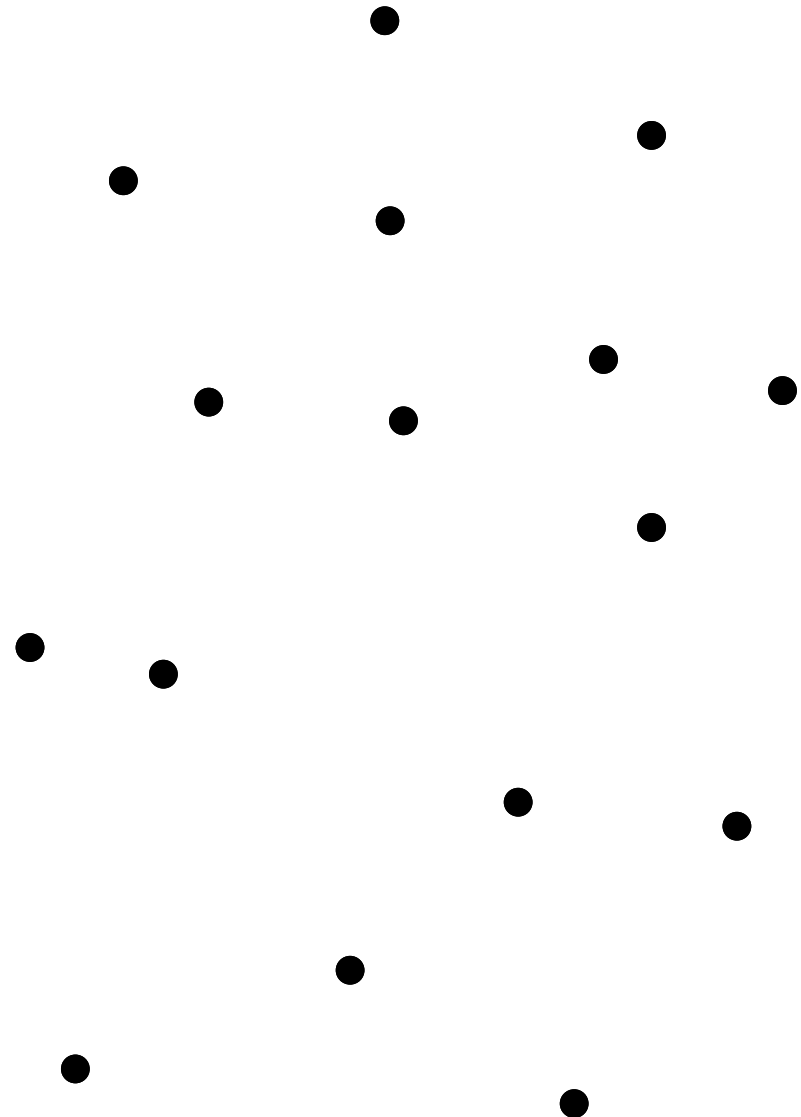
ONE COMMON SOLUTION:

Voronoi diagram

PROXIMITY

VORONOI DIAGRAM

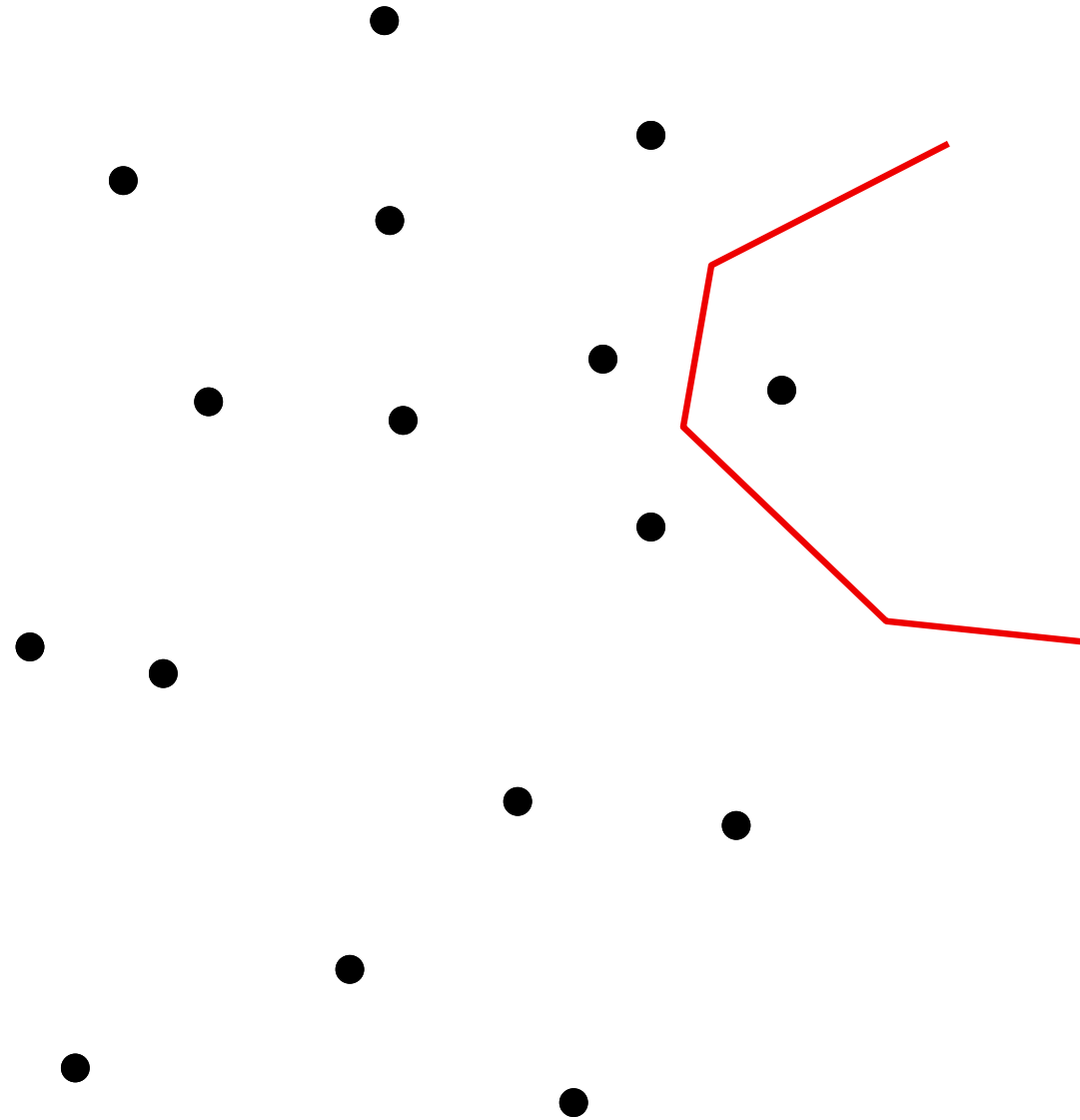
- Structure capturing the proximity information
- Decomposes the plane into regions, each one is associated by proximity to one of the points of P
- Given two points, the boundary separating the portion of the plane closer to one point than to the other is the perpendicular bisector of the segment connecting the two points



PROXIMITY

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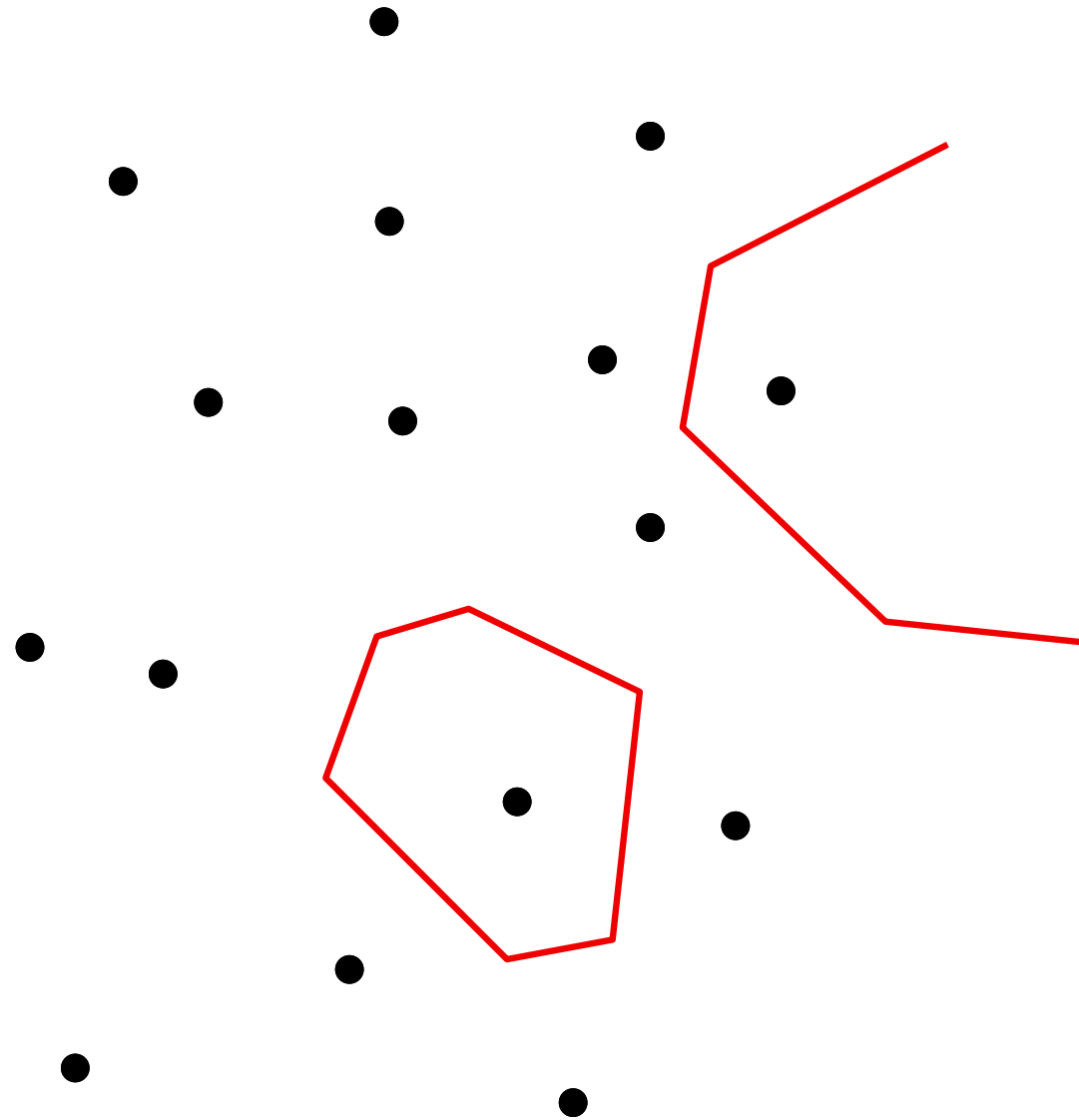
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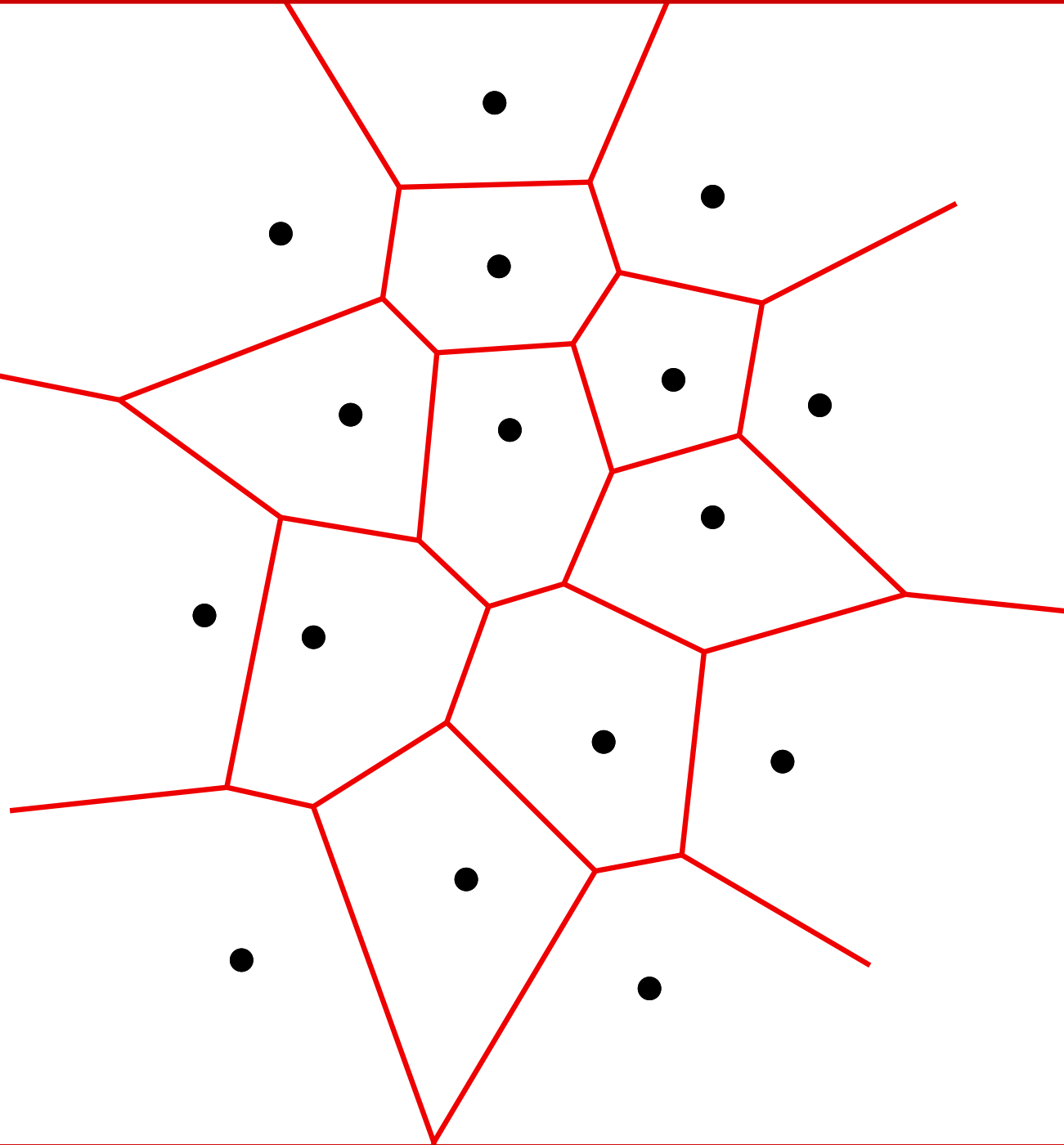
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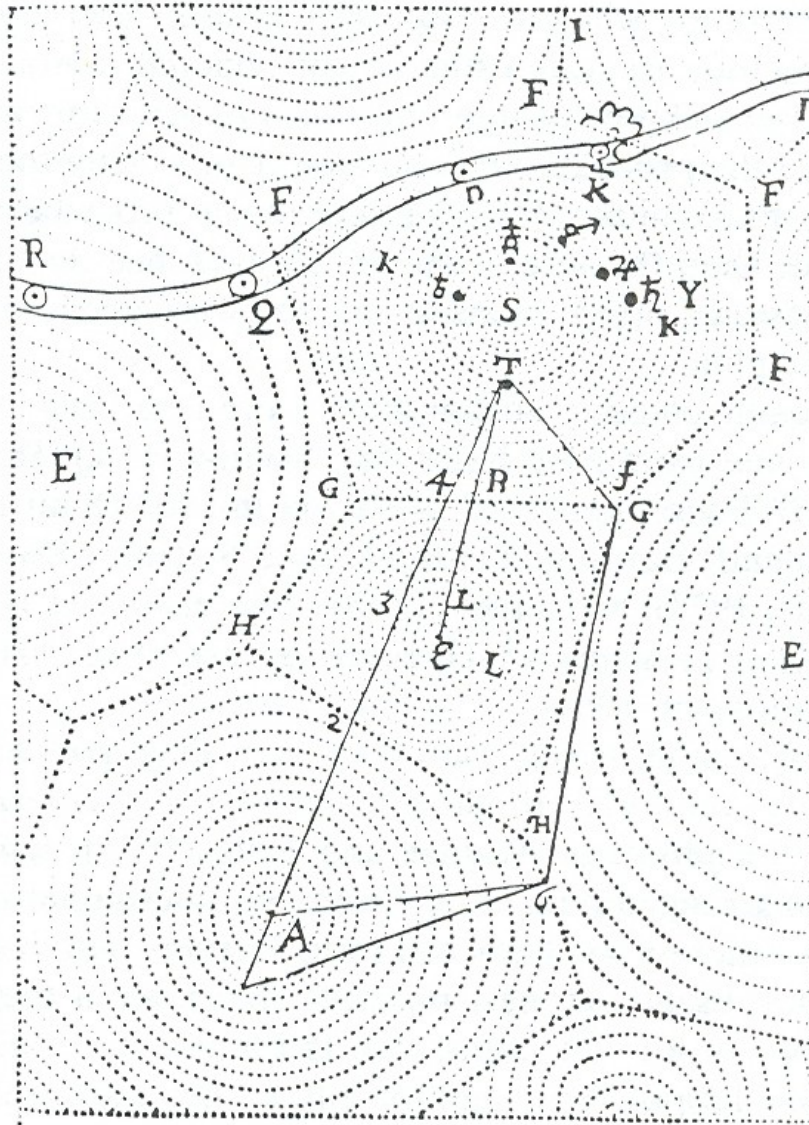
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PROXIMITY

History of the rediscoveries of the Voronoi diagram

- **Descartes:** distribution of matter in the solar system and surroundings (1644).



Copy from the original book of René Descartes *Le Monde, ou Traité de la Lumière* (1644), reproduced in the edition of M. S. Mahoney, 1979.

Quoted in A. Okabe, B. Boots, K. Sugihara, S. N. Chiu, *Spatial Tessellations*, 2nd ed., J. Wiley & Sons, 2000.

PROXIMITY

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- **Descartes:** distribution of matter in the solar system and surroundings (1644).
- **Dirichlet** in dim 2 and 3, and **Voronoi** in dim d . Study of positive definite quadratic forms. *Dirichlet tessellation* (1850), *Voronoi Diagram* (1908).
- **Crystallography:** regular distribution of the points *Wirkung Sbereich, action domain, activity area, influence area* (end of the XIX century).
- **Meteorology:** rainfall average estimation based on local data, *Thiessen's polygons* (1911).
- **Geology:** estimation of the presence of gold in a sediment, based on trial pits, *influence area polygons* (1909).
- **Physics and Chemistry:** study of the chemical properties of some elements, *Wigner-Seitz's regions*(1933); alloy structure modelling, *atom domain* (1958).
- **Ecology:** survival of organisms in competition for nourishment or light, like trees in a forest, *area potentially available* (1965), *plants polygons* (1966)
- **Medicine:** region of muscle tissue supplied by a capillary, *capillarity domains* (1985)

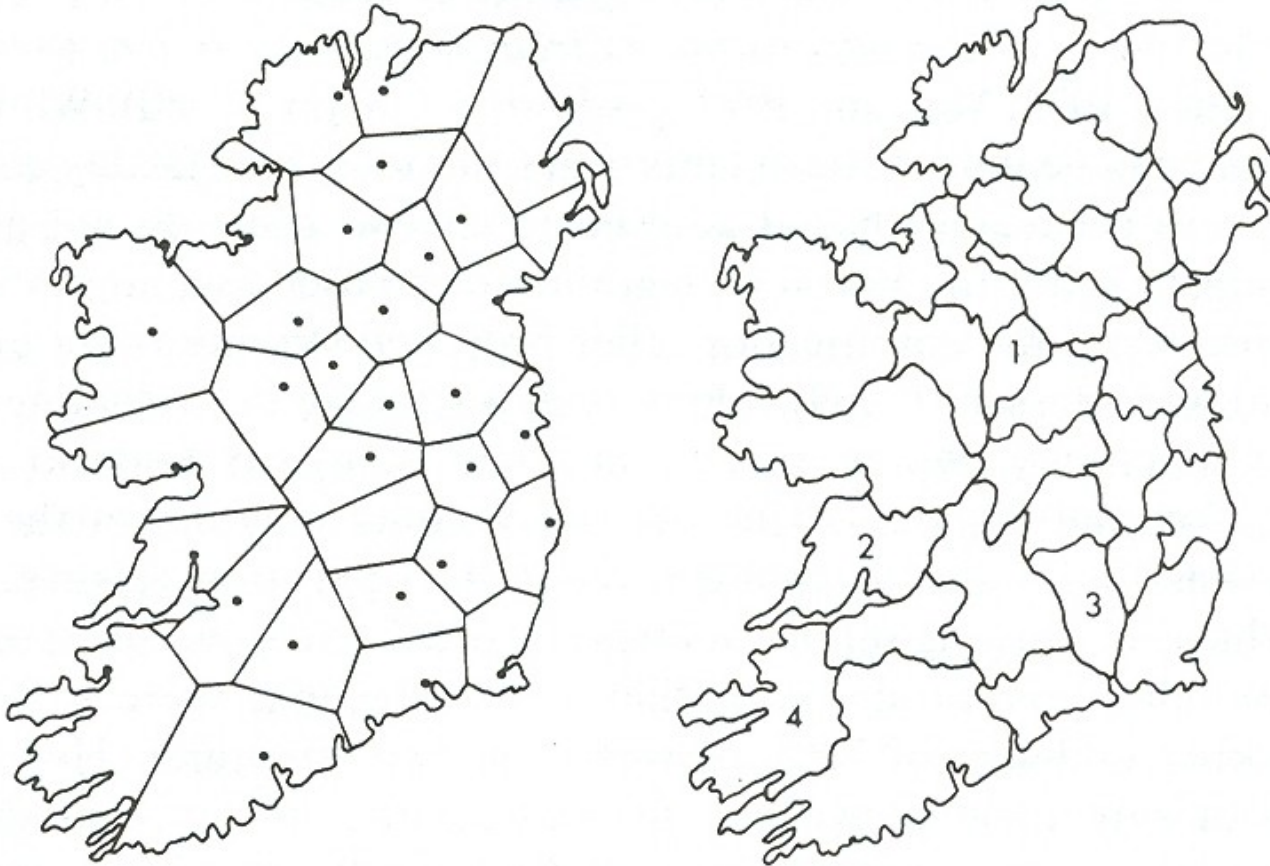
PROXIMITY

History of the rediscoveries of the Voronoi diagram (continued)

- **Economics:** market areas of centers in competition, under various conditions of market prices and transportation costs (since the mid XIX century)
- **Archaeology:** study of the diffusion of the use of tools in order to analyze the influence of the civilization centers (sixties-seventies)
- **Anthropology:** modelling territorial human organization under several aspects (sixties-seventies)

PROXIMITY

More recent applications of the Voronoi diagram

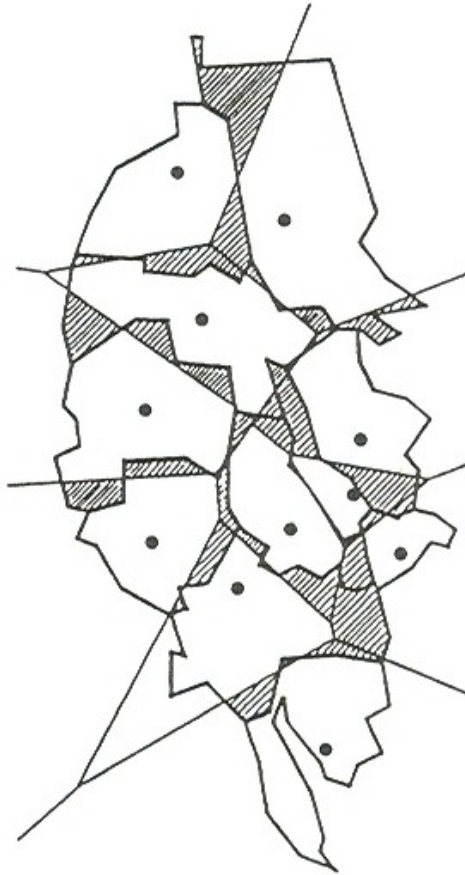


Ireland counties. On the left, theoretical partition created with the Voronoi diagram of the counties capitals. On the right, the real counties.

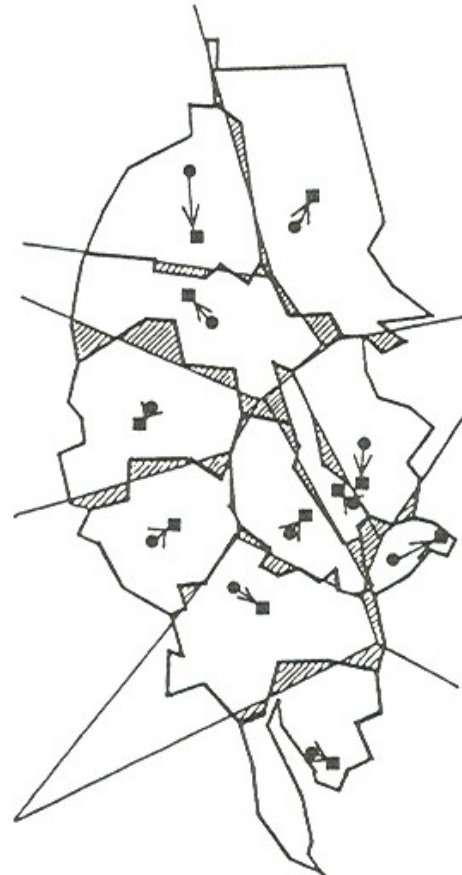
K. R. Cox, J. A. Agnew, *The spatial correspondence of theoretical partitions*, Discussion paper, Syracuse University, Dept. of Geography, 1976. Quoted in A. Okabe, B. Boots, K. Sugihara, S. N. Chiu, *Spatial Tessellations*, 2nd ed., J. Wiley & Sons, 2000.

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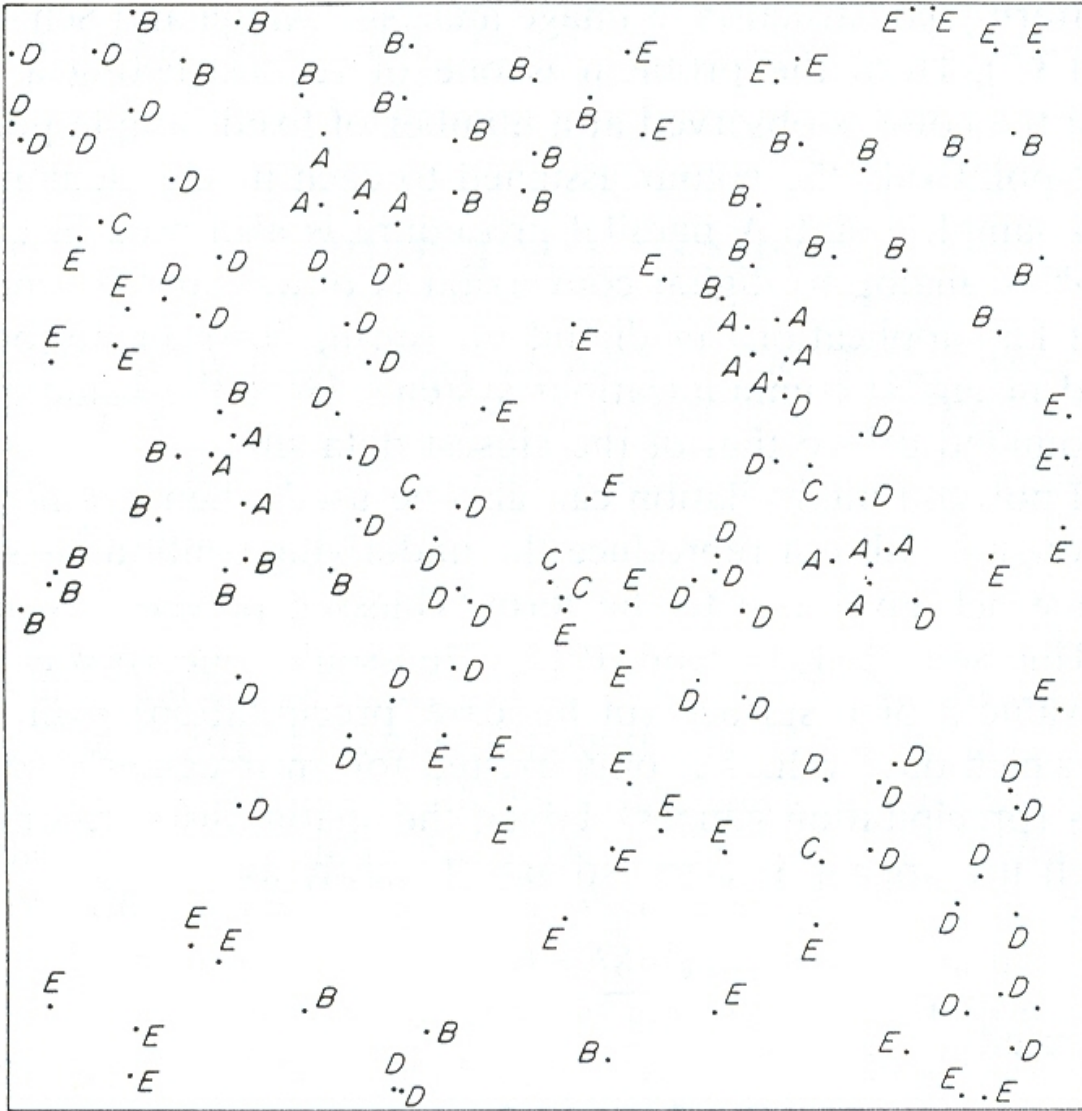
The school districts at Tsukuba. Left, real districts and the location of the schools (circles). Right, relocation of the schools (squares) as to minimizing the area of the students that do not attend their closest school.



A. Suzuki, M. Iri, *Aproximation on a tessellation of the plane by a Voronoi diagram*, journal of the Operation Research Society of Japan, 29, 1986. Quoted in A. Okabe, B. Boots, K. Sugihara, S. N. Chiu, *Spatial Tessellations*, 2nd ed., J. Wiley & Sons, 2000.

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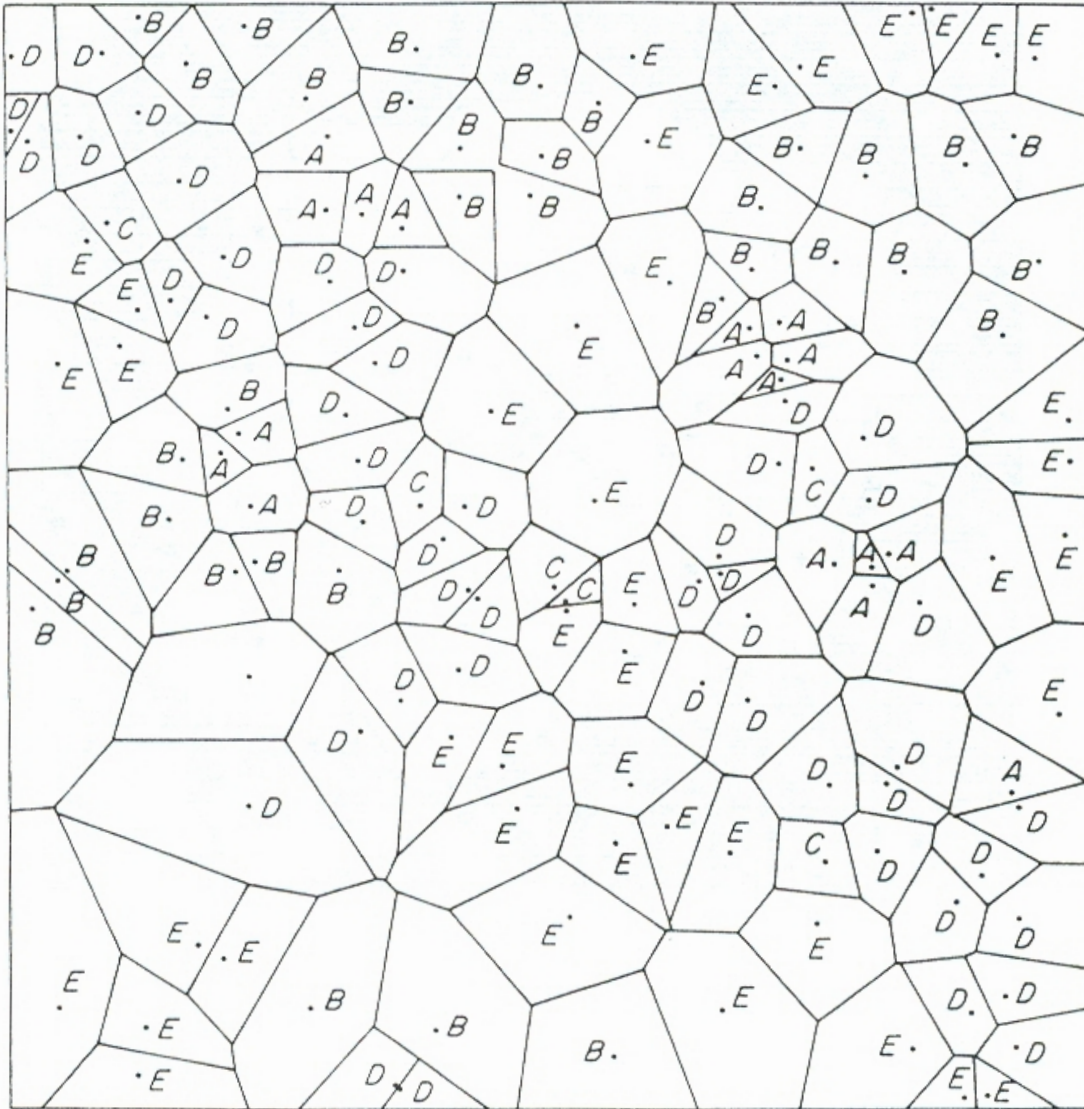


Interpolation of discrete sample values

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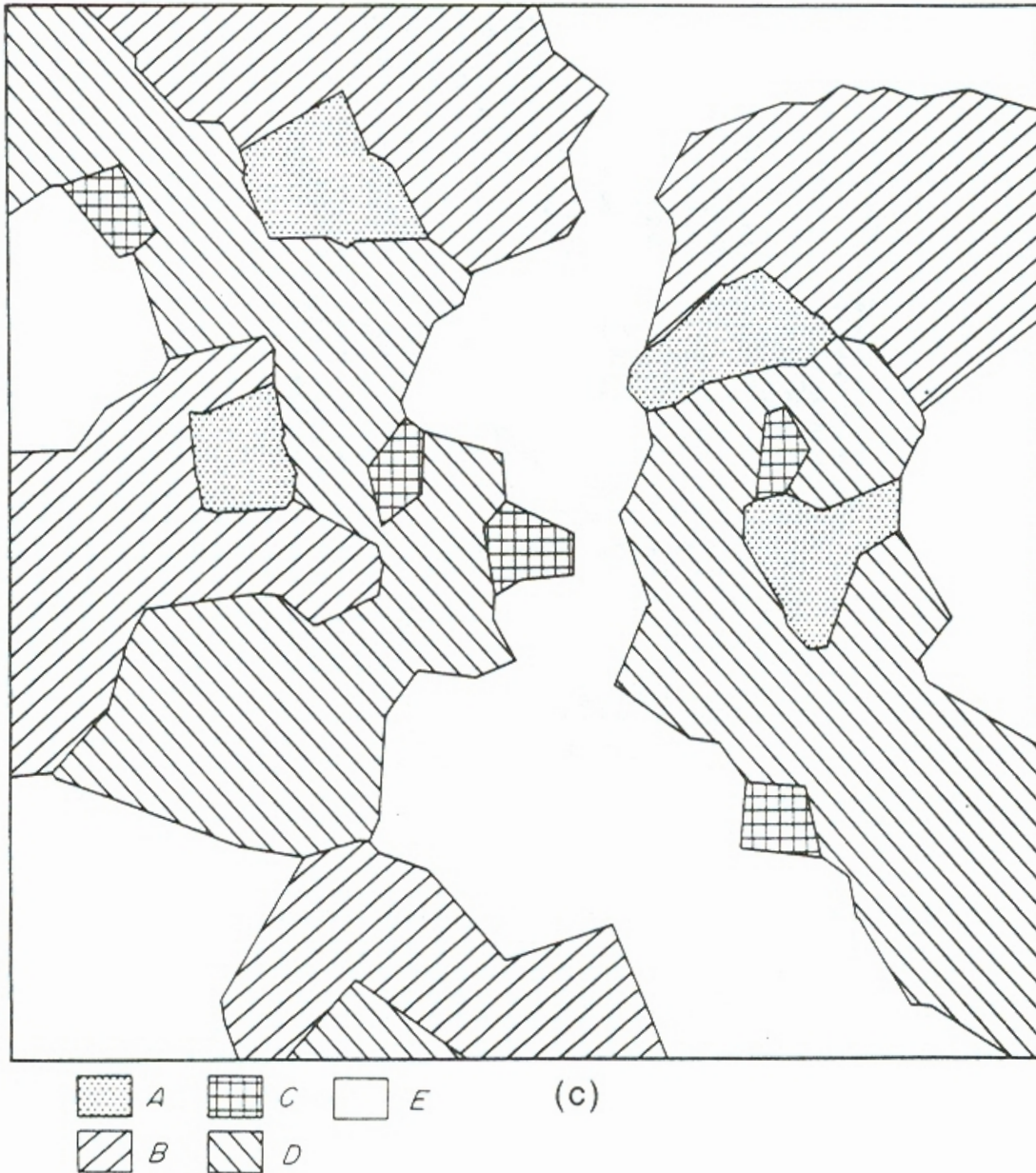


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PROXIMITY

Constructing the Voronoi diagram

Sixties

- Application concepts clear
- Lack of tools to build the diagrams

Seventies

- Start of the development of algorithm to compute the diagrams, thanks to computer science
- Discovery of new properties associated to the diagrams

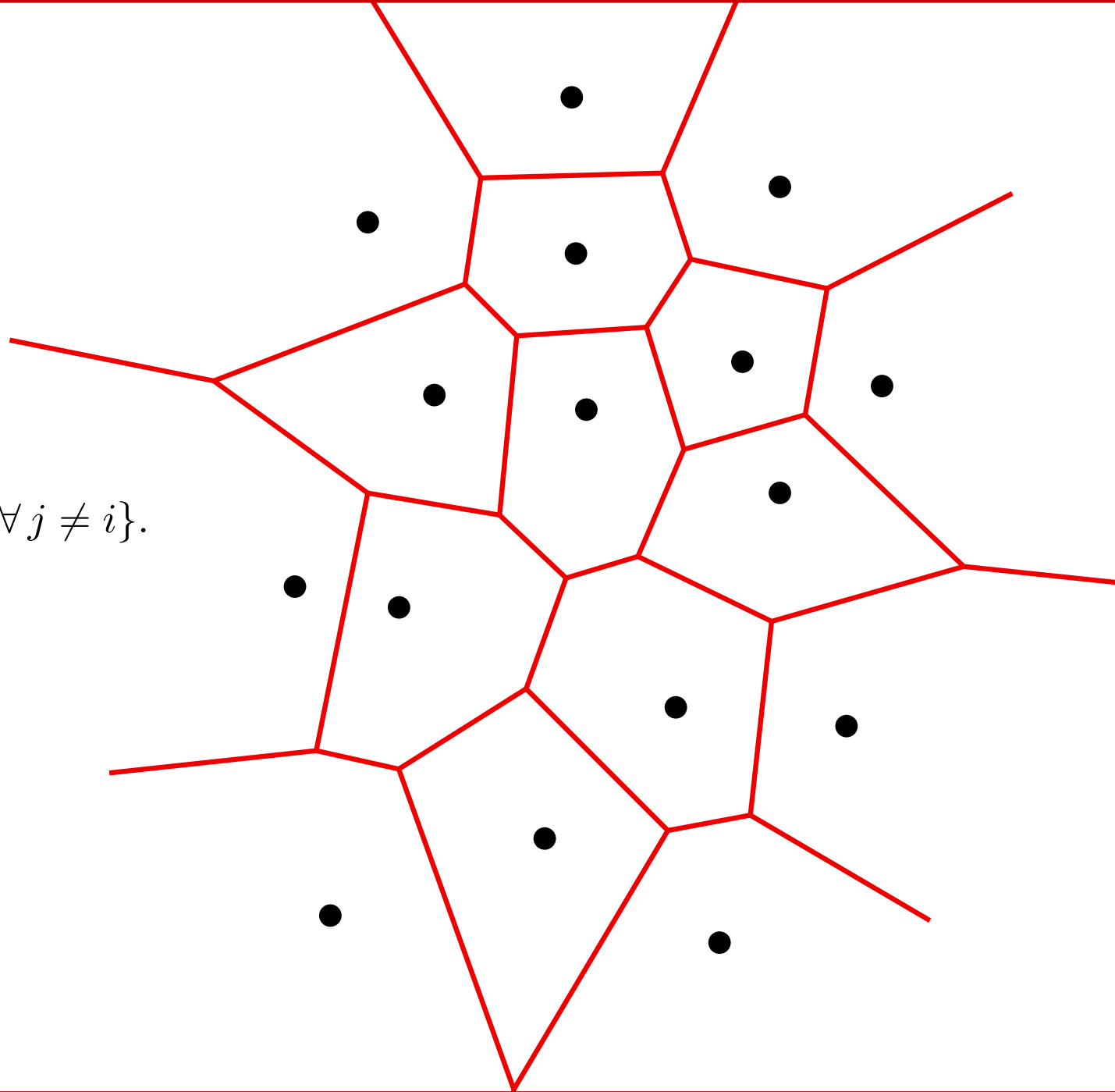
PROXIMITY

DEFINITION

Let $P = \{p_1, \dots, p_n\}$ be a finite set of points in the plane.

The *Voronoi diagram* of P is the decomposition of the plane $V(P) = \{V(p_1), \dots, V(p_n)\}$ such that

$$V(p_i) = \{x \mid d(x, p_i) \leq d(x, p_j) \ \forall j \neq i\}.$$



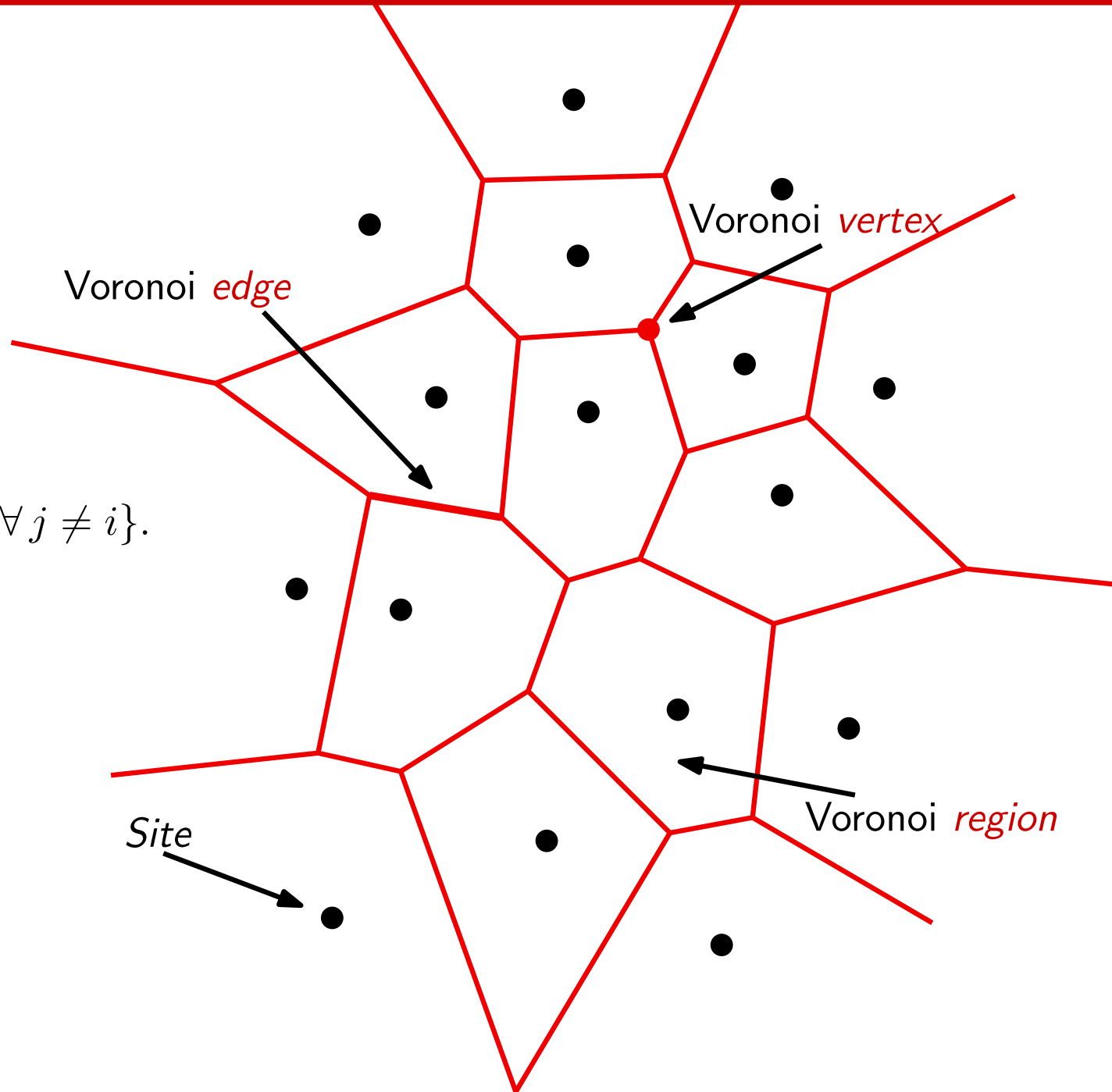
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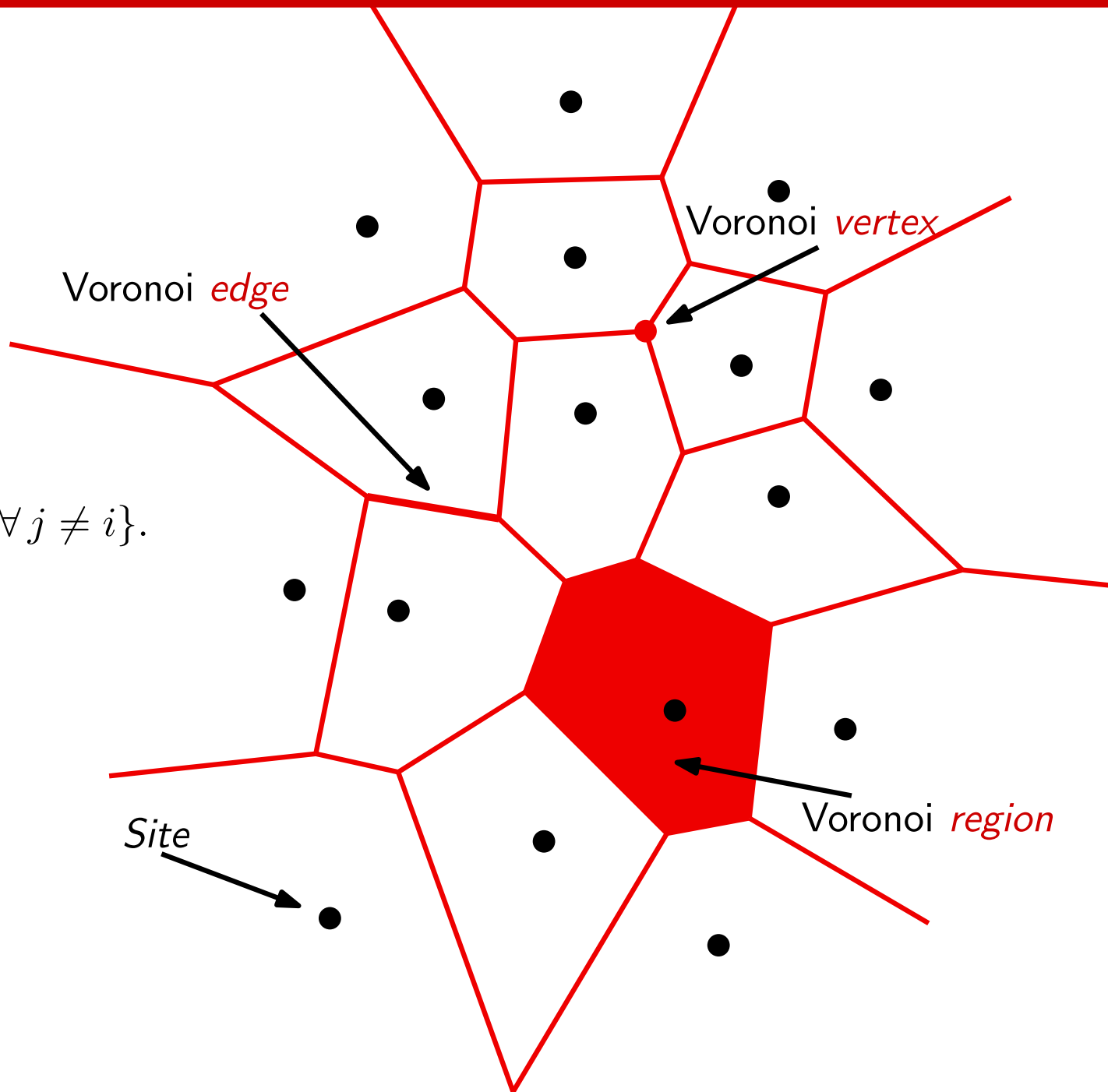
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CHARACTERIZATION

If b_{ij} is the perpendicular bisector of the segment $p_i p_j$, and H_{ij} is the halfplane defined by b_{ij} enclosing p_i , then

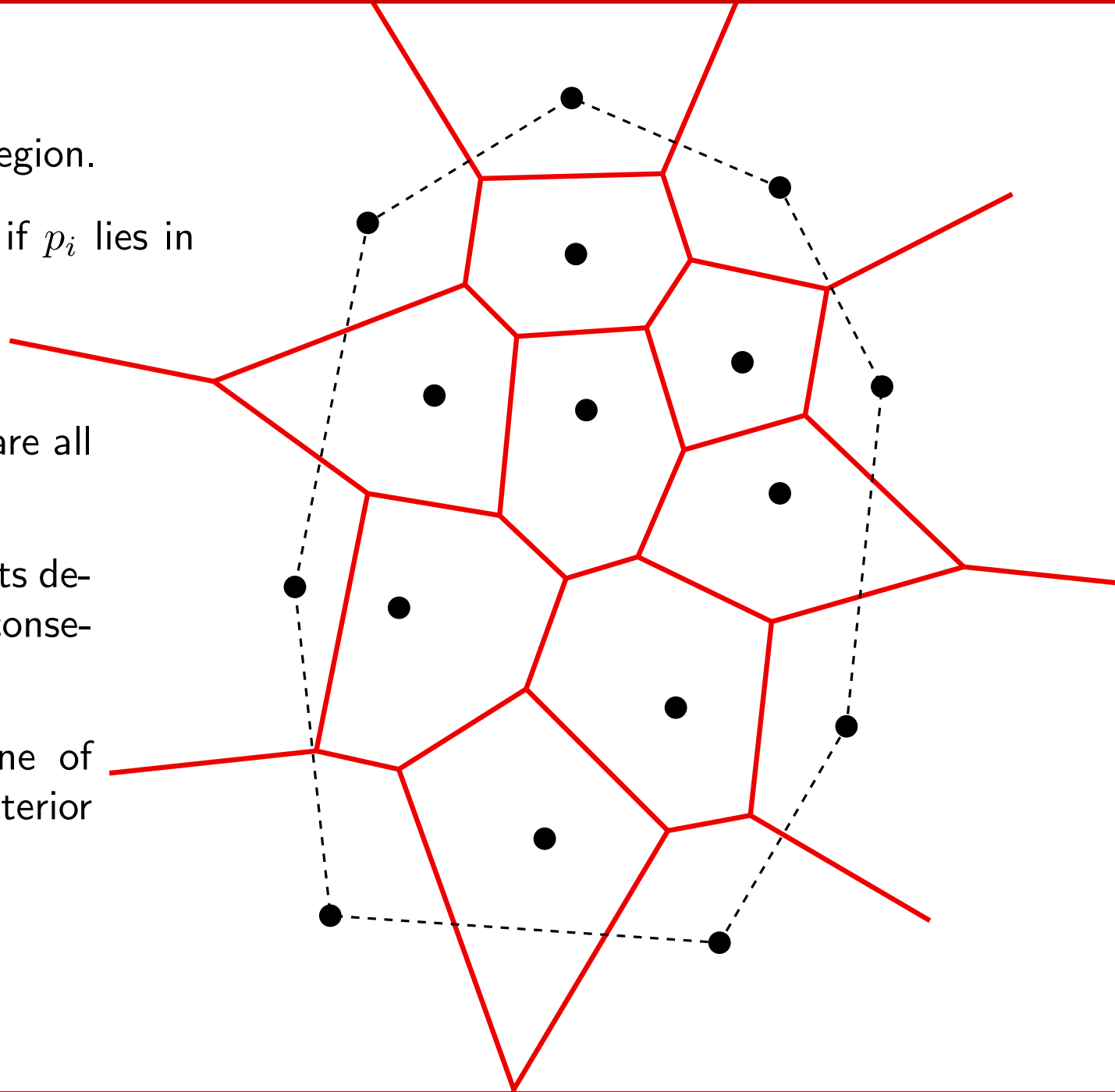
$$V(p_i) = \bigcap_{j \neq i} H_{ij}.$$



PROXIMITY

PROPERTIES

1. $V(p_i)$ is a convex polygonal region.
2. $V(p_i)$ is bounded if and only if p_i lies in the interior of $ch(P)$.
3. Voronoi edges can be:
 - lines, if the points of P are all aligned;
 - half-lines, if the two points determining the edge are consecutive vertices of $ch(P)$;
 - segments, if at least one of the points lies in the interior of $ch(P)$.



PROXIMITY

PROPERTIES (continued)

4. The planarity of the Voronoi diagram allows applying Euler's formula to the graph adding a vertex at infinity:

$$v + n = e + 1.$$

Since each vertex is incident to at least 3 edges, and each edge has exactly 2 endpoints,

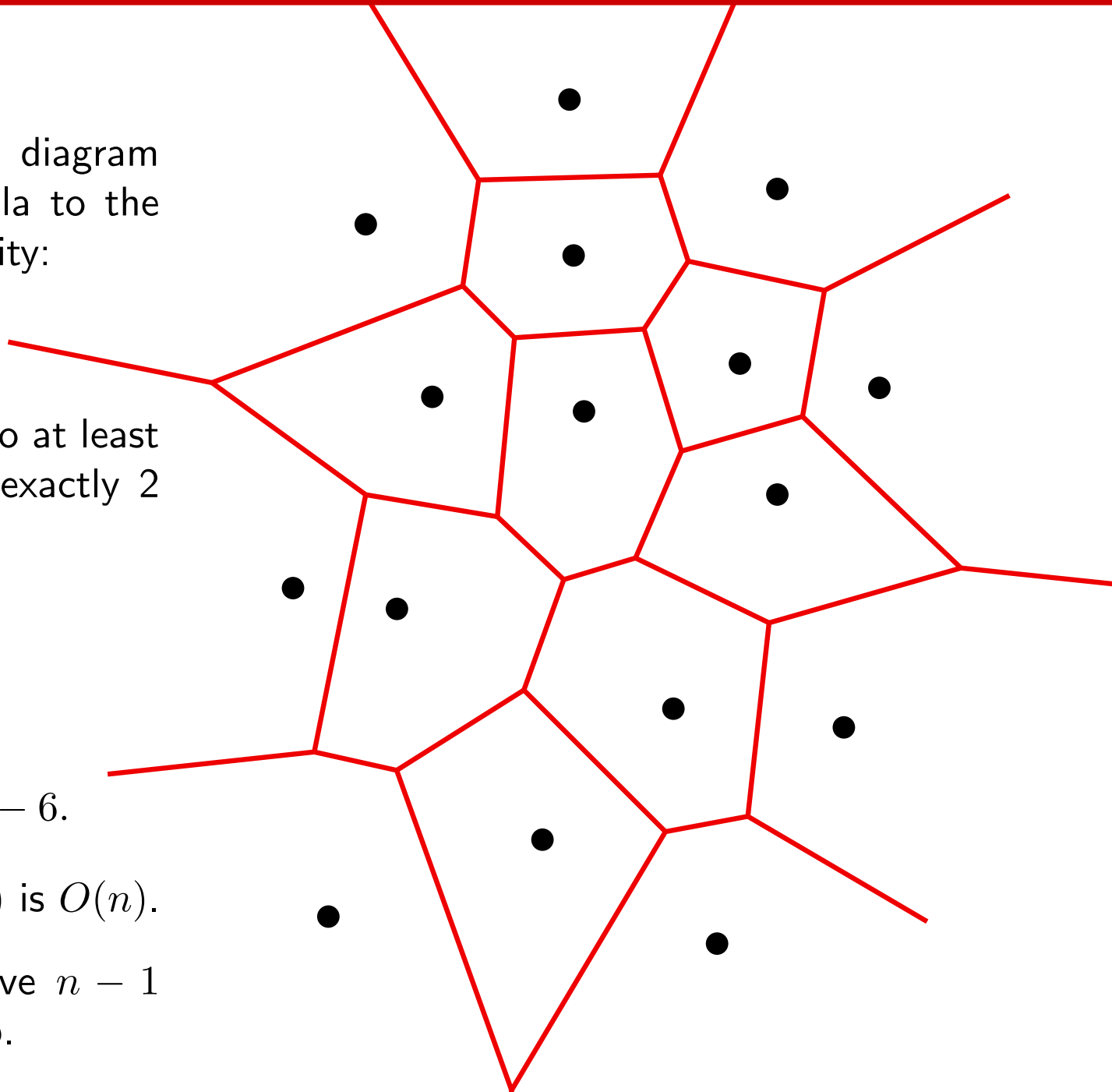
$$2e \geq 3(v + 1),$$

from where we get:

$$v \leq 2n - 5, \quad e \leq 3n - 6.$$

Thus, the complexity of $V(P)$ is $O(n)$.

5. Although each $V(p_i)$ can have $n - 1$ edges, on average they have 6.



PROXIMITY

THE DUAL GRAPH

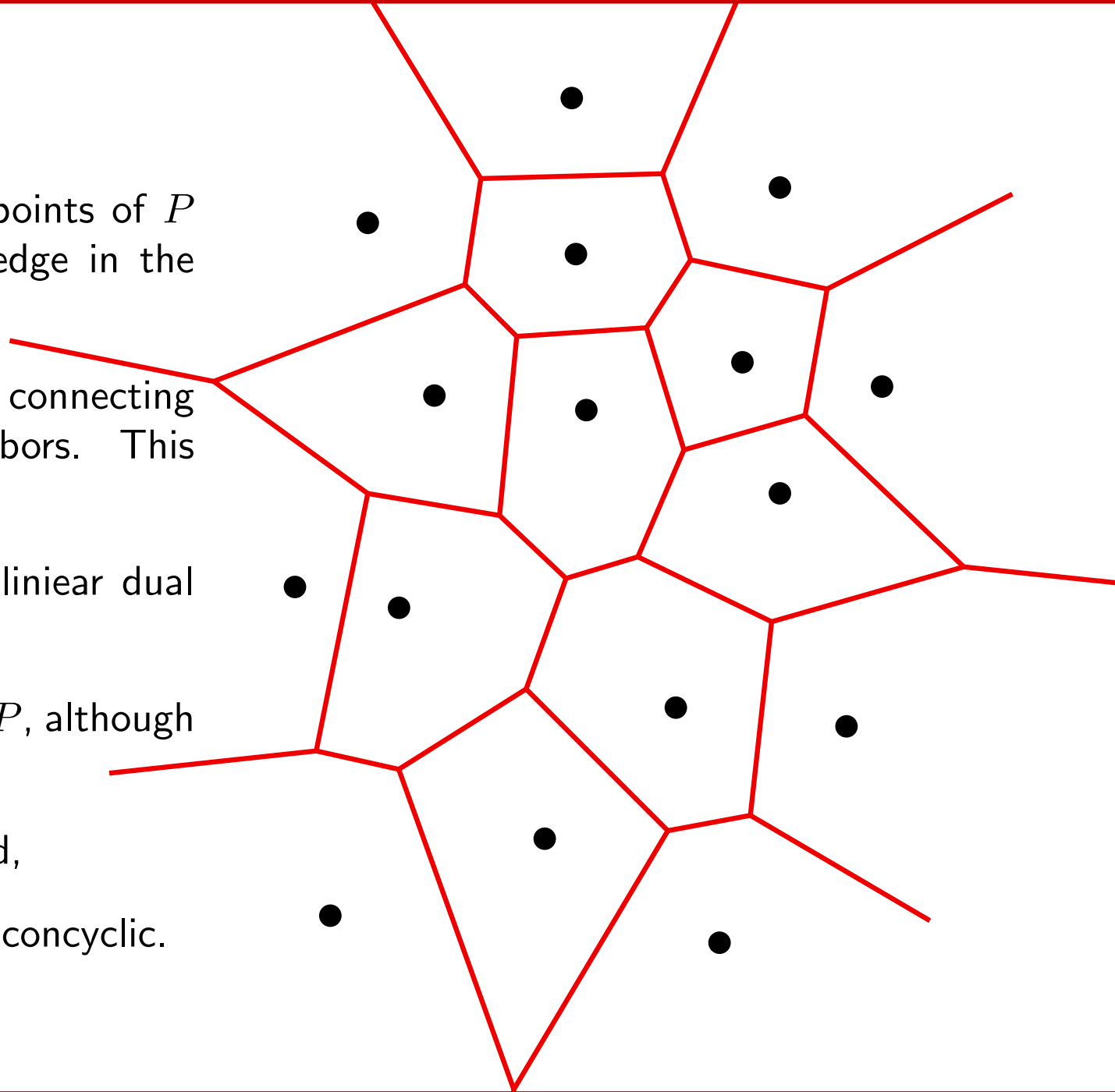
Proximity between points: two points of P are neighbors if they share an edge in the Voronoi diagram of P .

A proximity graph is obtained by connecting each point to its Voronoi neighbors. This graph is called *Delaunay graph*.

The Delaunay graph is the rectilinear dual of the Voronoi diagram!

In general, it is a triangulation of P , although sometimes it is not:

- when the points are aligned,
- when 4 or more points are concyclic.



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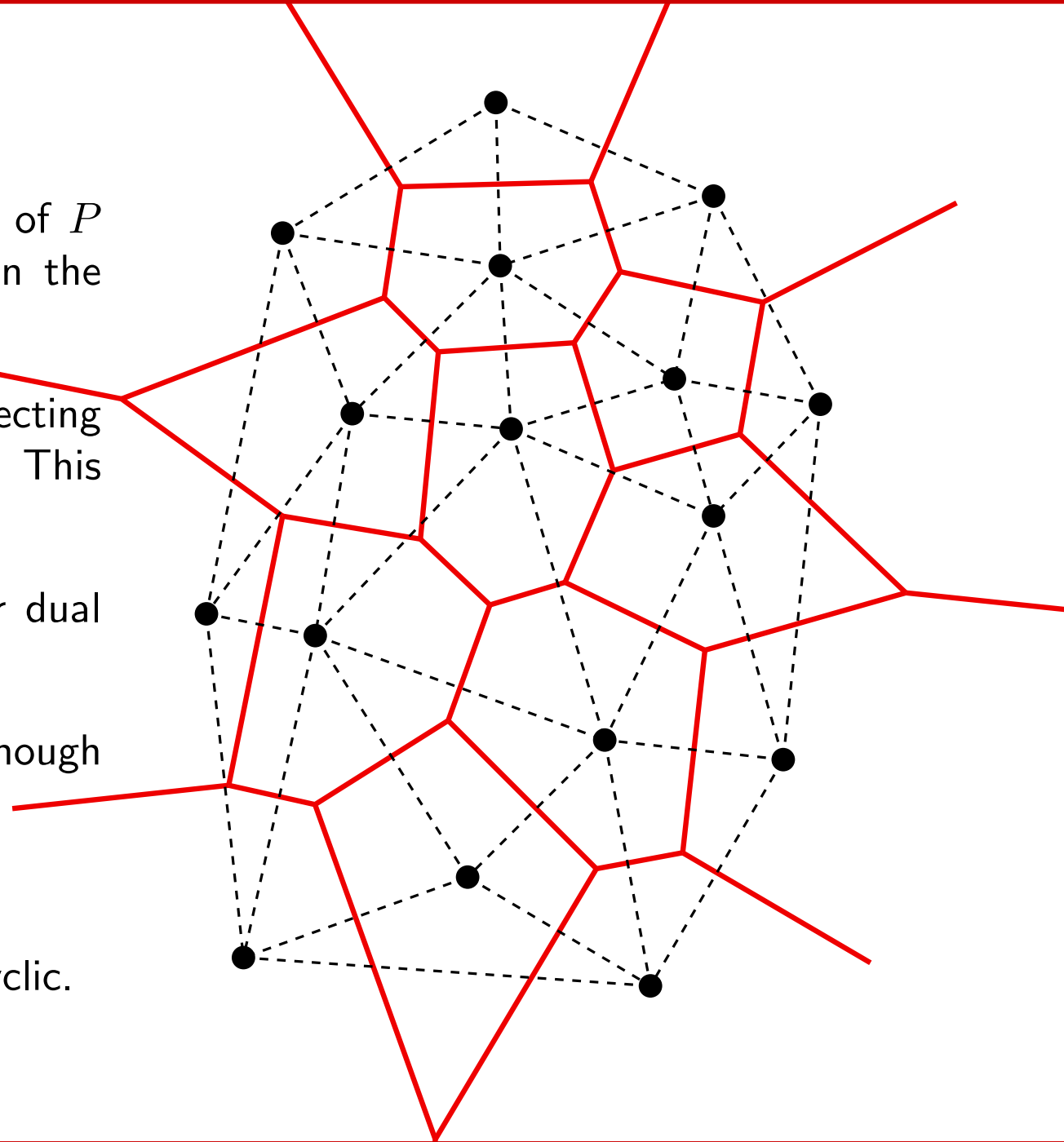
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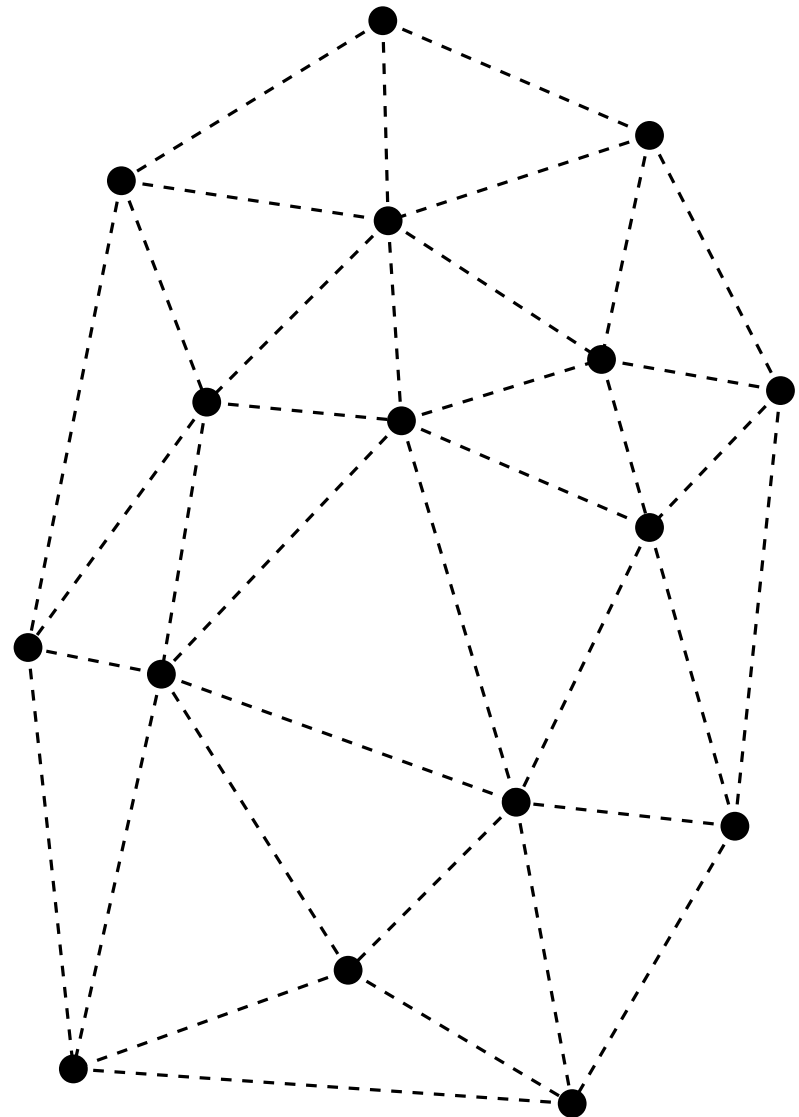
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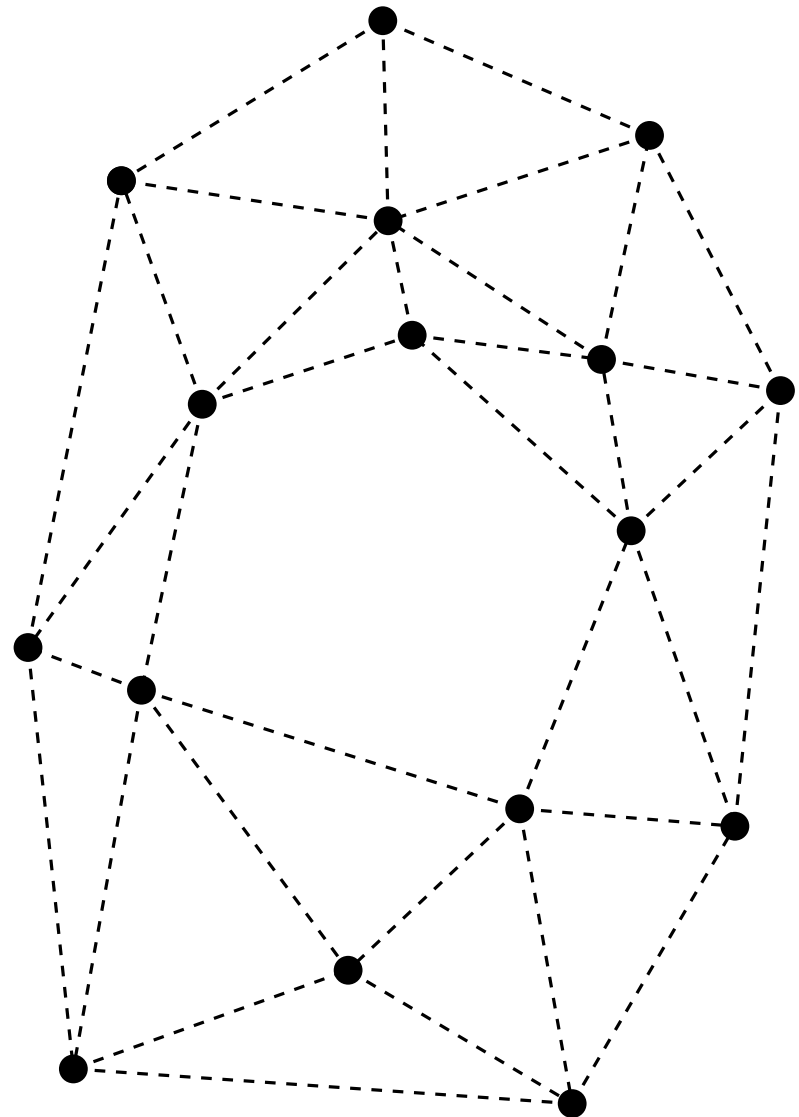
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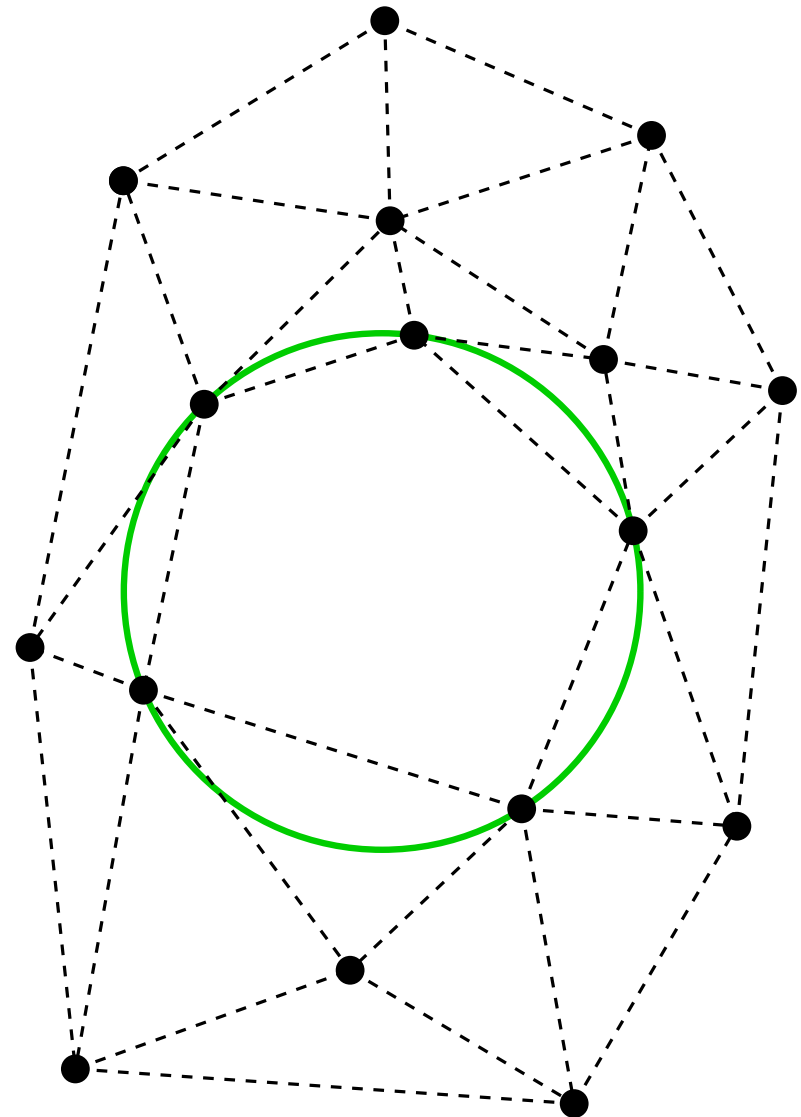
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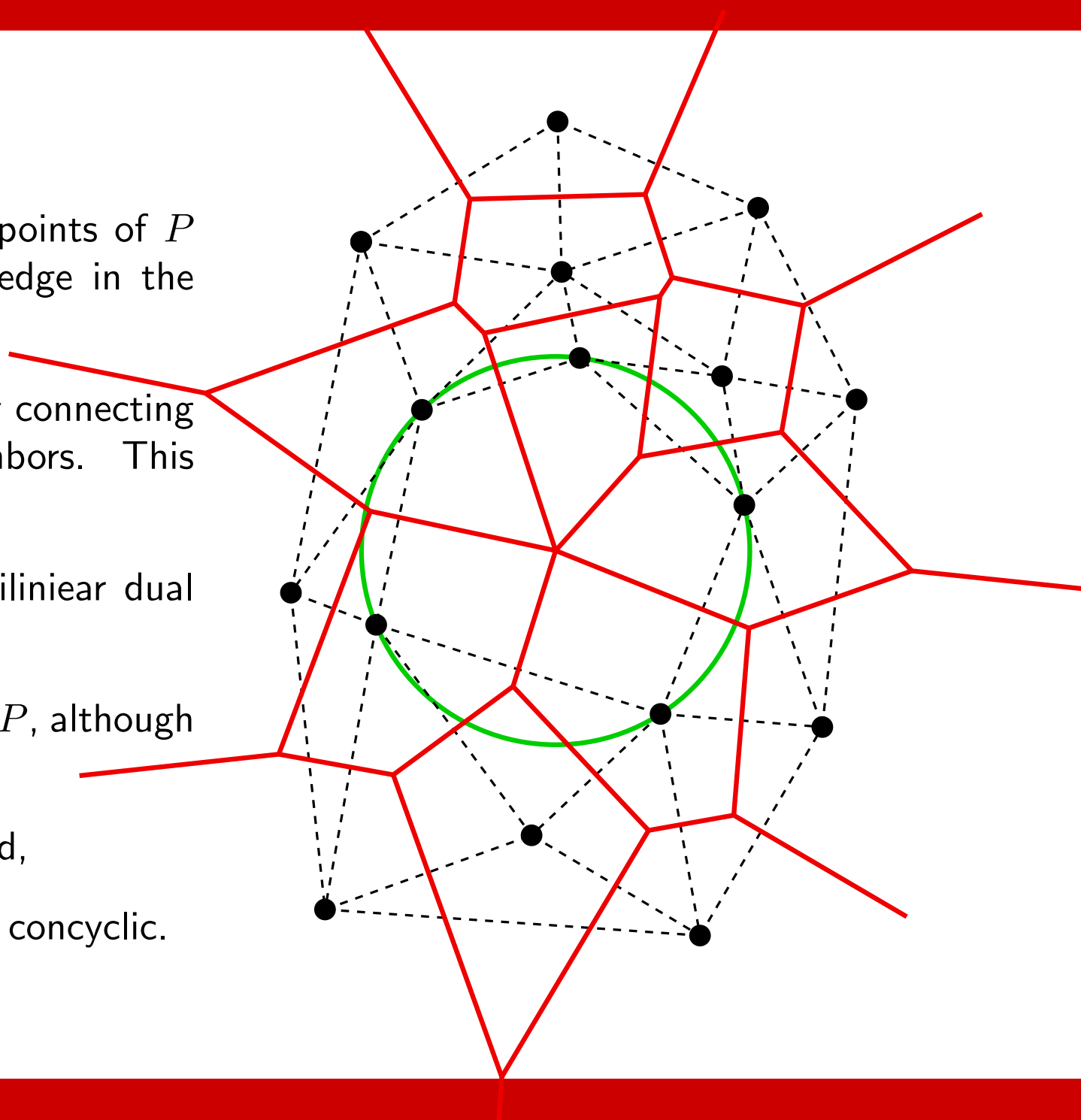
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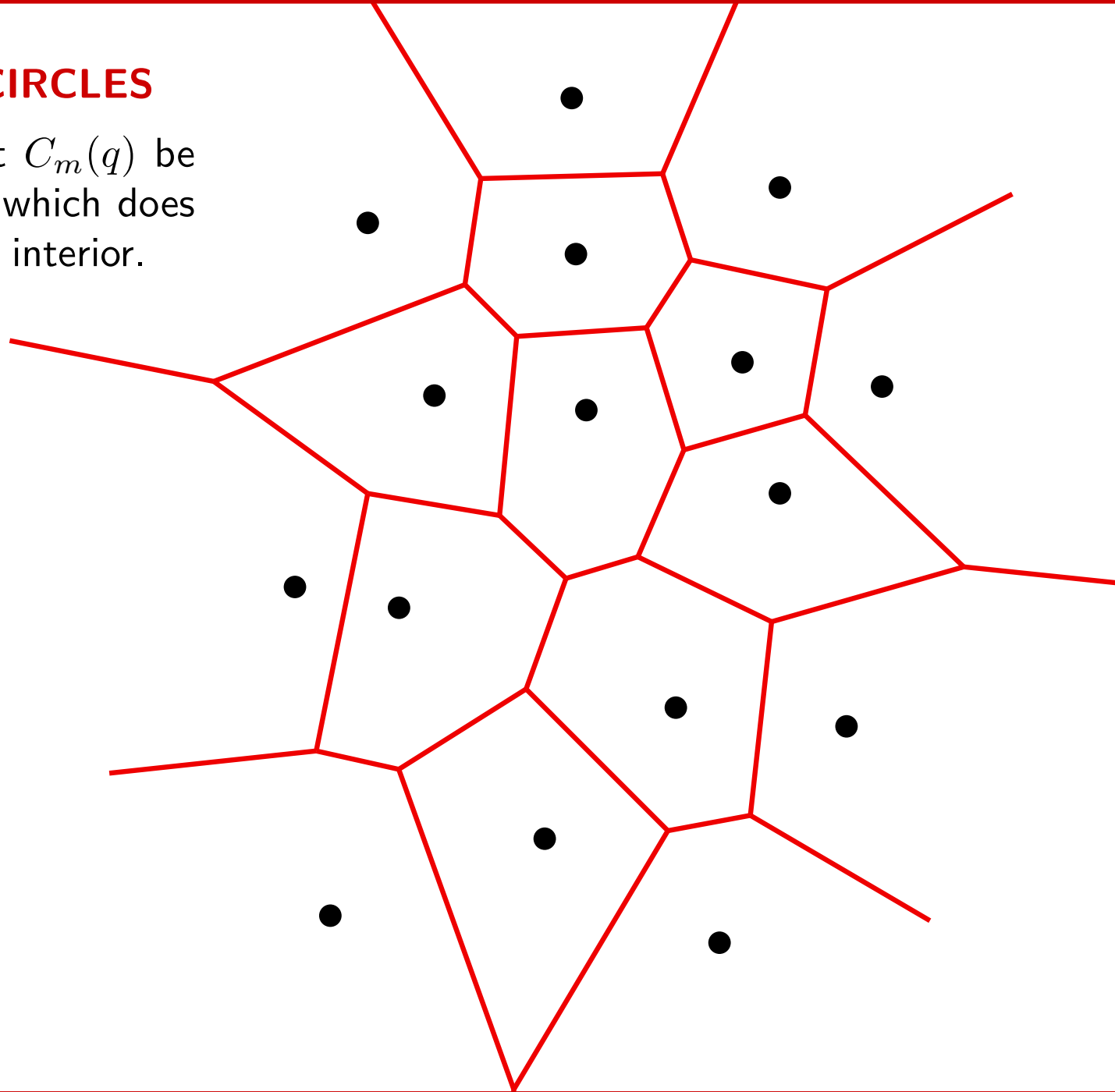
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PROXIMITY

PROXIMITY AND EMPTY CIRCLES

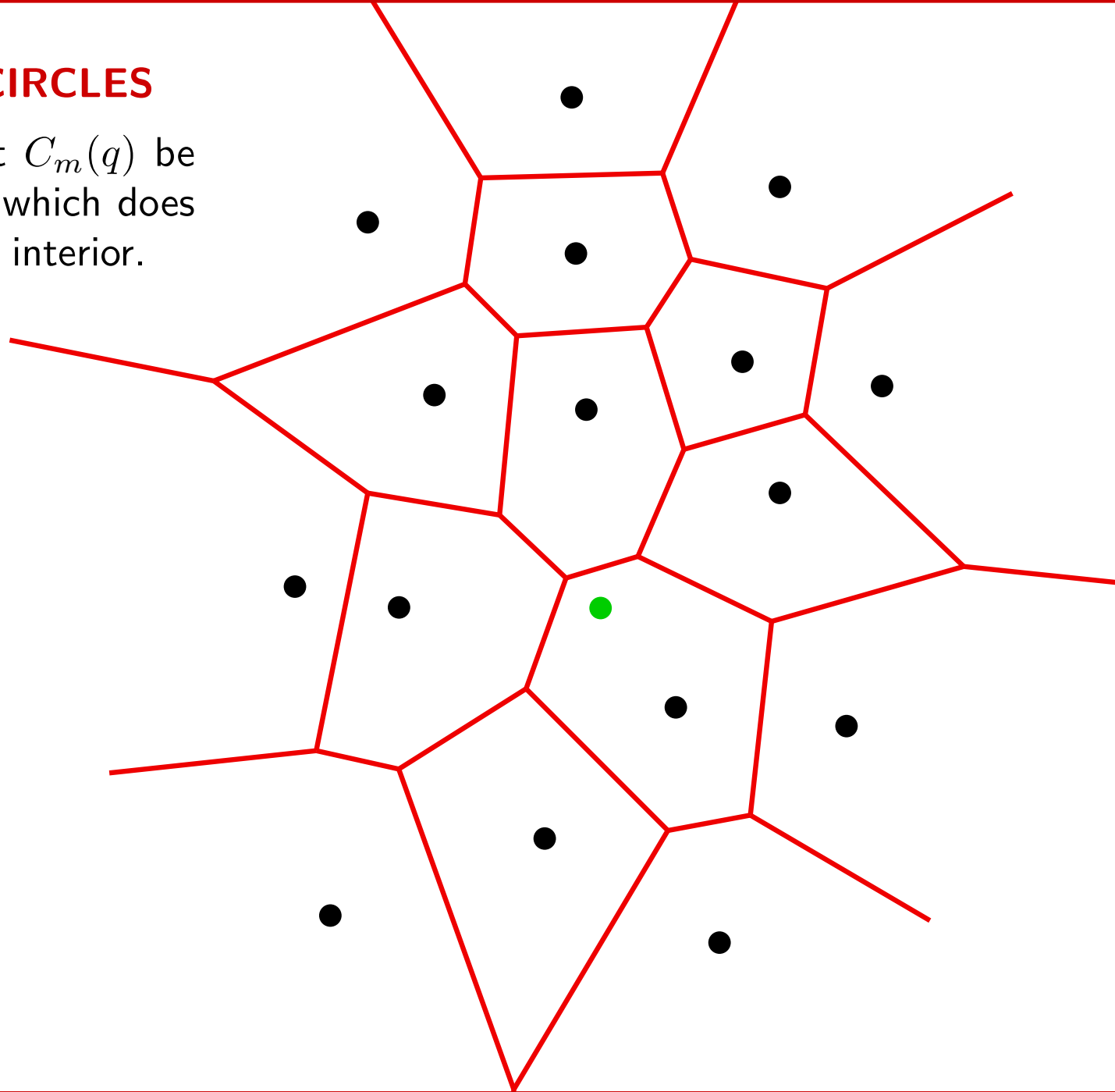
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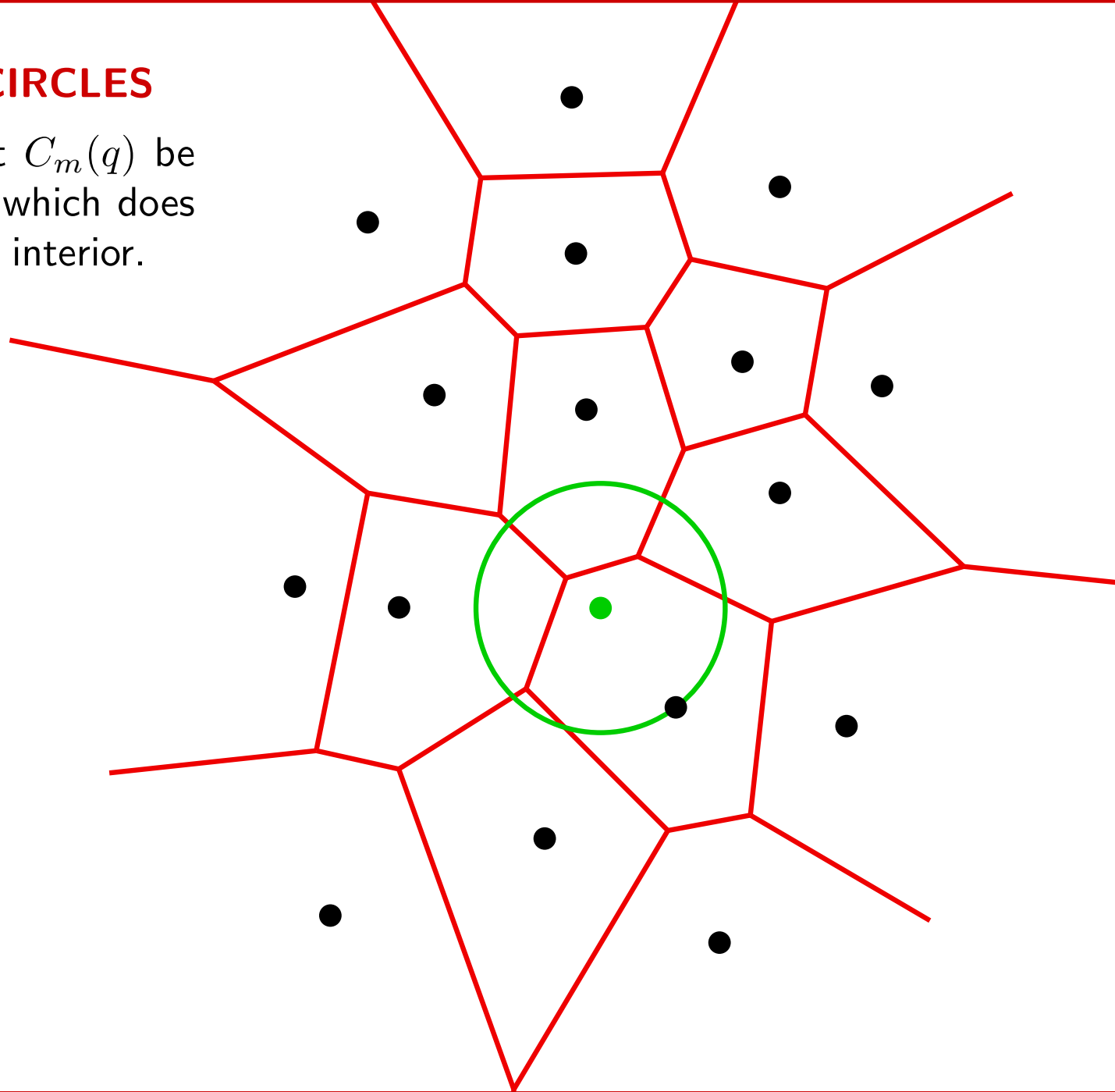
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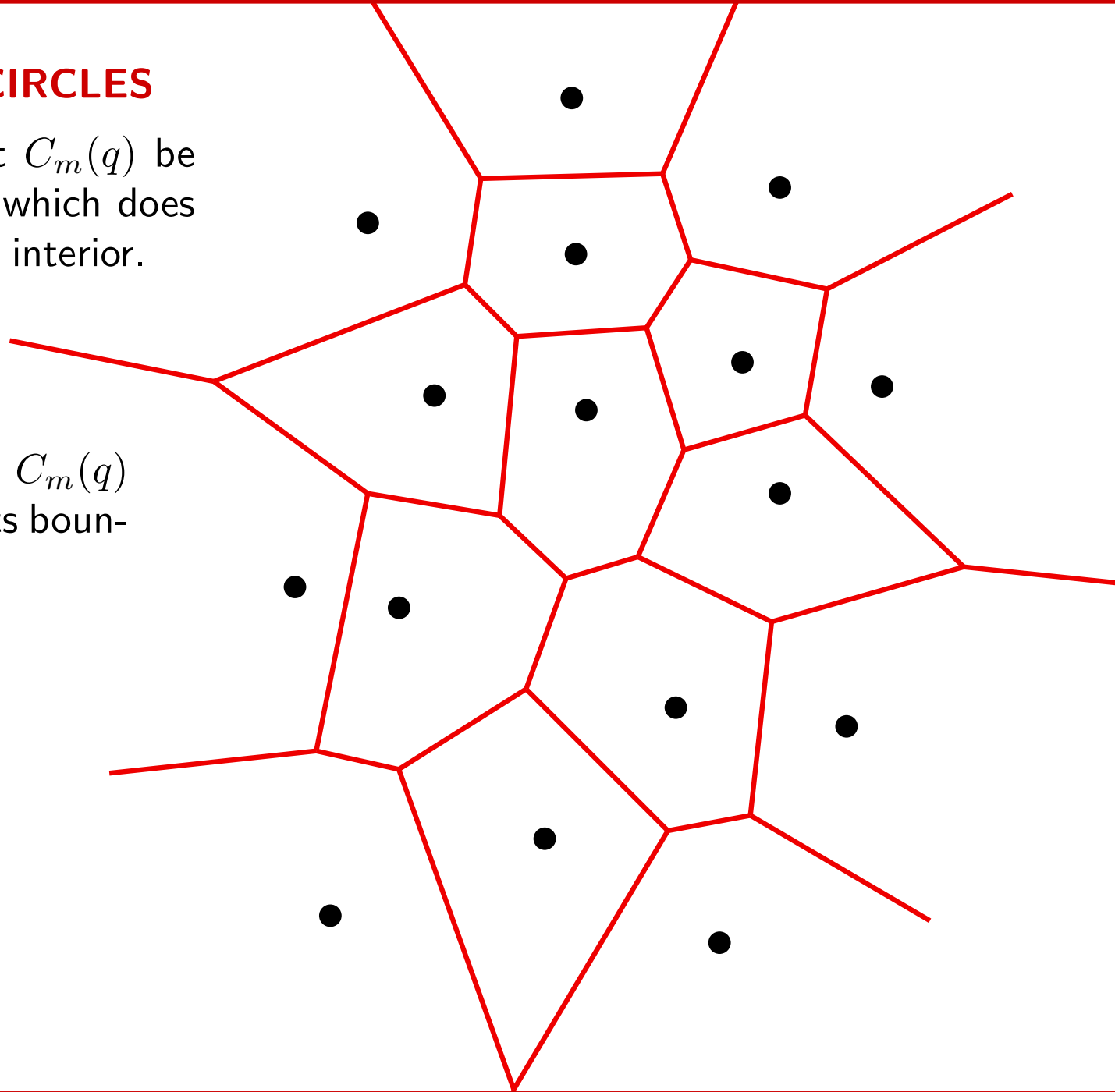
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A point q is a Voronoi vertex iff $C_m(q)$ has three or more points of P in its boundary.



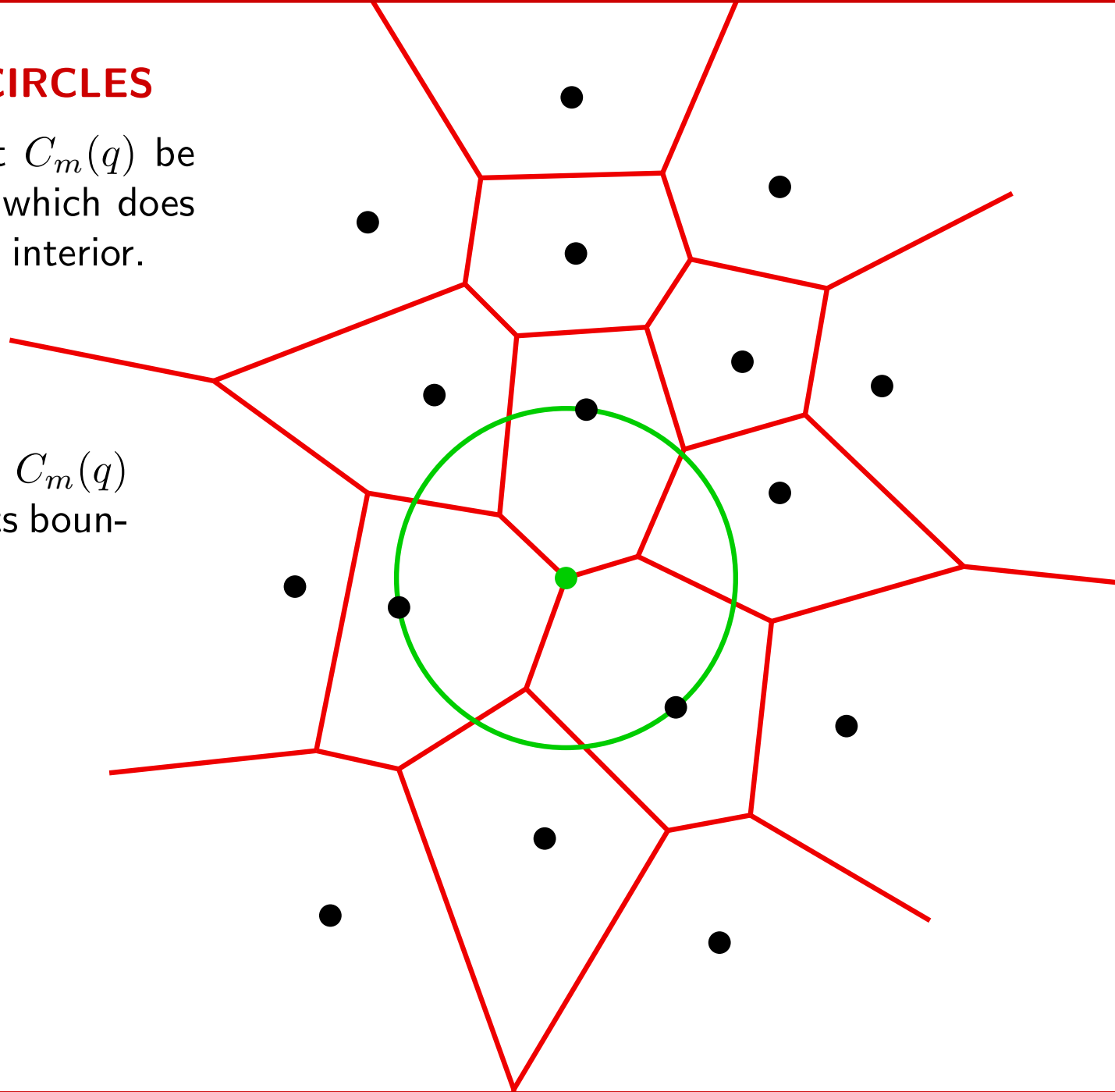
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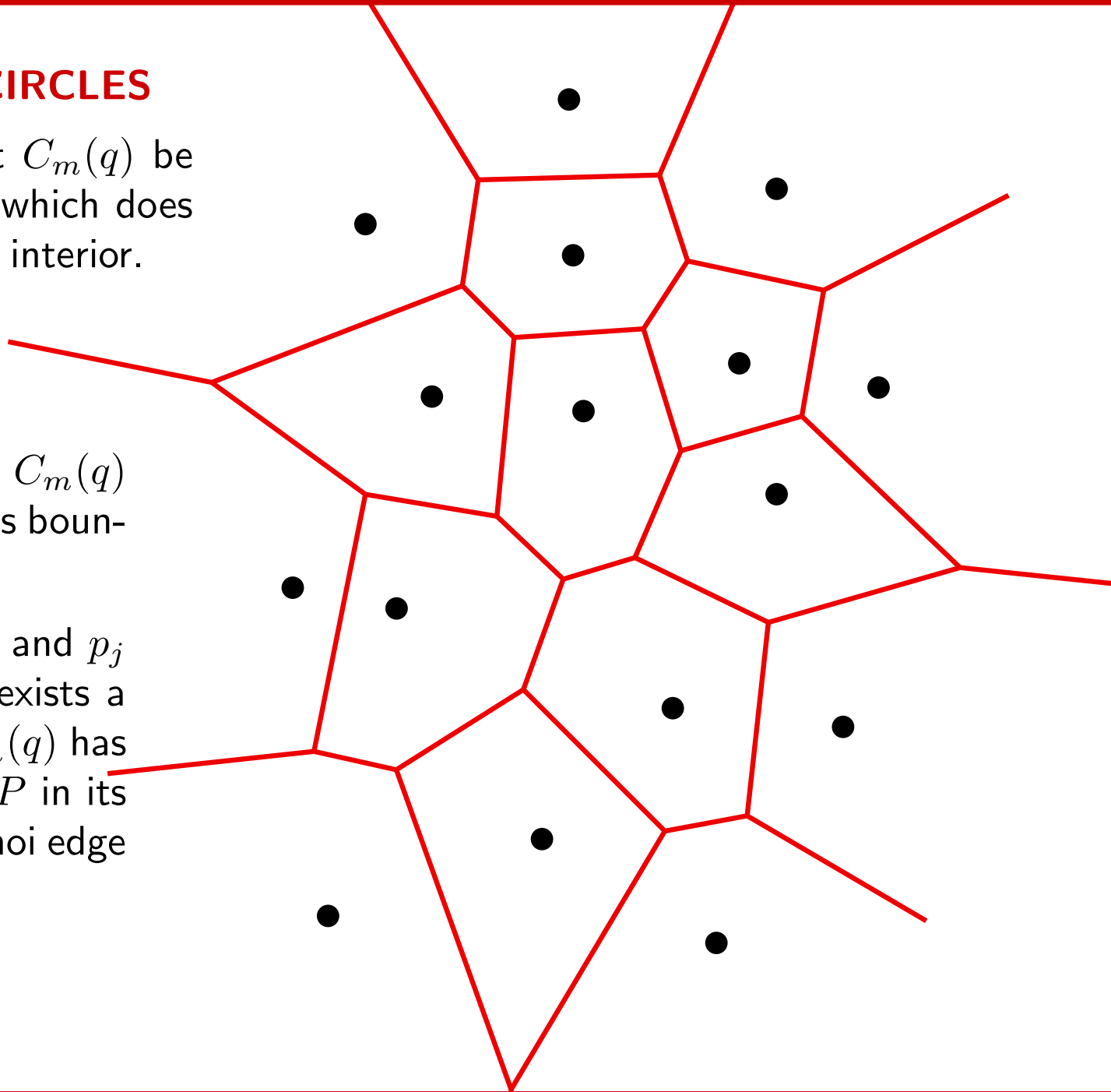
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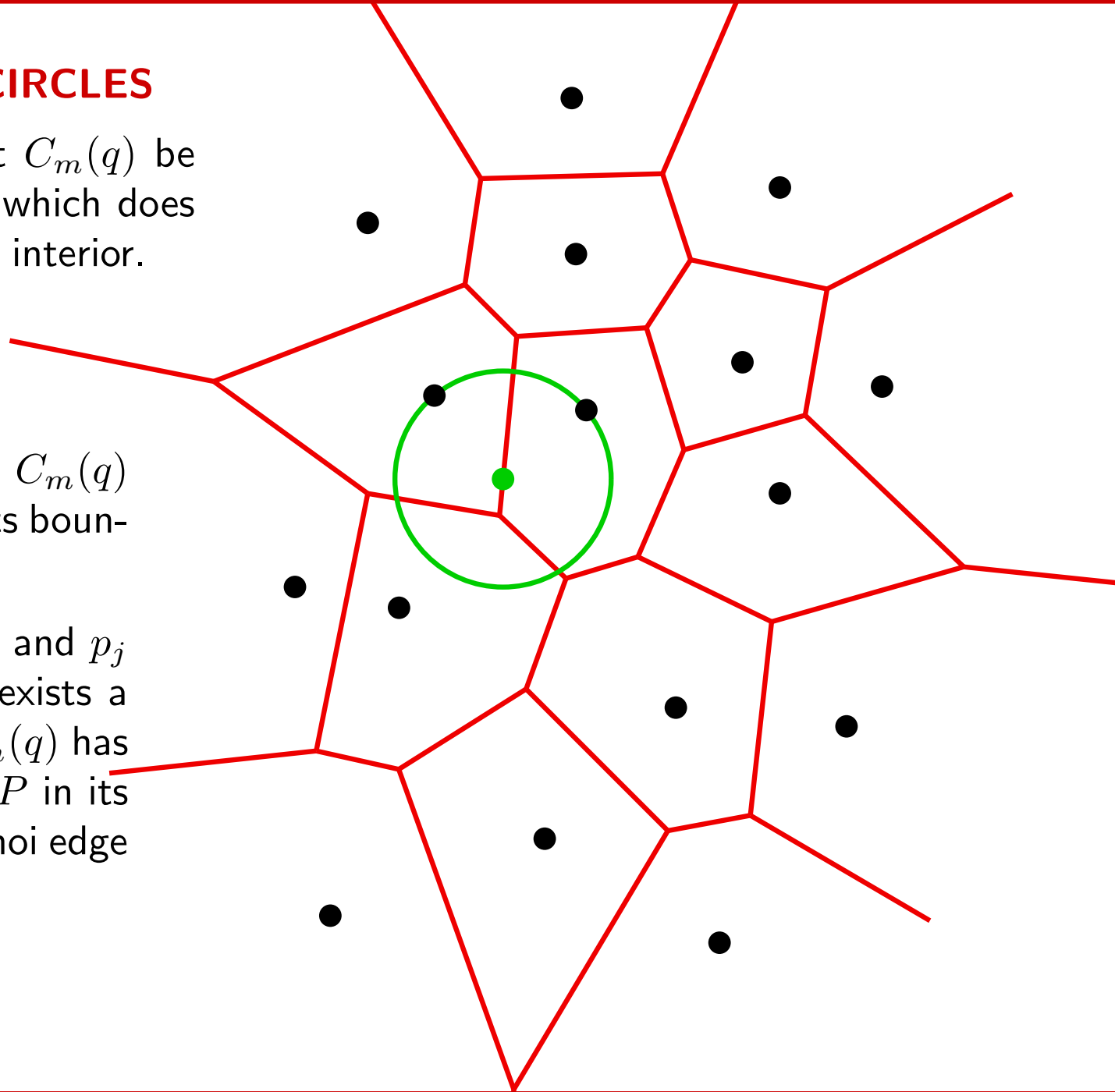
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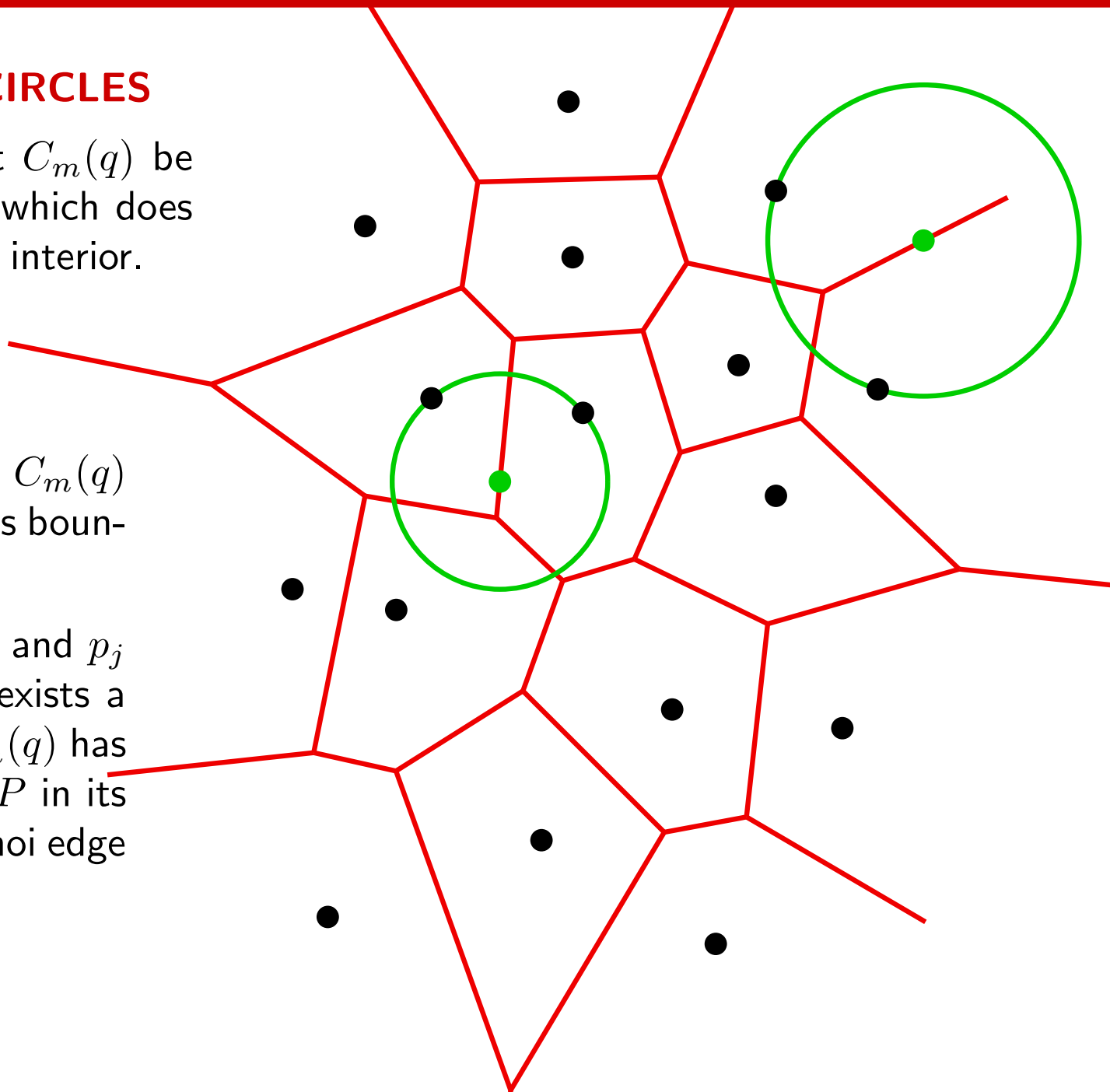
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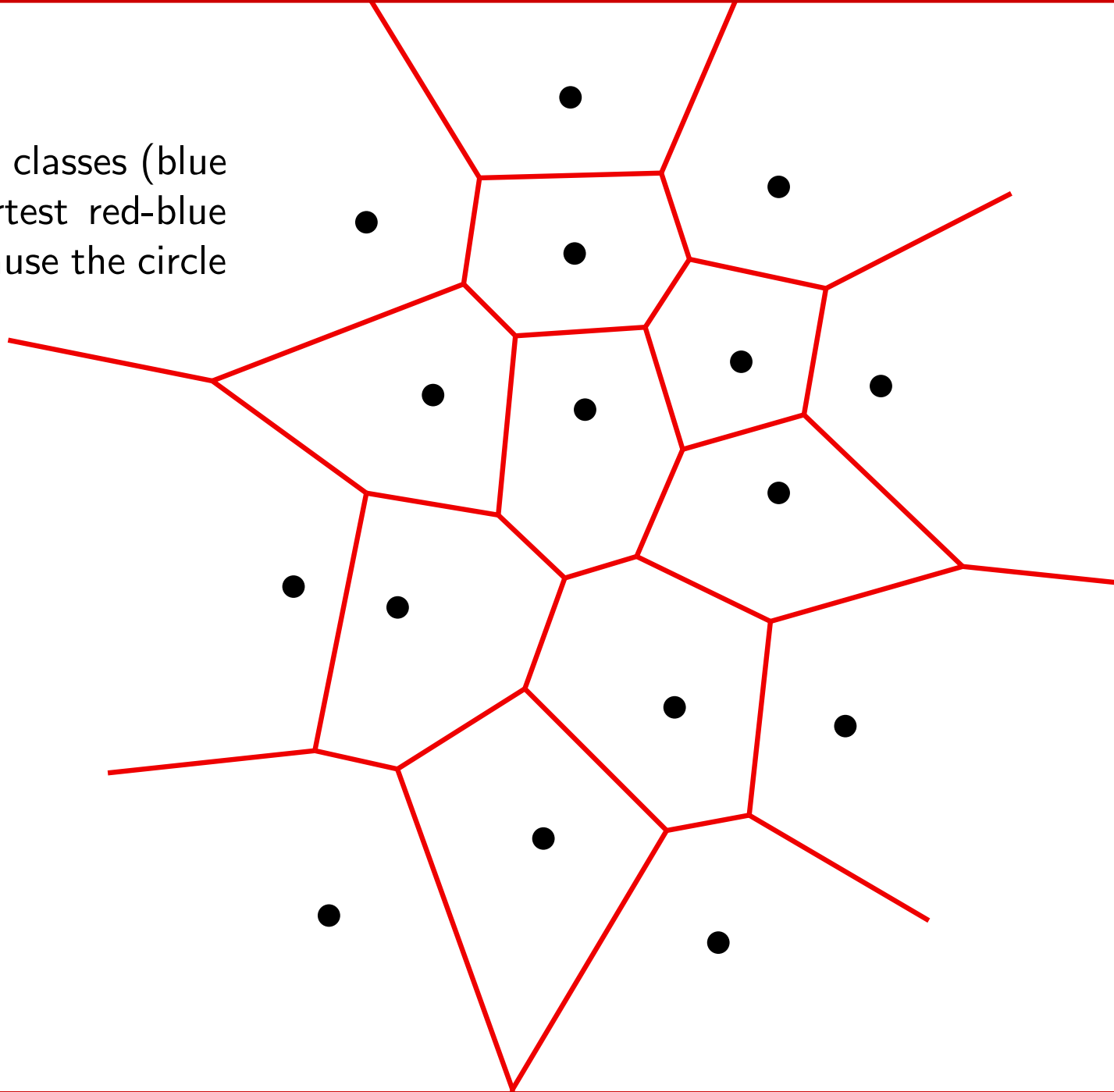
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PROXIMITY

Corollary (blue and red)

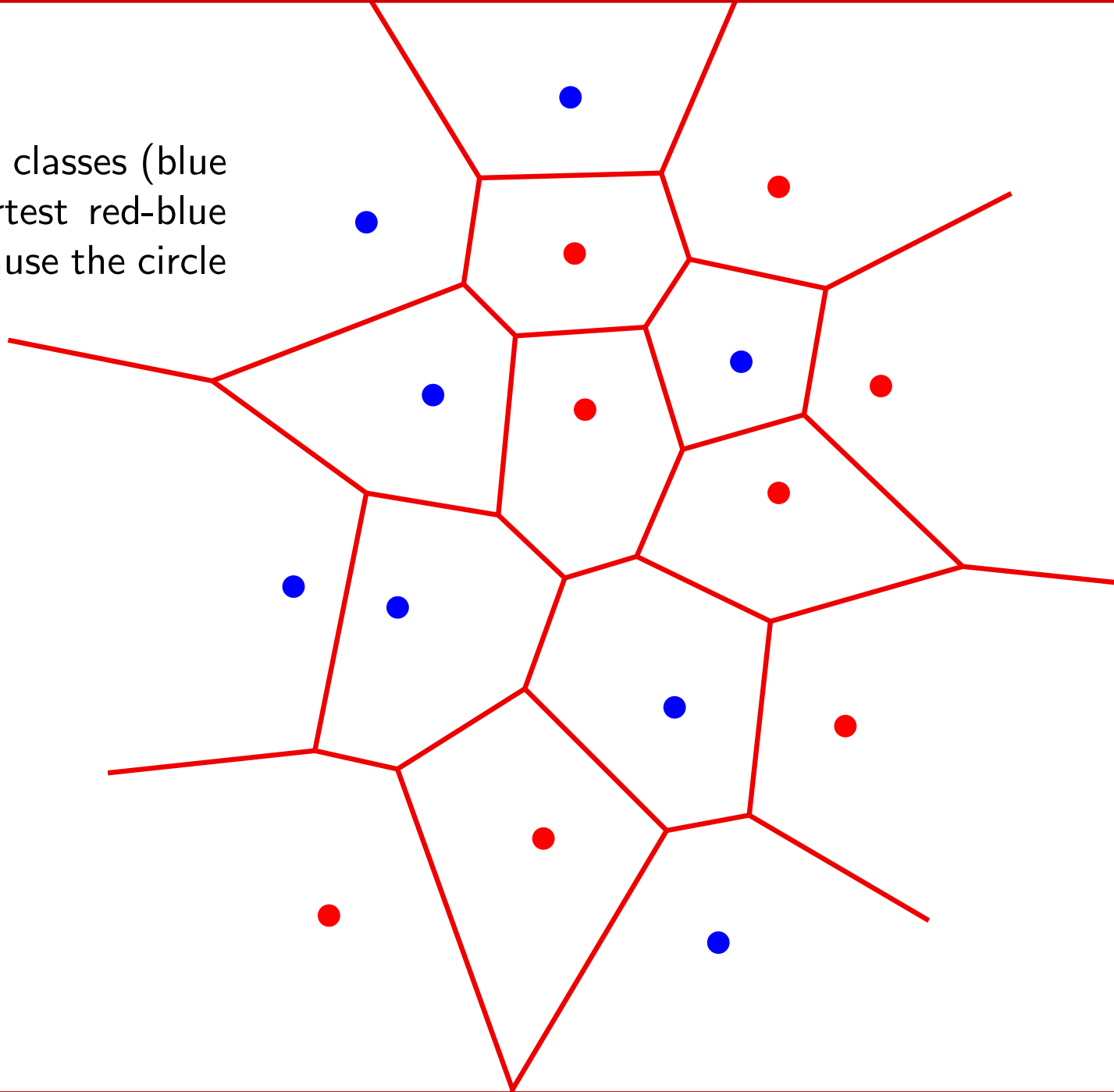
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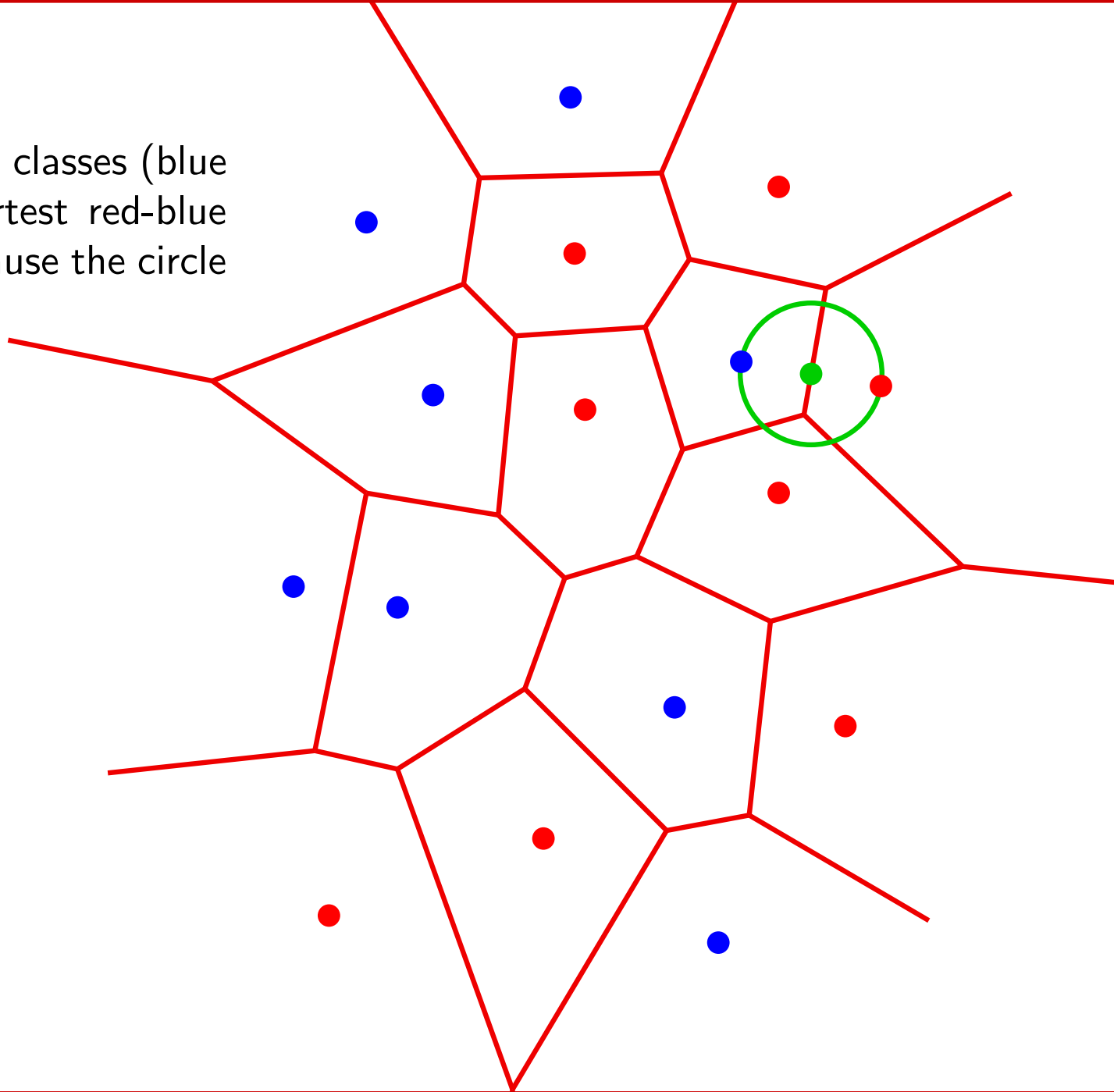
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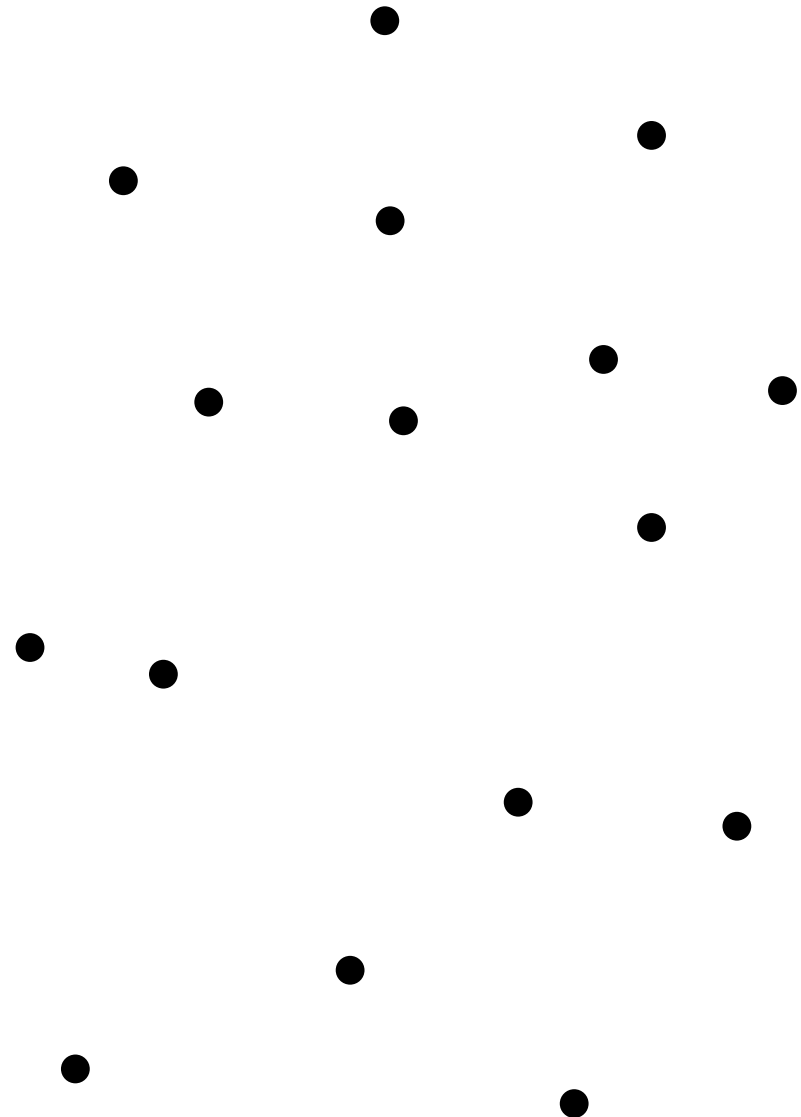
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Closest neighbors digraph

For each point $p_i \in P$, the closest point of P is a Voronoi neighbor. Therefore, the closest neighbors digraph is a subgraph of the Delaunay graph.



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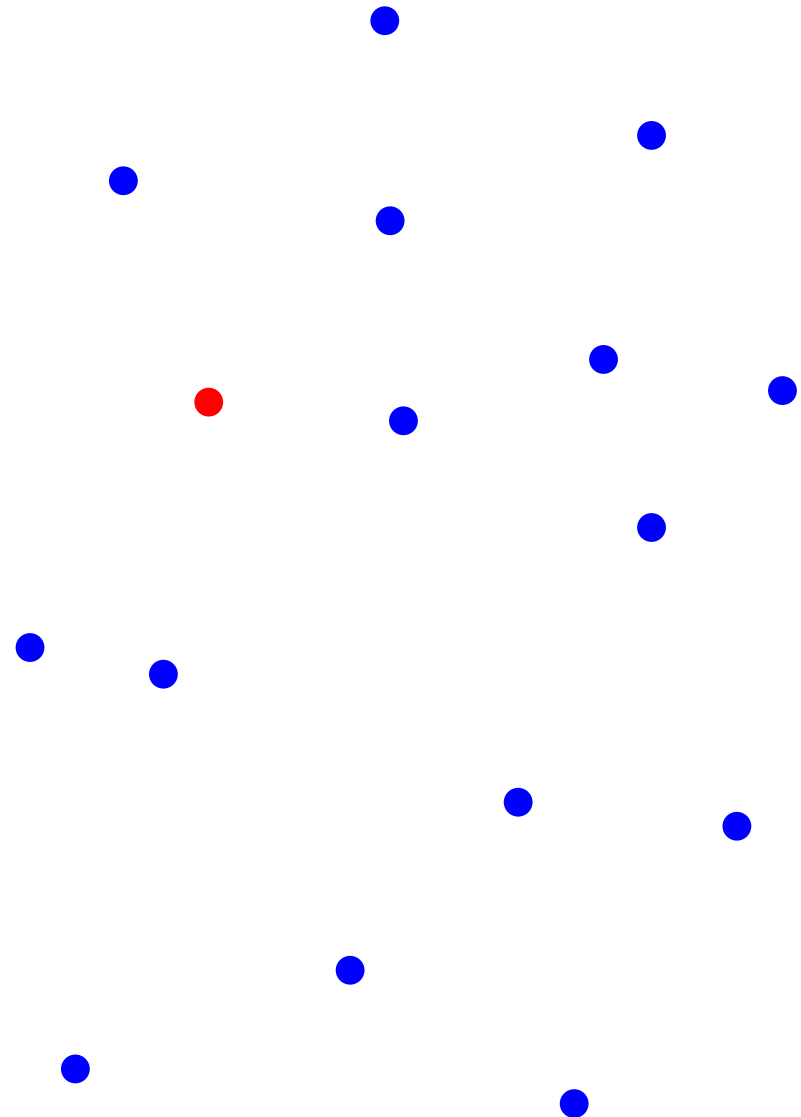
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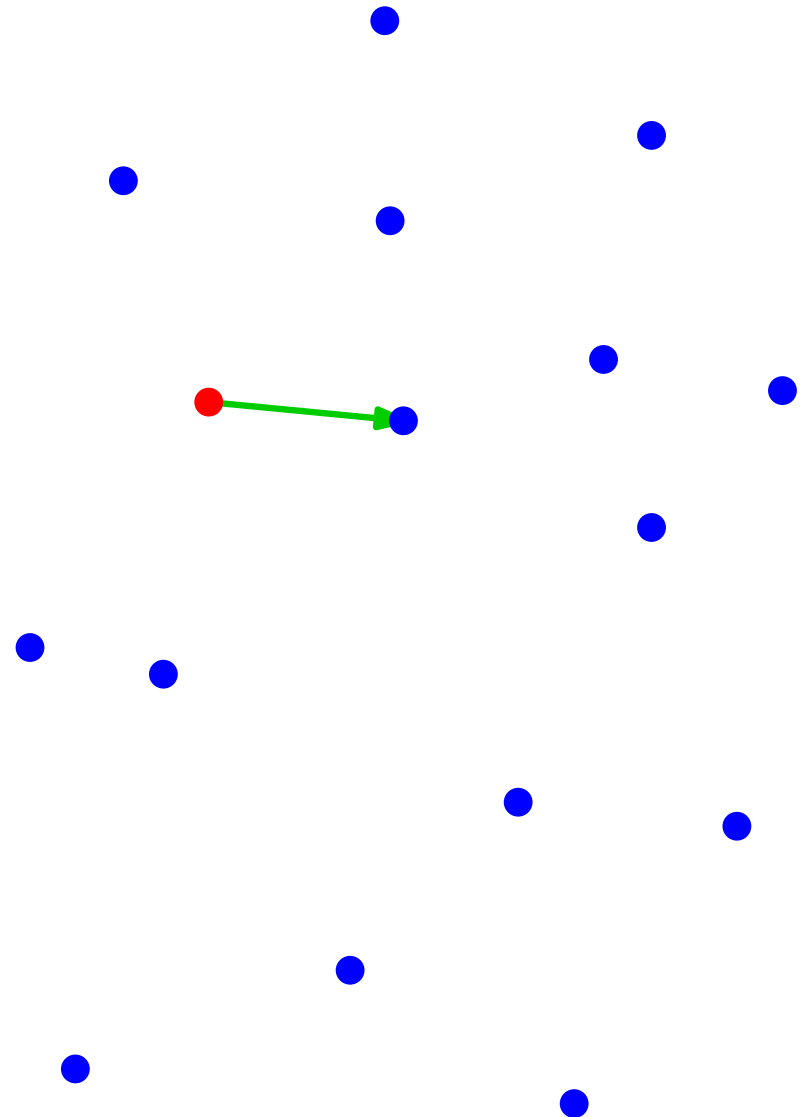
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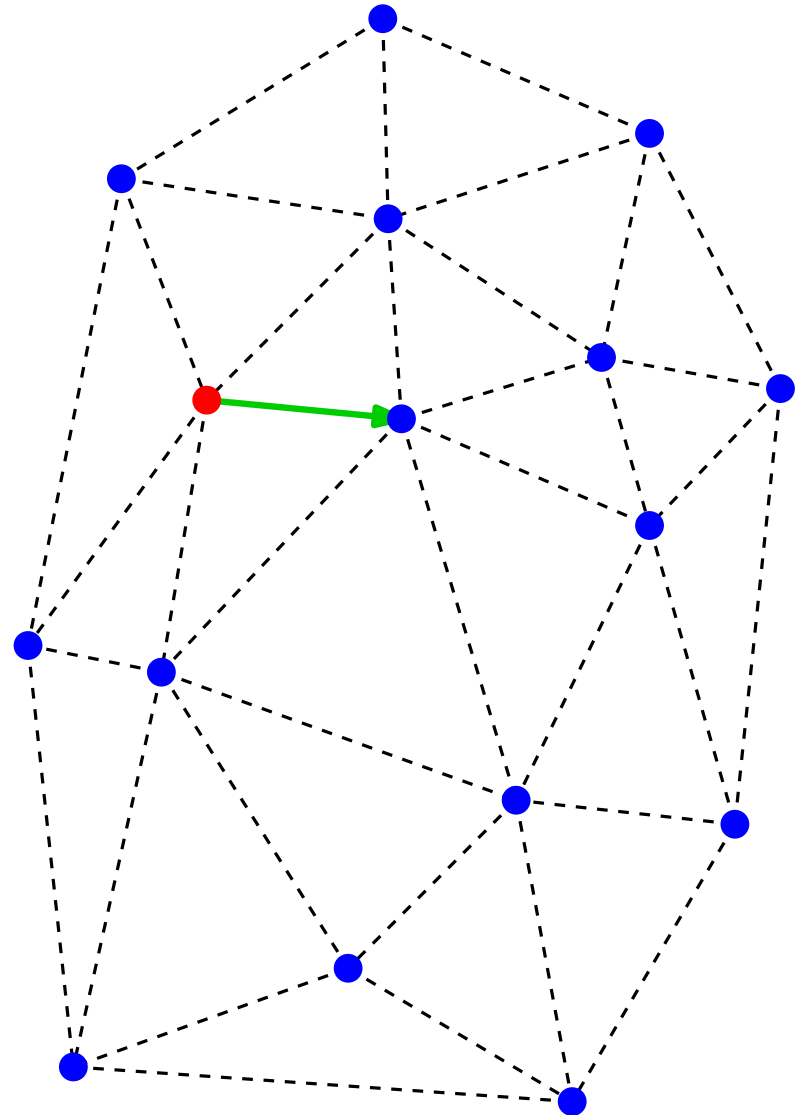
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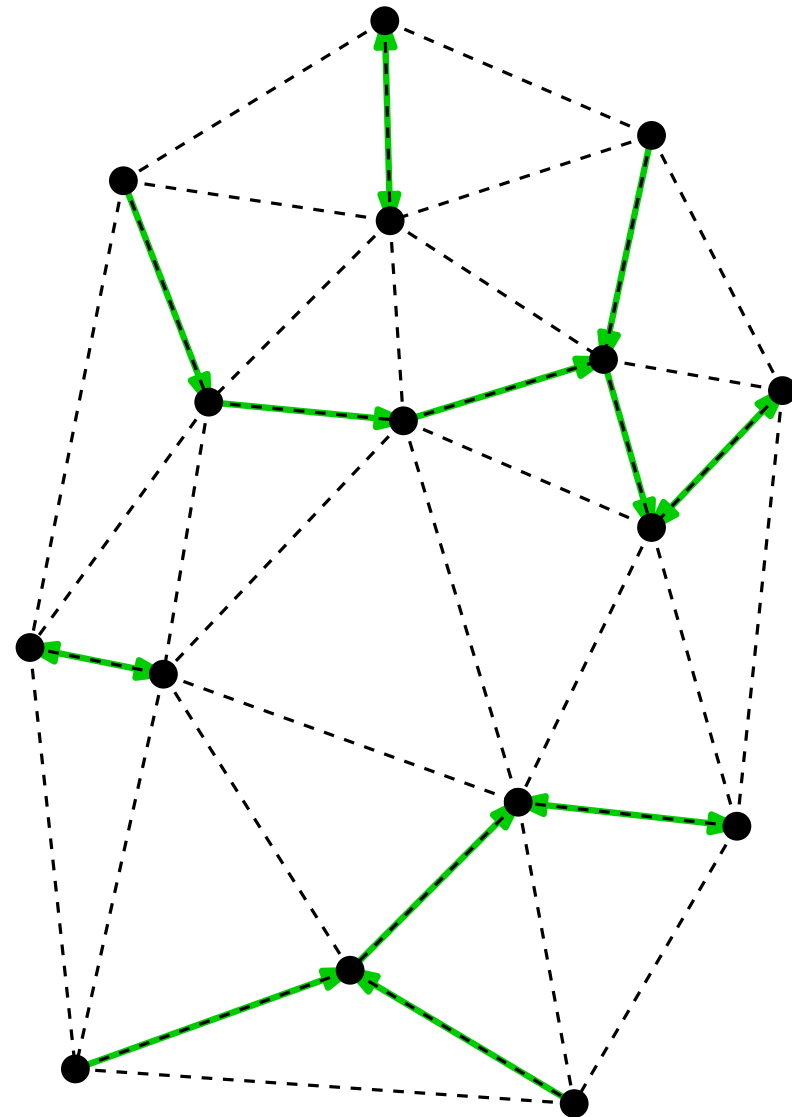
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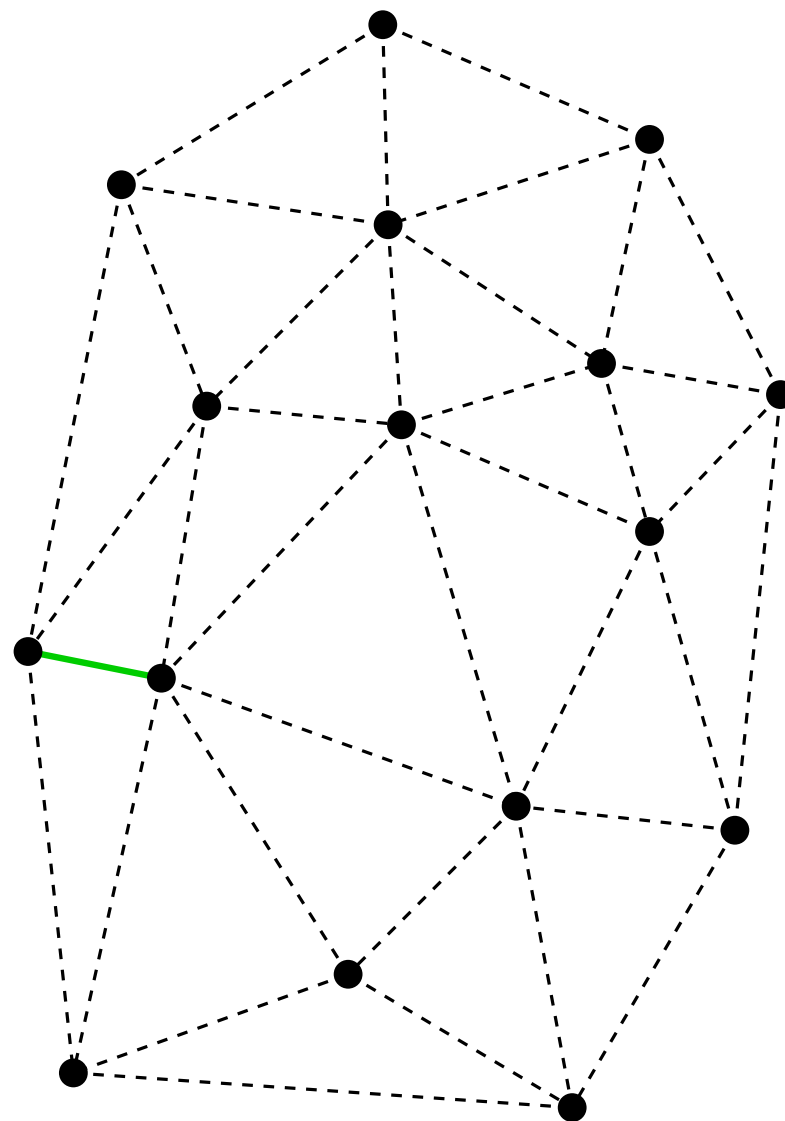
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Application 2

The closest pair

In particular, the shortest edge $p_i p_j$ belongs to the Delaunay graph.



PROXIMITY

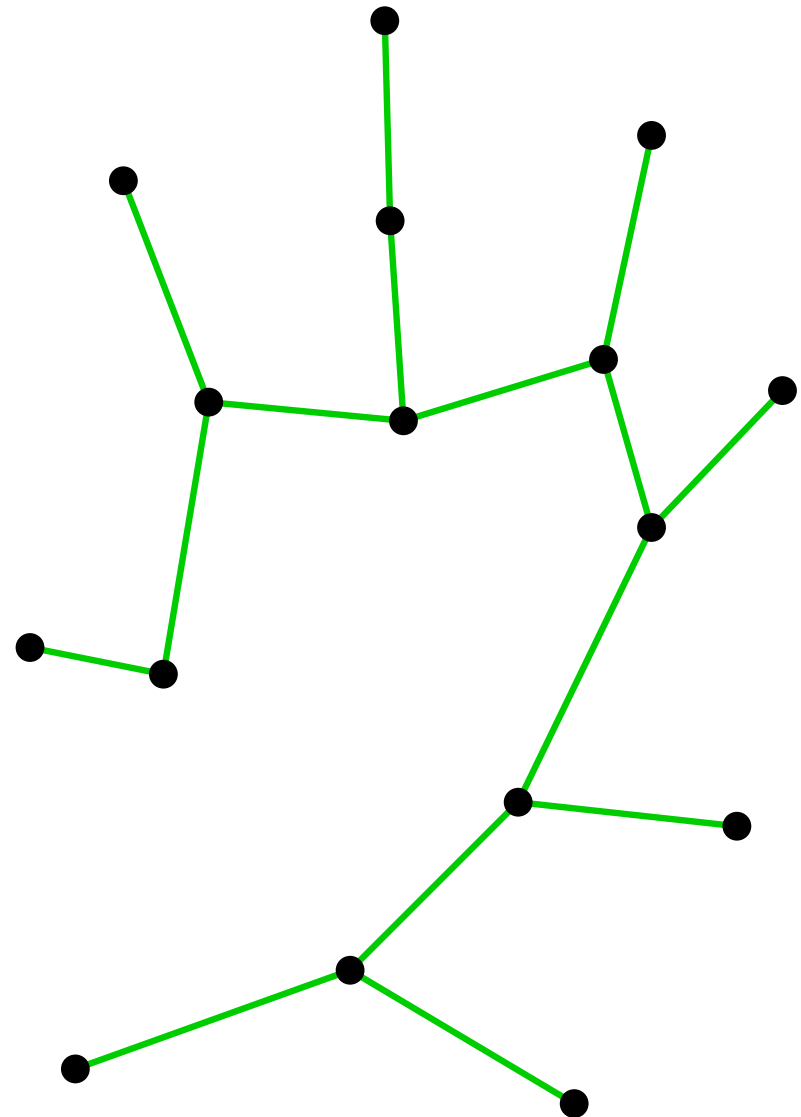
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Euclidean minimum spanning tree

The euclidean minimum spanning tree is a subgraph of the Delaunay graph.



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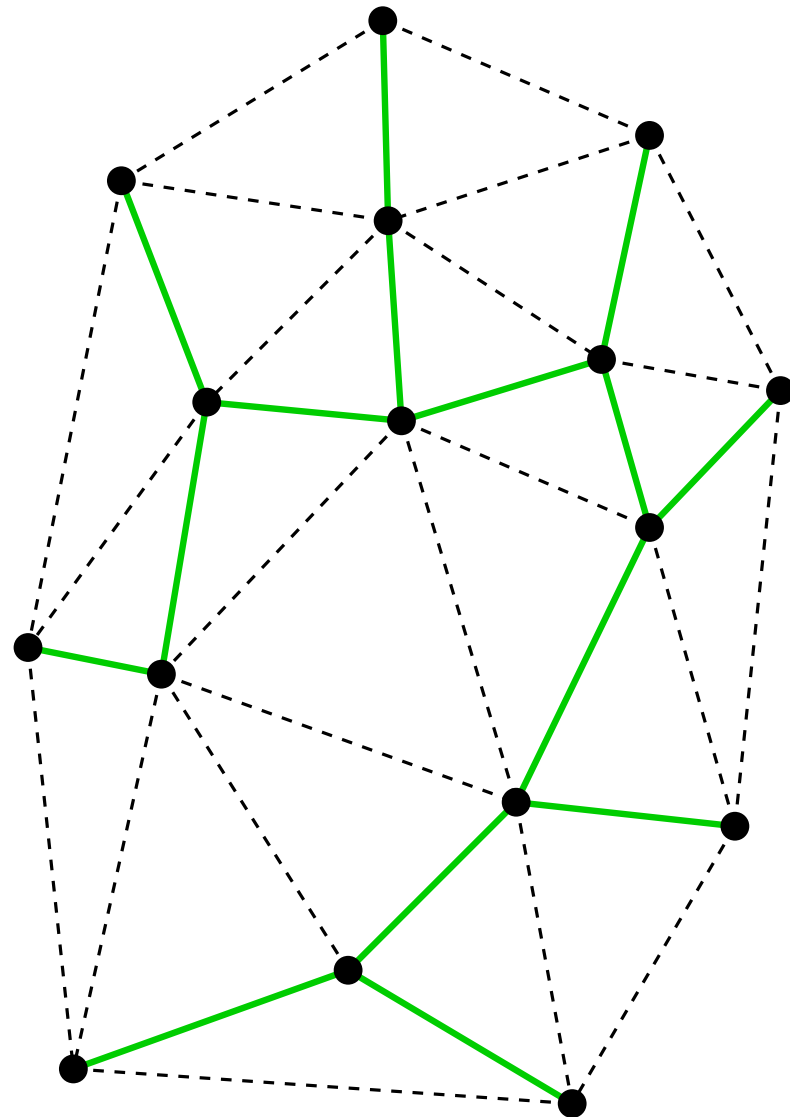
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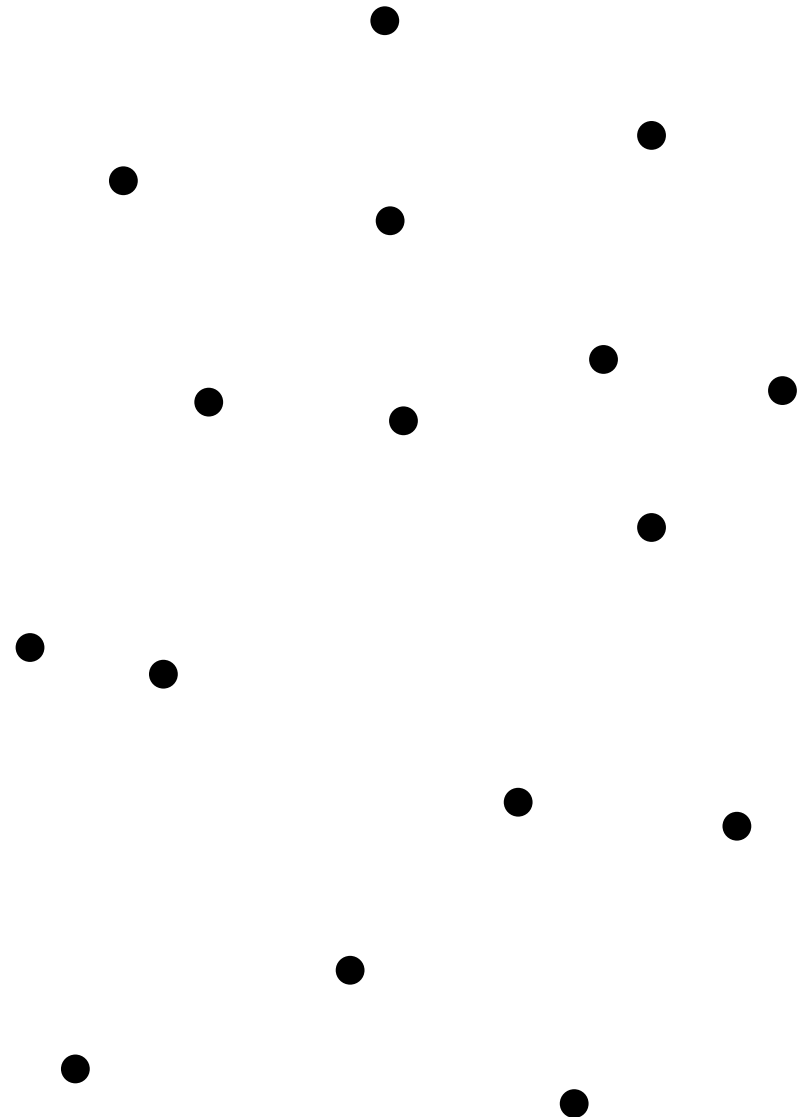
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Prim's algorithm to compute the minimum spanning tree in a weighted graph builds the tree T by adding, at each iteration, the edge of minimum weight incident to T which does not produce a cycle when added to T .



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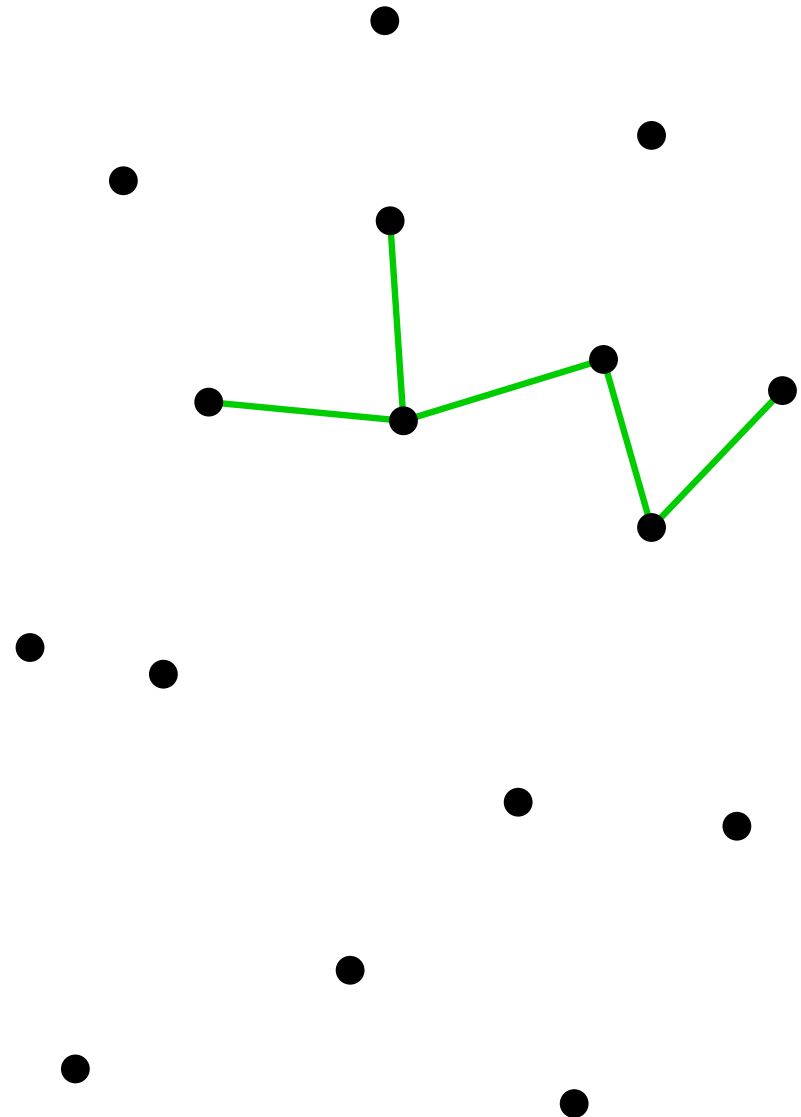
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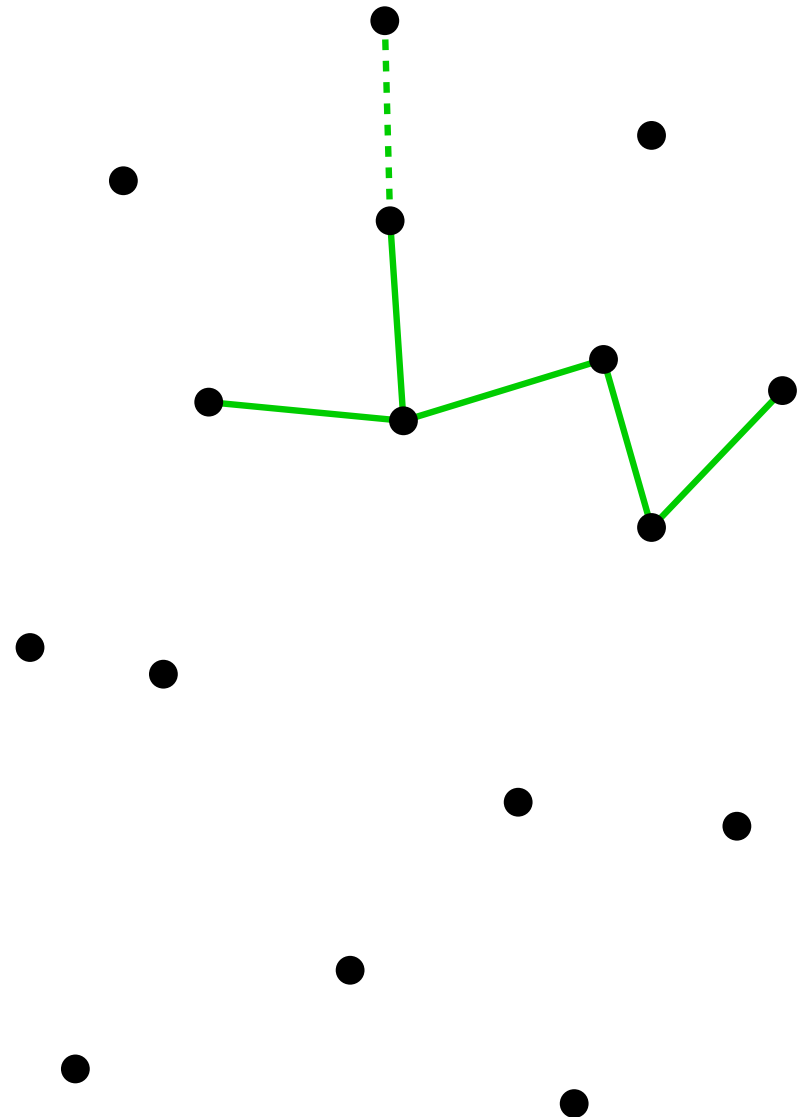
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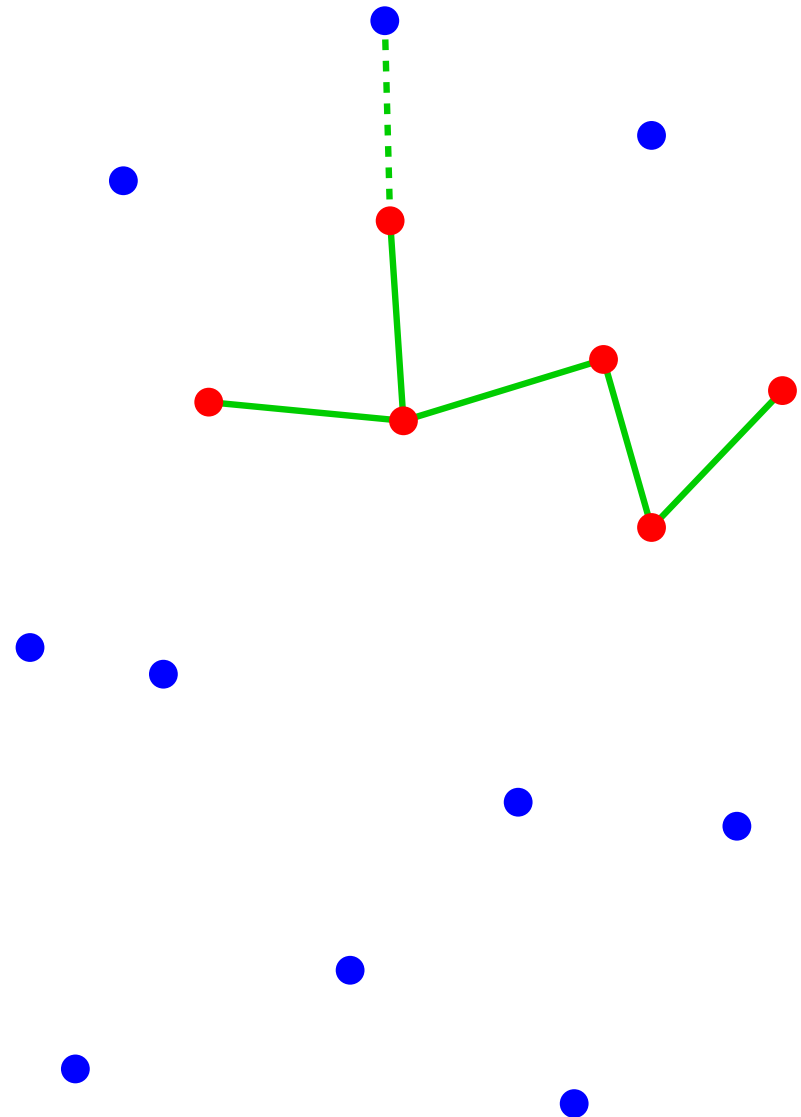
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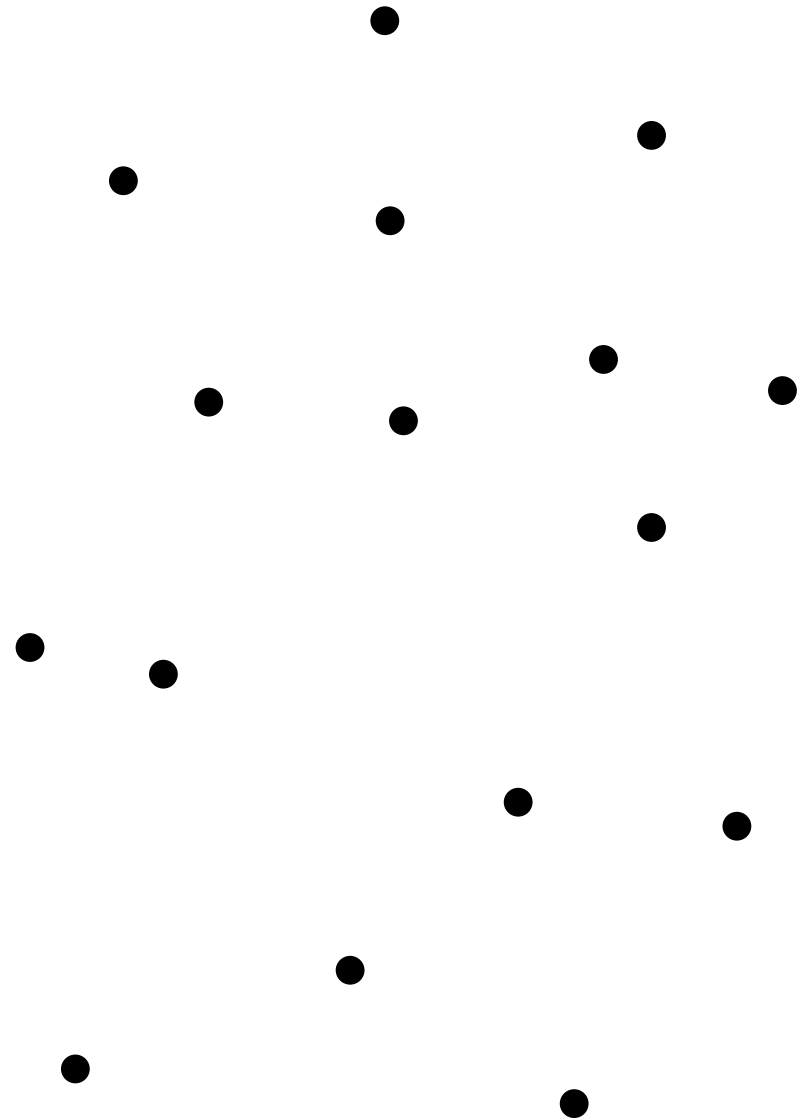


PROXIMITY

Application 4

Finding the closest site

Given any point q in the plane, finding its closest site $p_i \in P$ means detecting the Voronoi region $V(p_i)$ containing q .

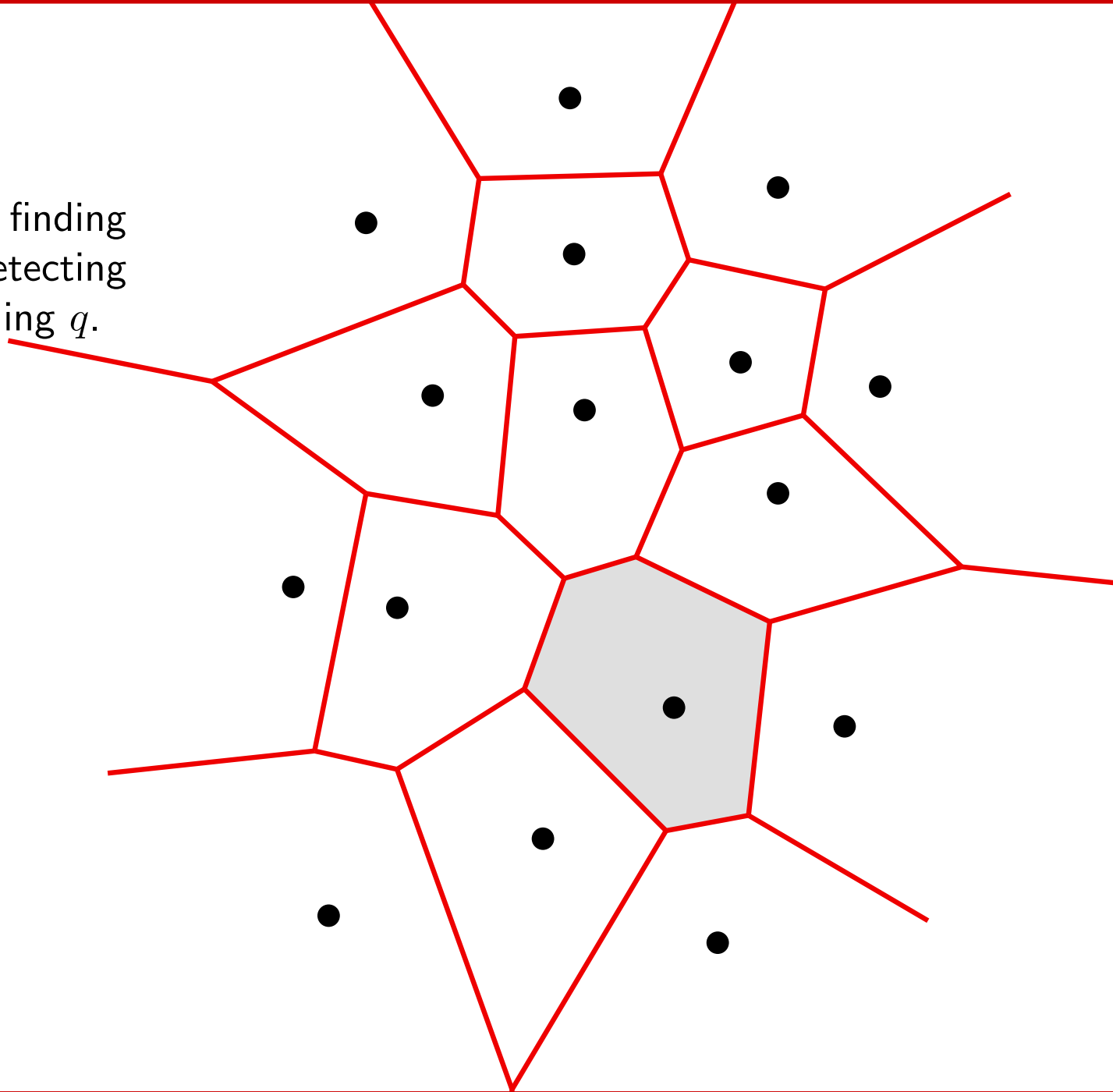


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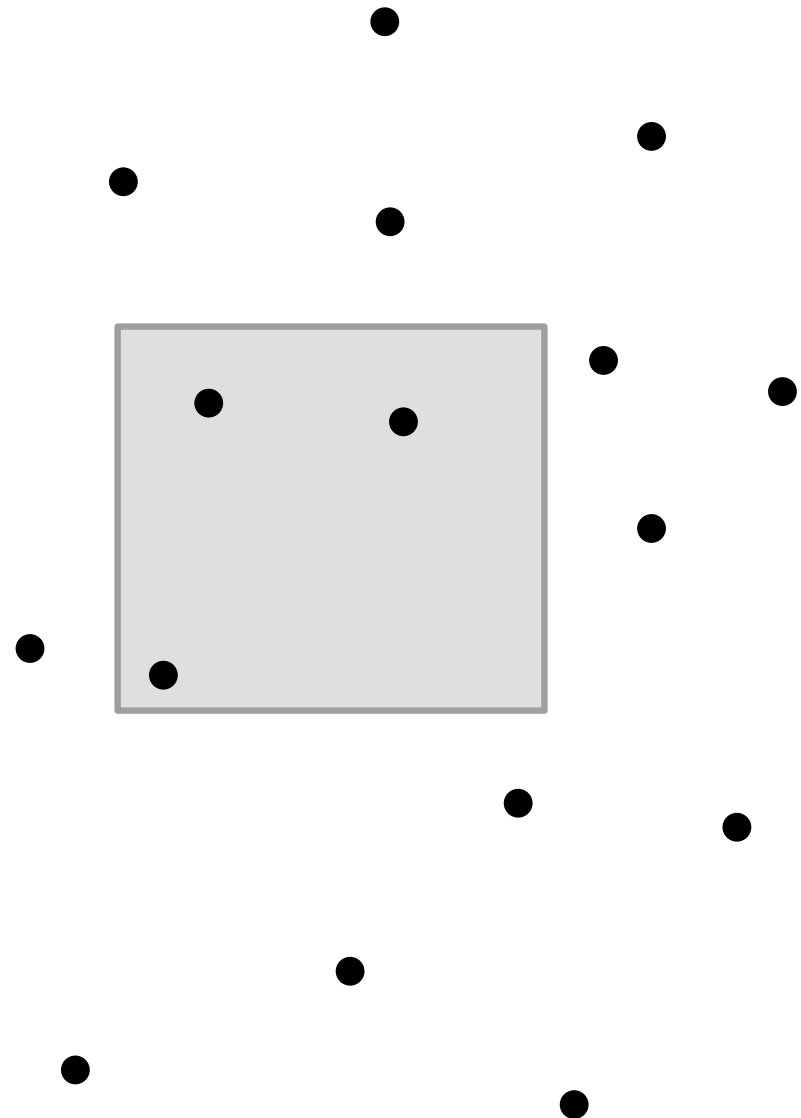
PROXIMITY

Application

Max-min facility location

The center of the largest empty circle, restricted to a given region A , can be located at:

- a Voronoi vertex,
- the intersection of a Voronoi edge with the boundary of A .
- a vertex of A .



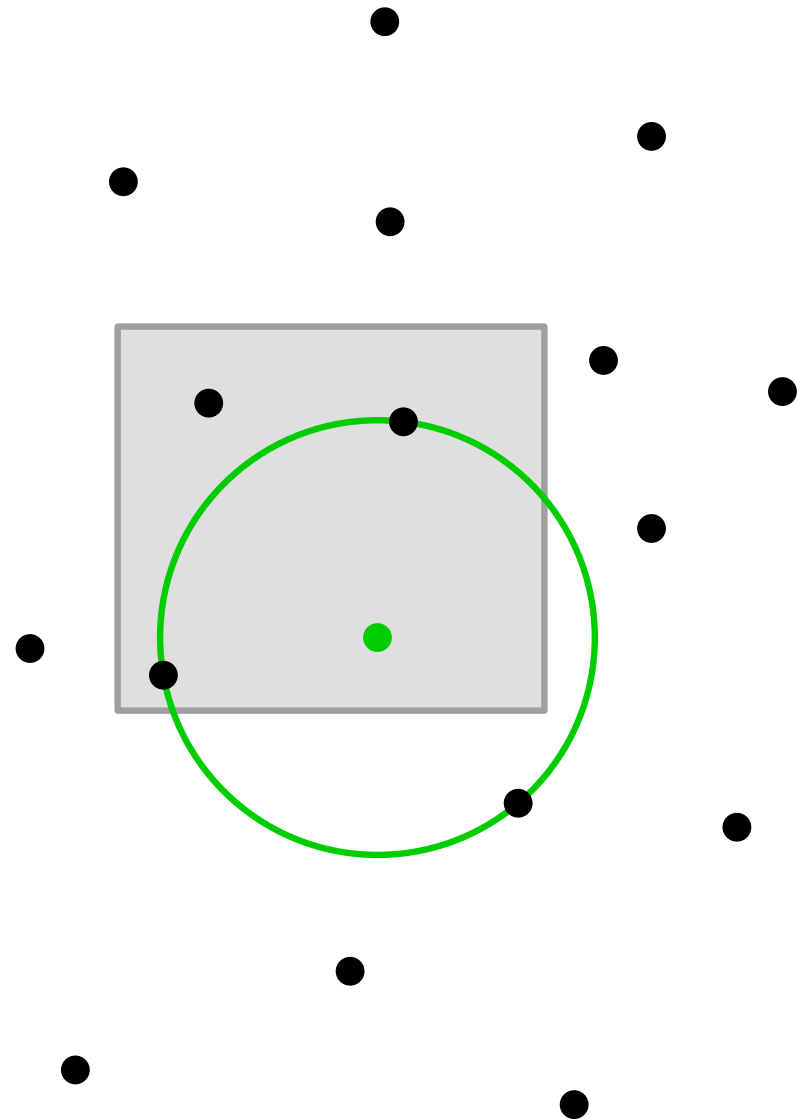
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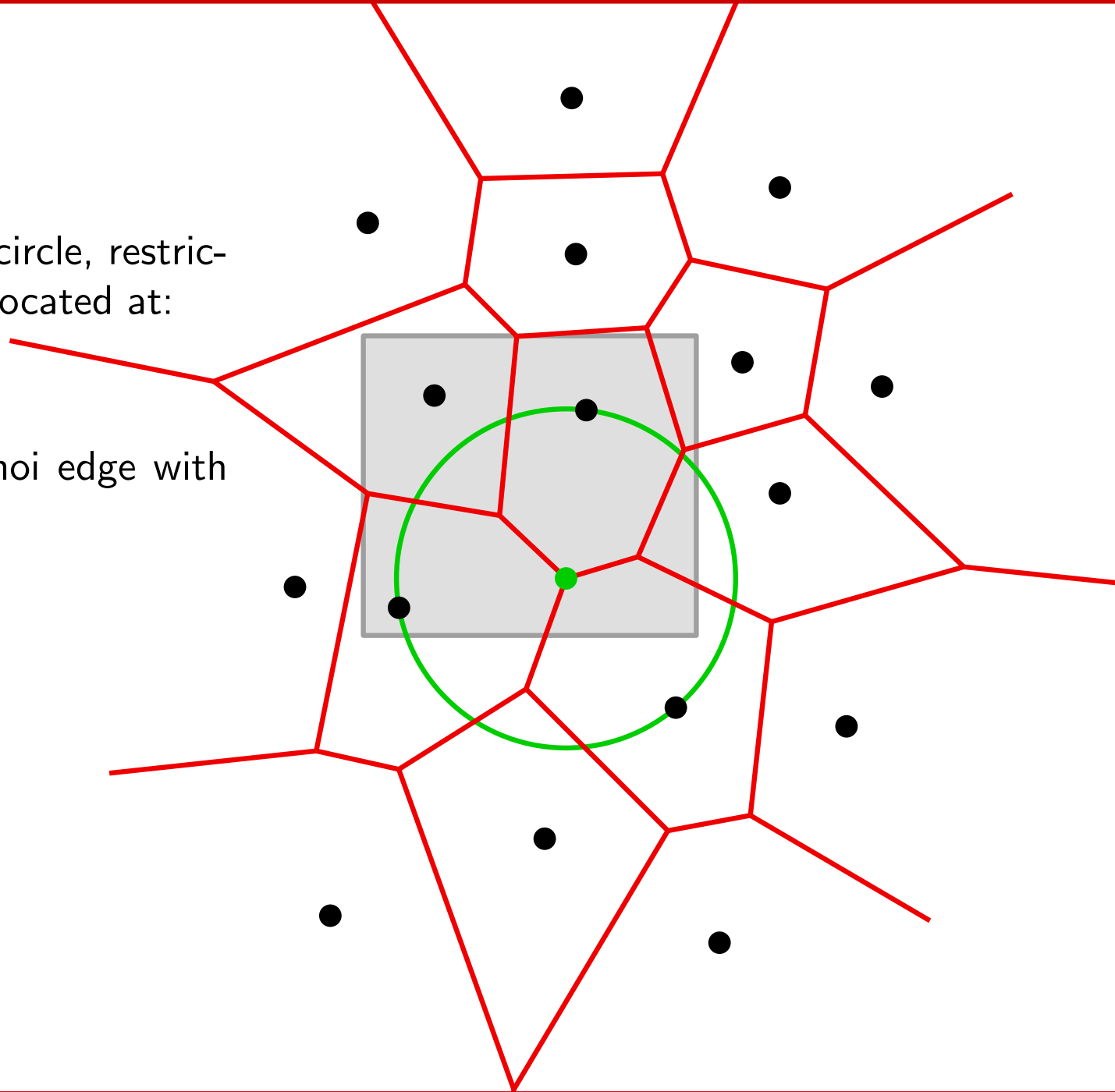
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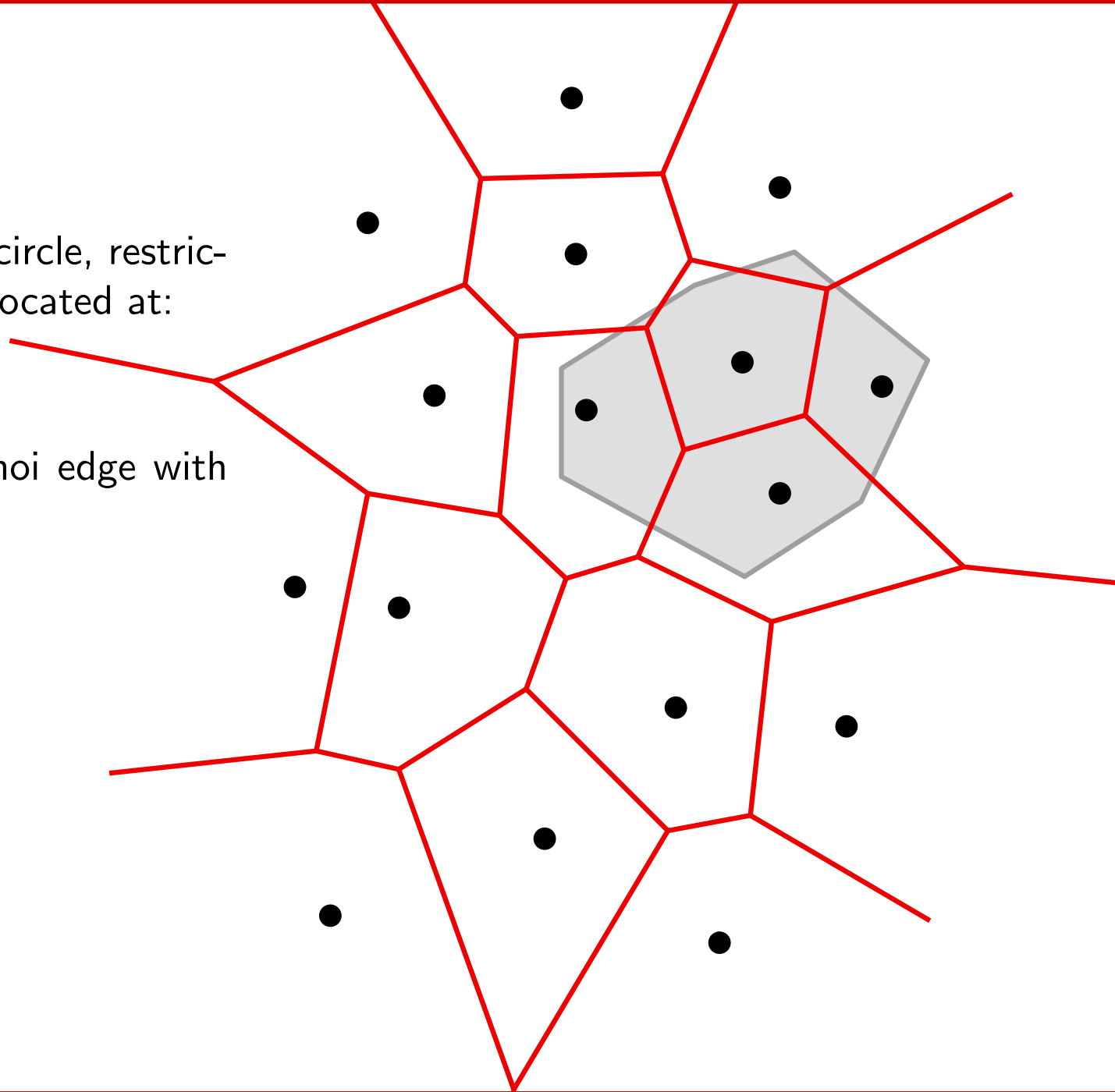
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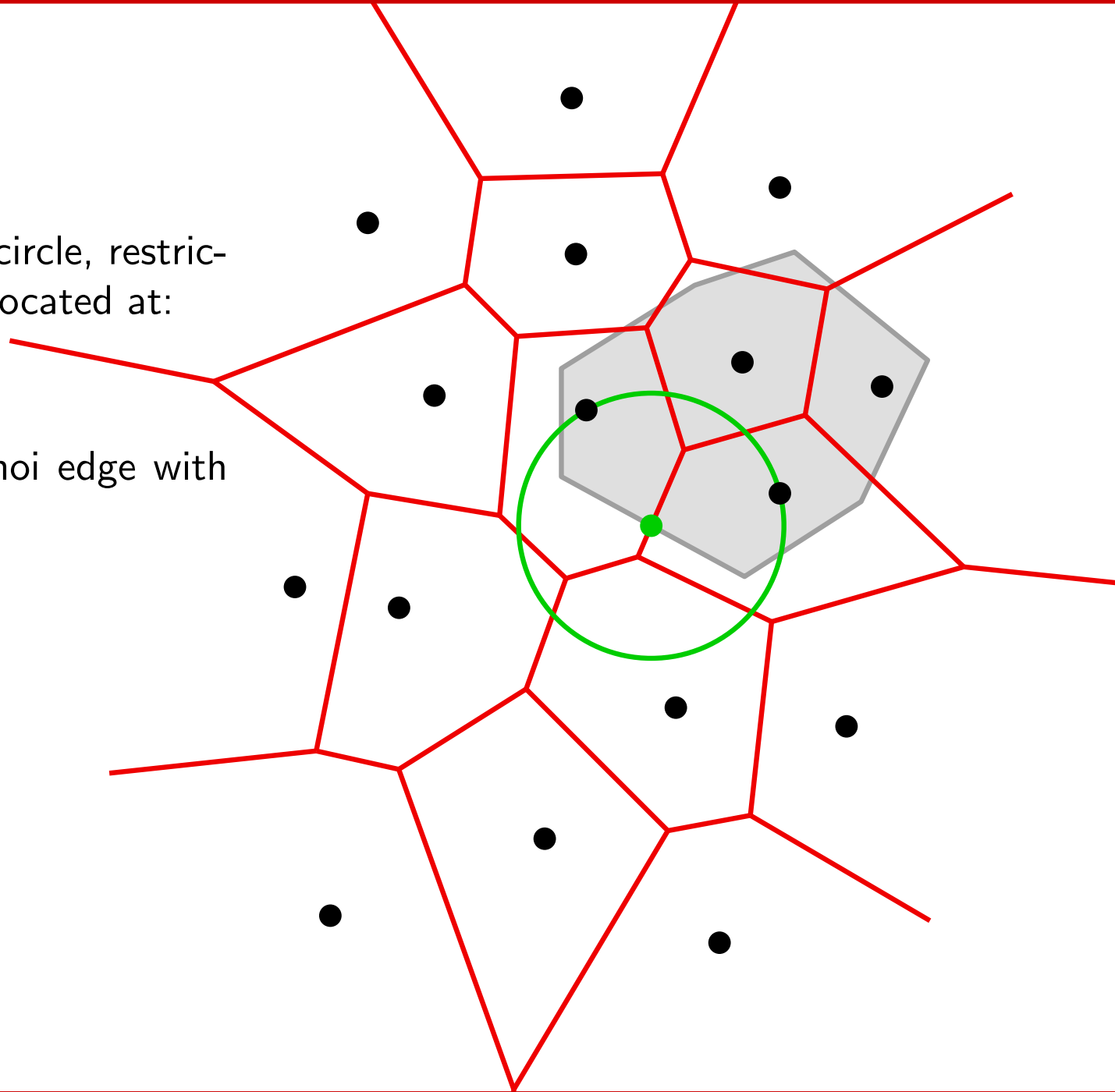
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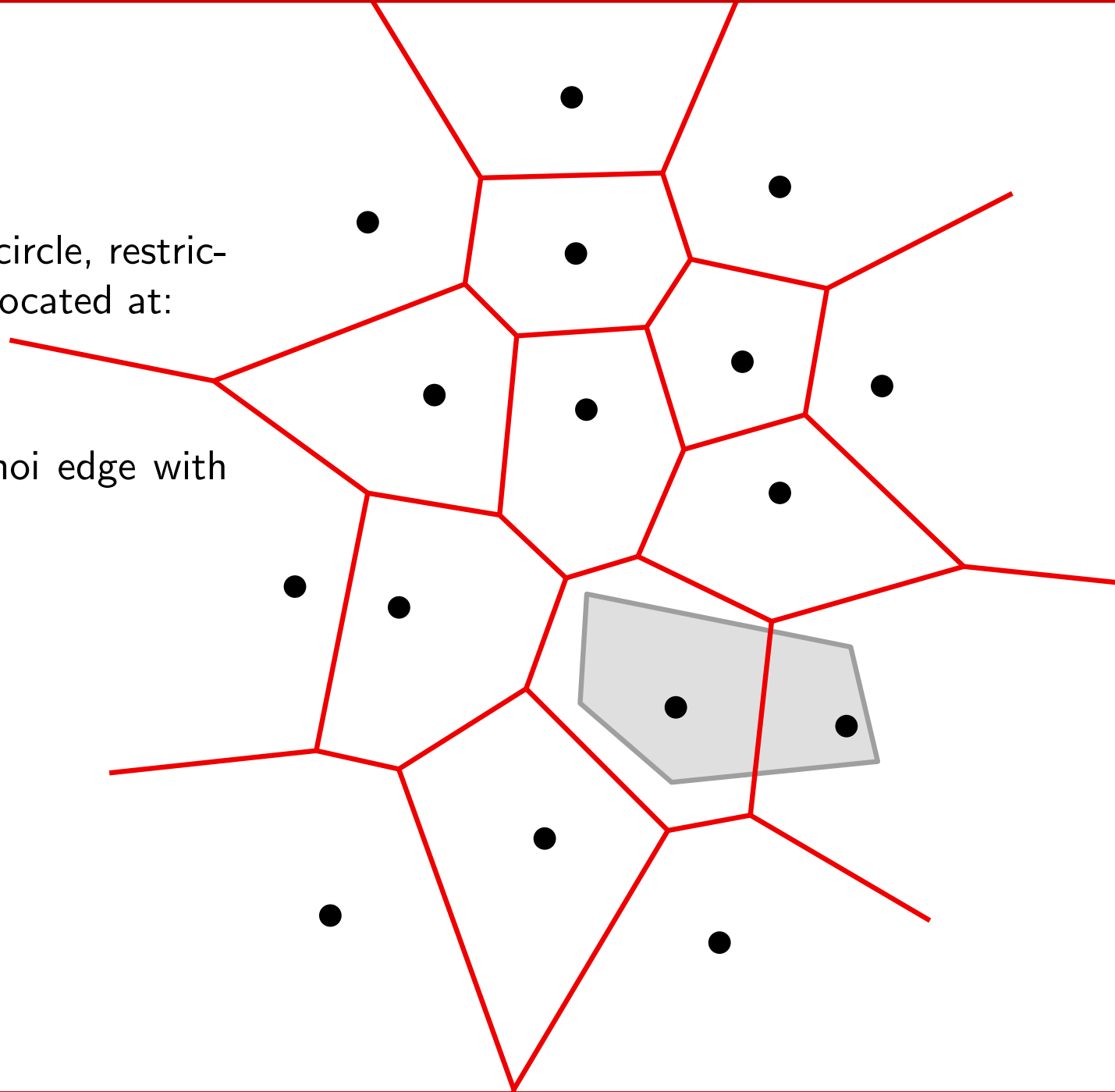
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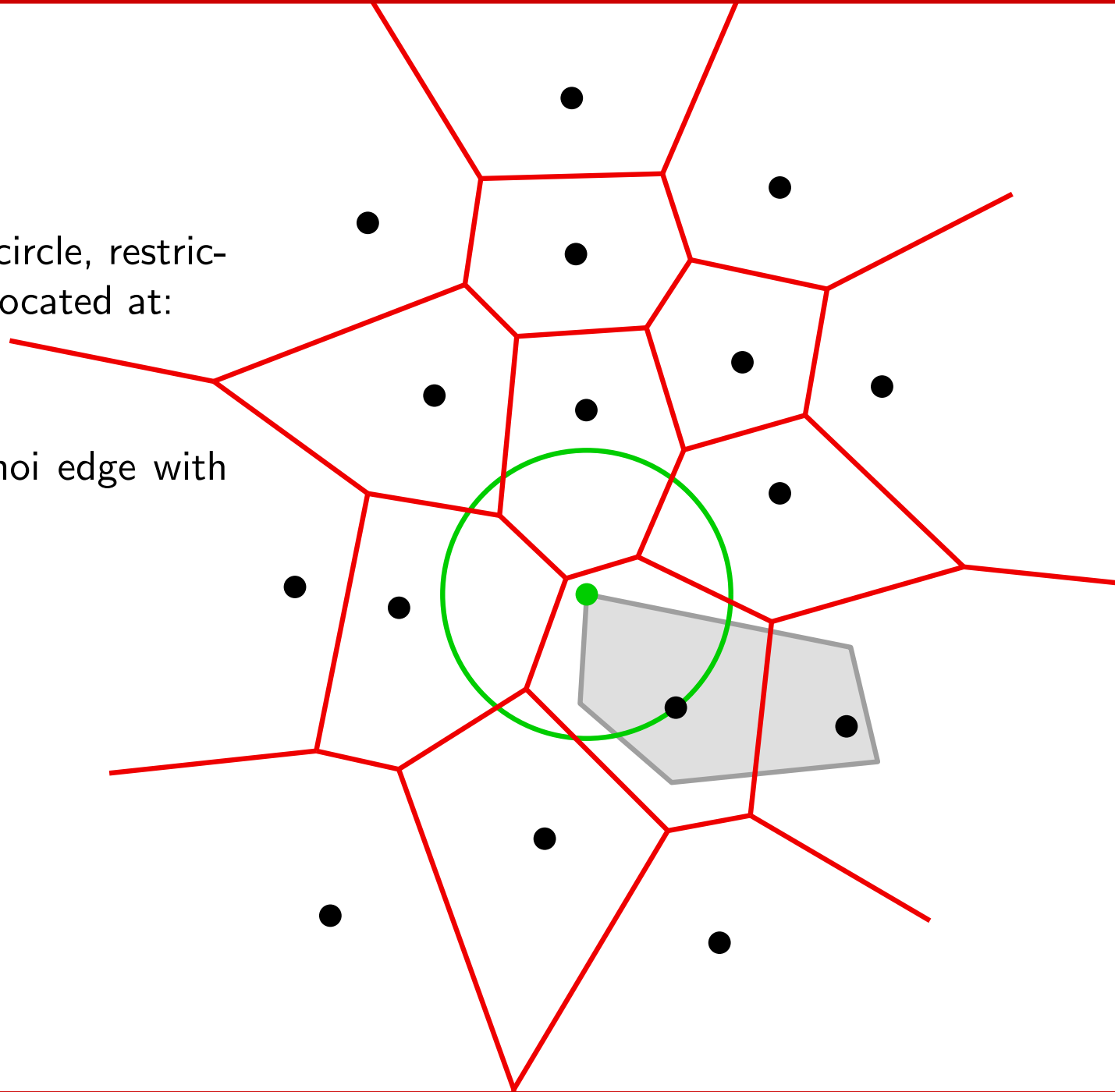
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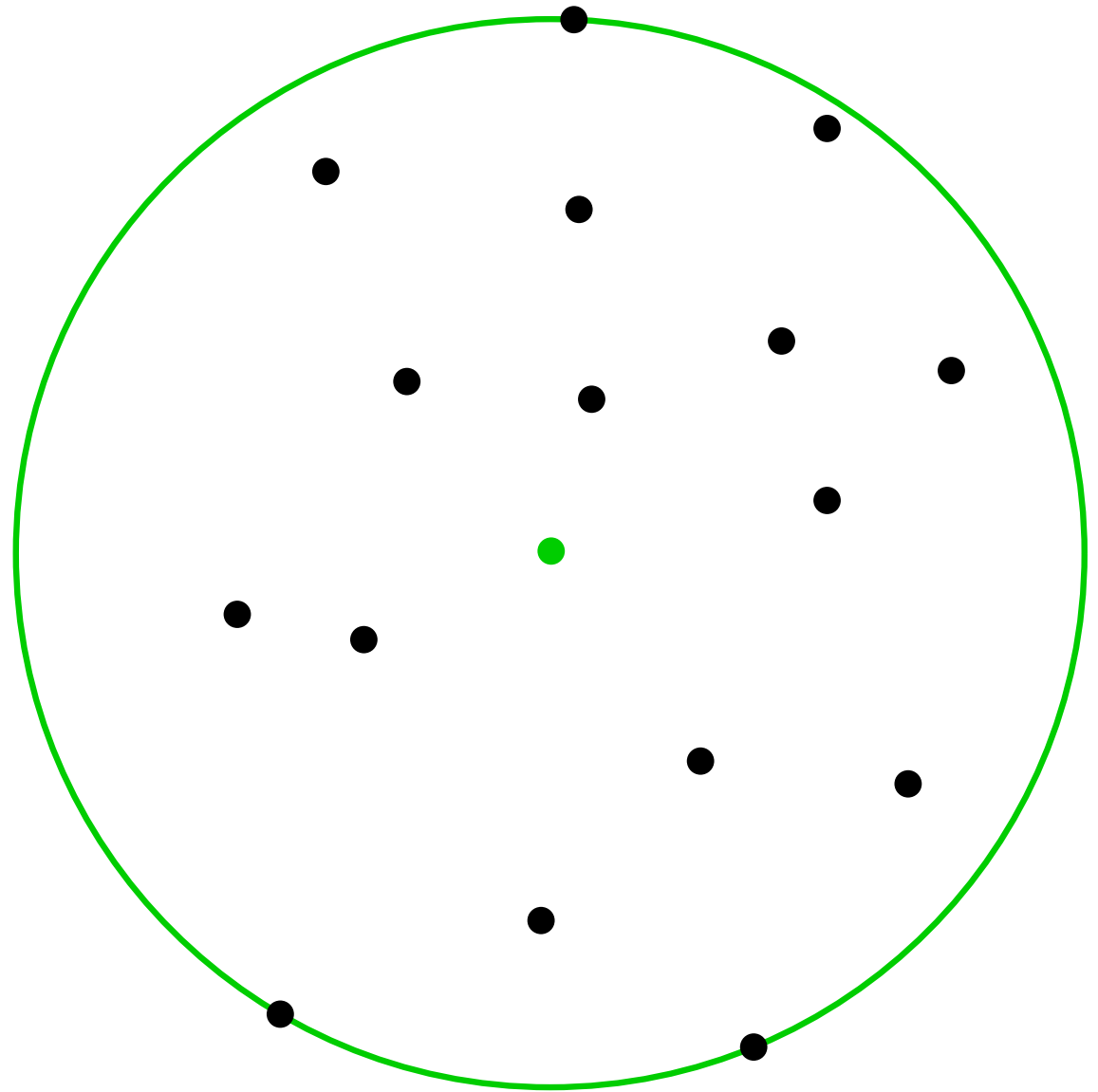


PROXIMITY

Application 6

Min-max facility location

The center of the minimum spanning circle

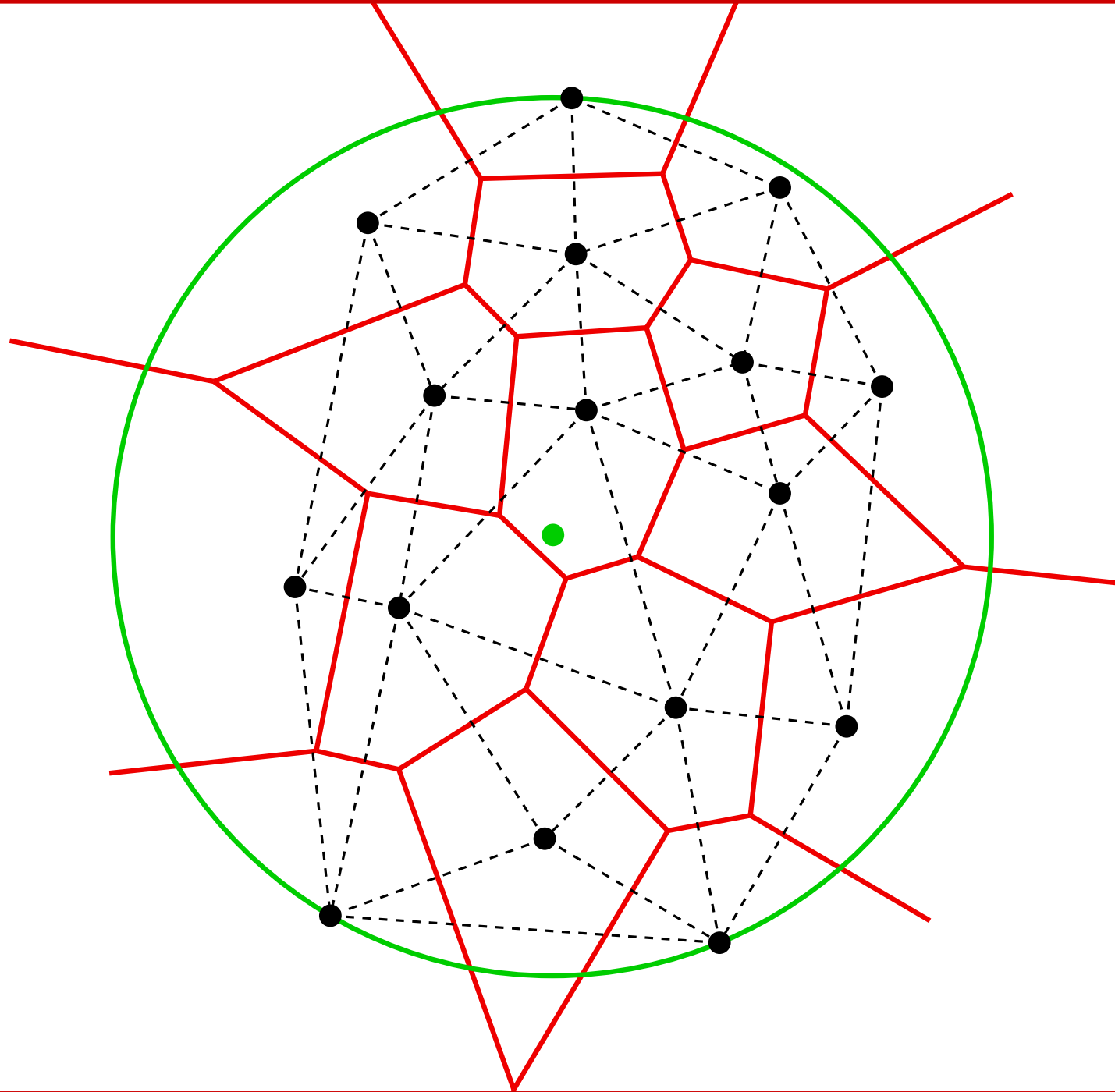


PROXIMITY

Application 6

Min-max facility location

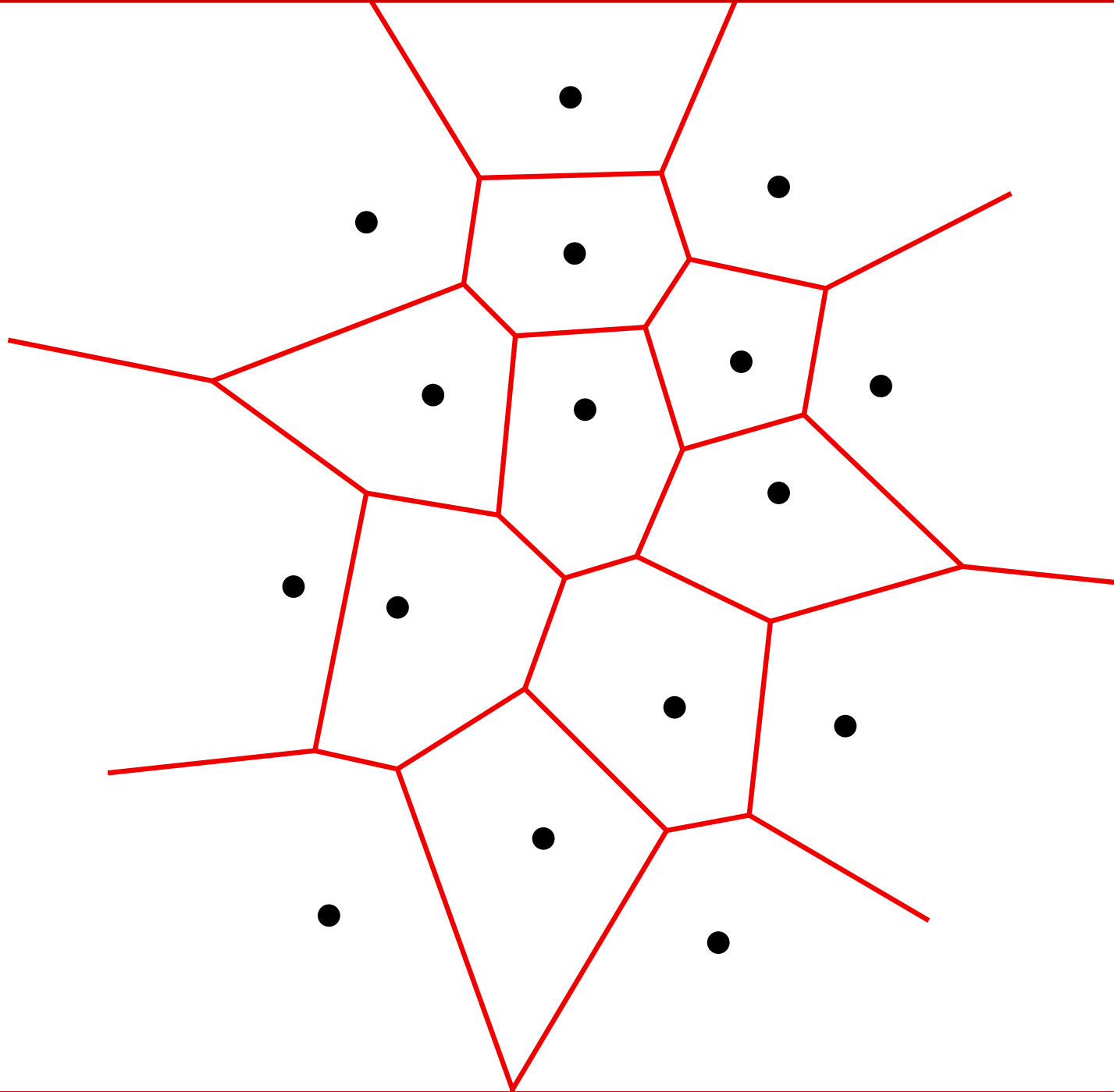
The center of the minimum spanning circle does not seem to be related with the Voronoi diagram! How strange!



PROXIMITY

GENERALIZATIONS OF THE VORONOI DIAGRAM

The Voronoi diagram of a finite set P of **points** in the **plane** is a partition of the plane such that each region is the locus of the points that lie **closer** to **one** element of P than to the remaining ones.

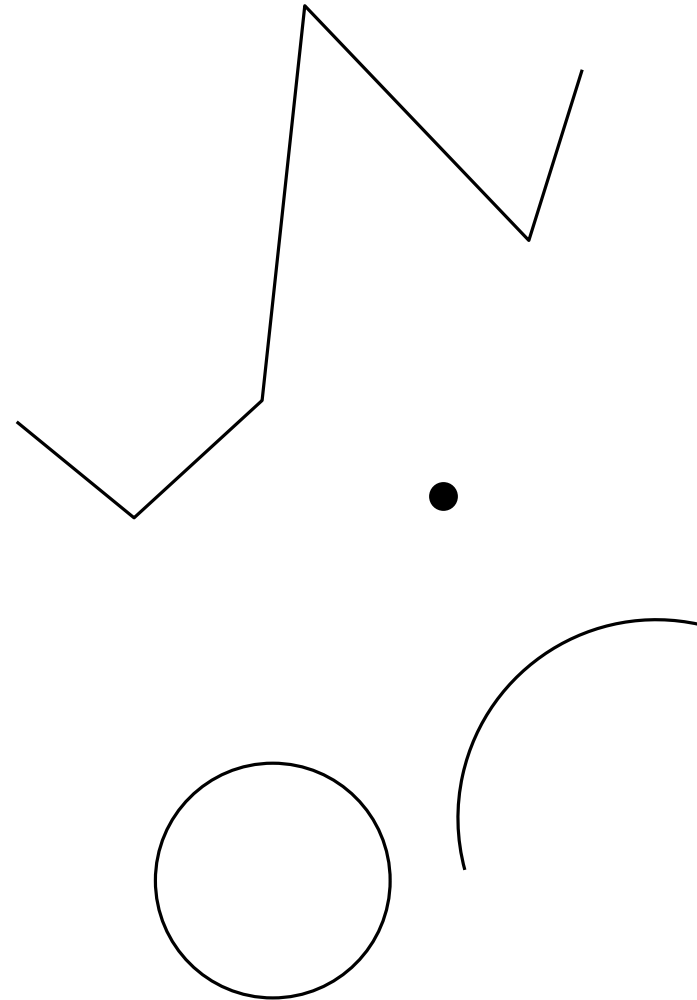


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segments
disks
⋮
barriers

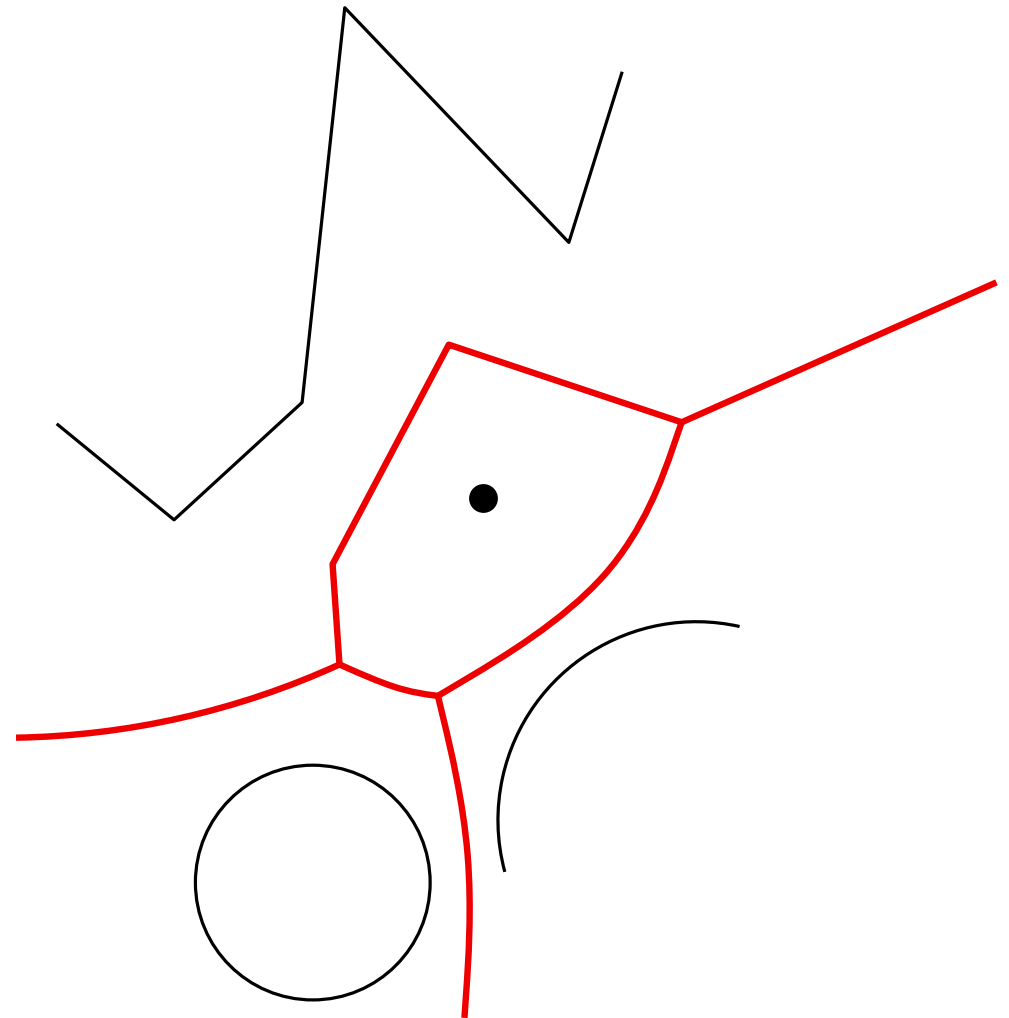


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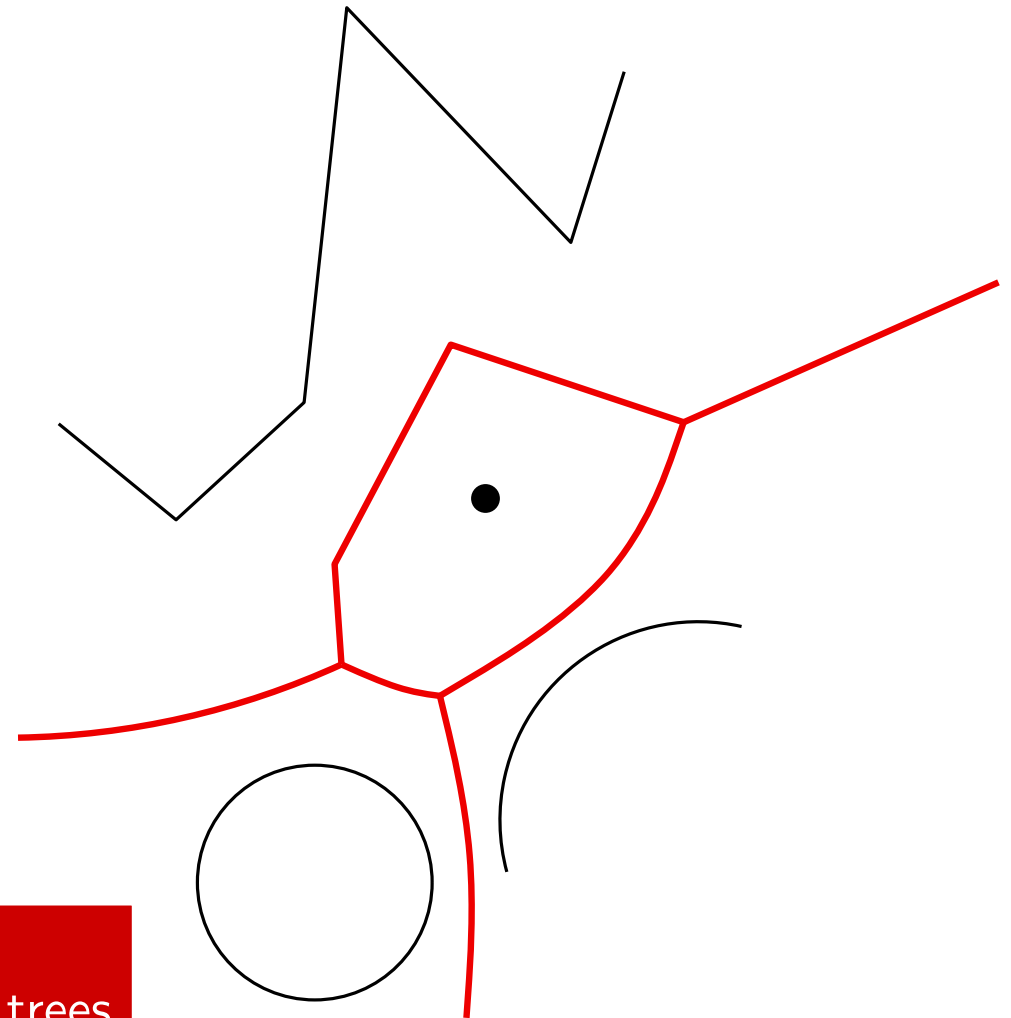
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segments
disks
⋮
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APPLICATIONS

- How do car emissions affect the trees along the highways.
- Rivers and mountains influence in determining school districts.



PROXIMITY

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sphere
cylinder

⋮

dimension d

PROXIMITY

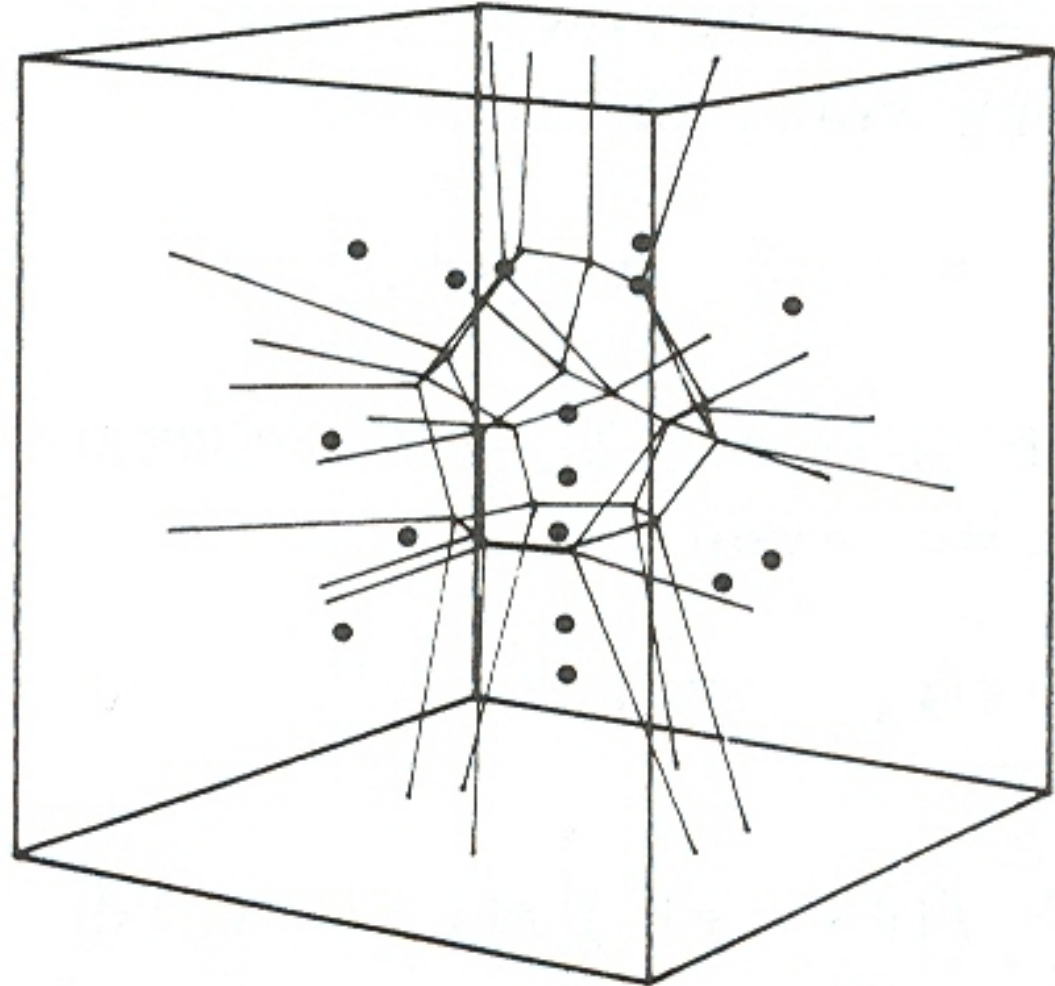
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The Voronoi diagram of a finite set P of **points** in the ~~plane~~ is a partition of the plane such that each region is the locus of the points that lie **closer** to **one** element of P than to the remaining ones.

sphere
cylinder

⋮

dimension d



PROXIMITY

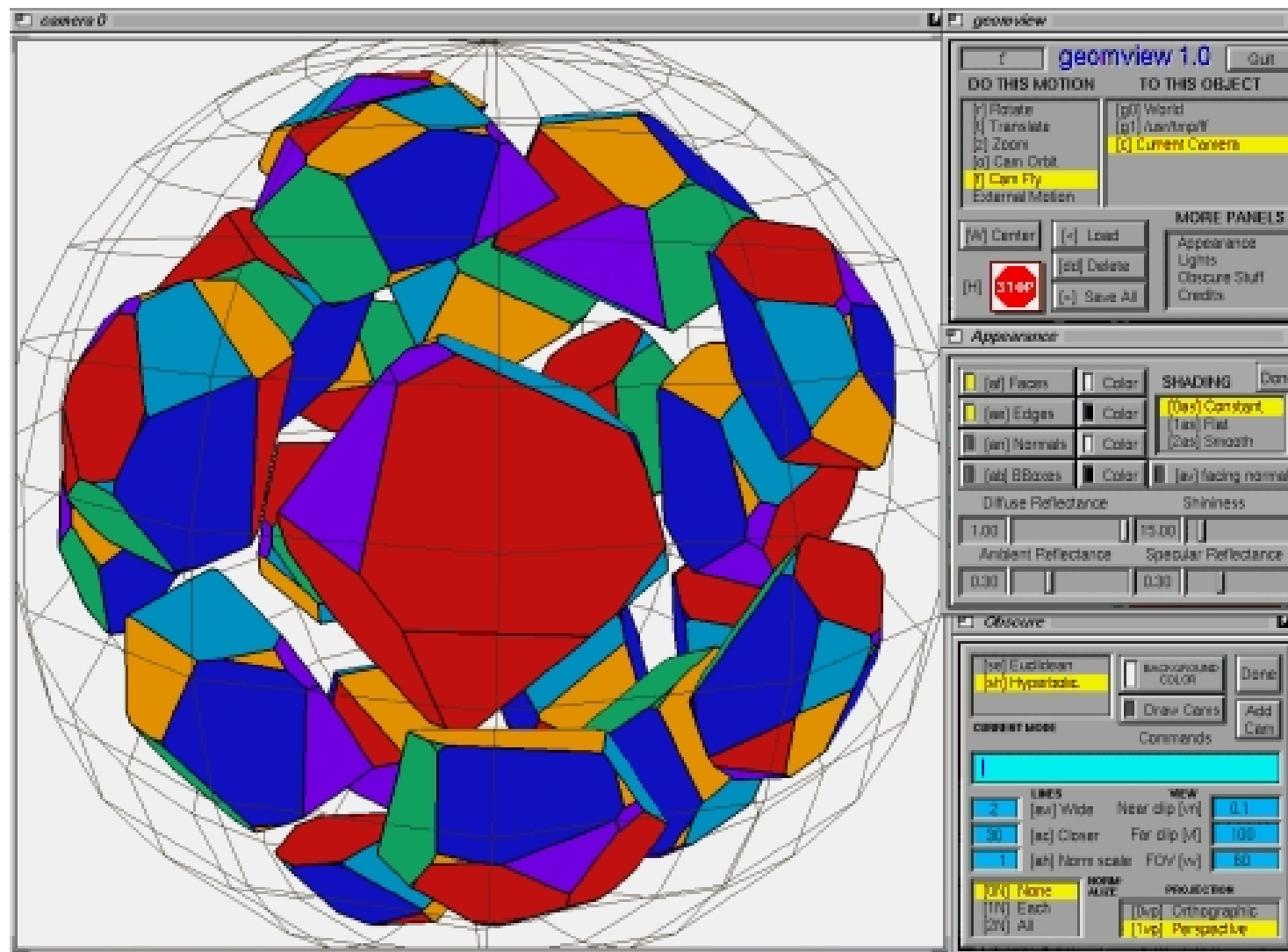
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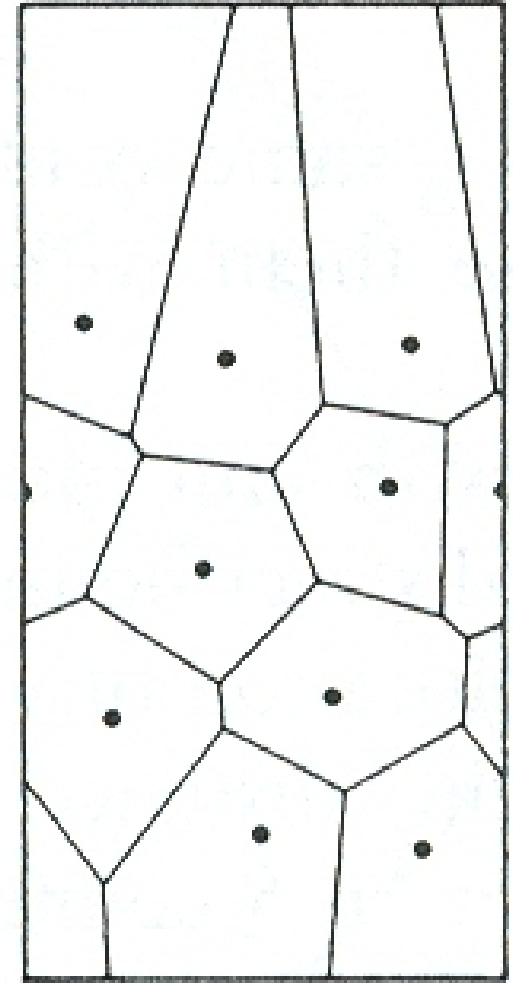
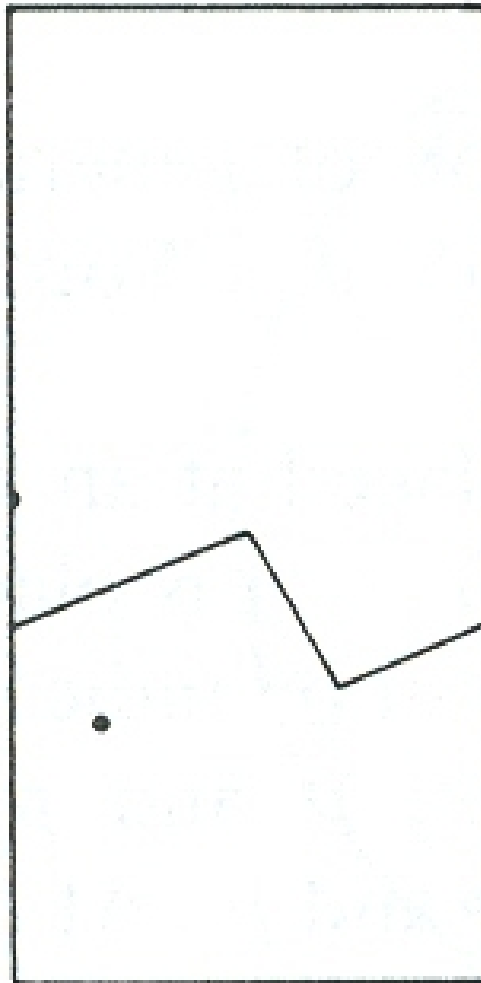
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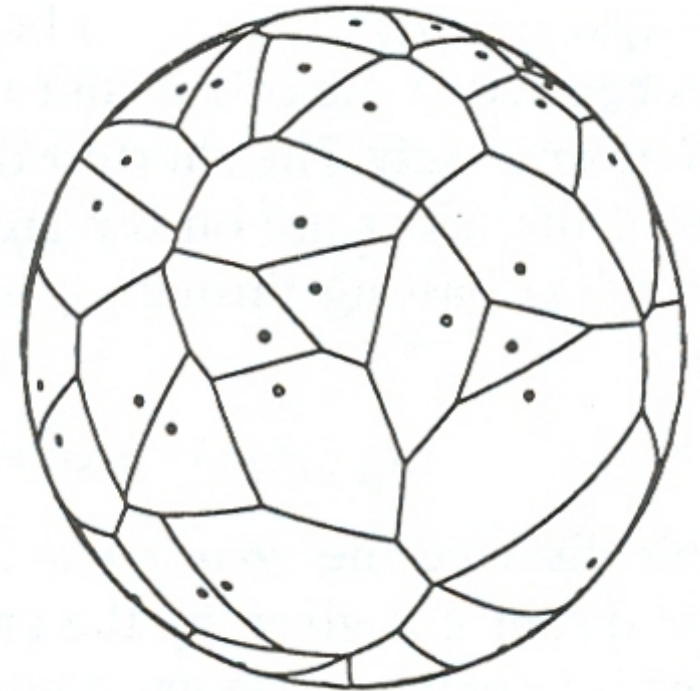
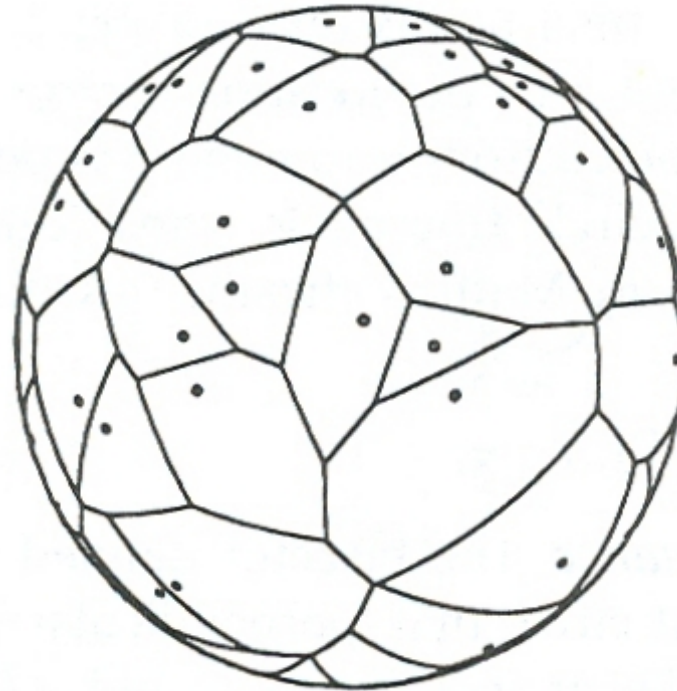
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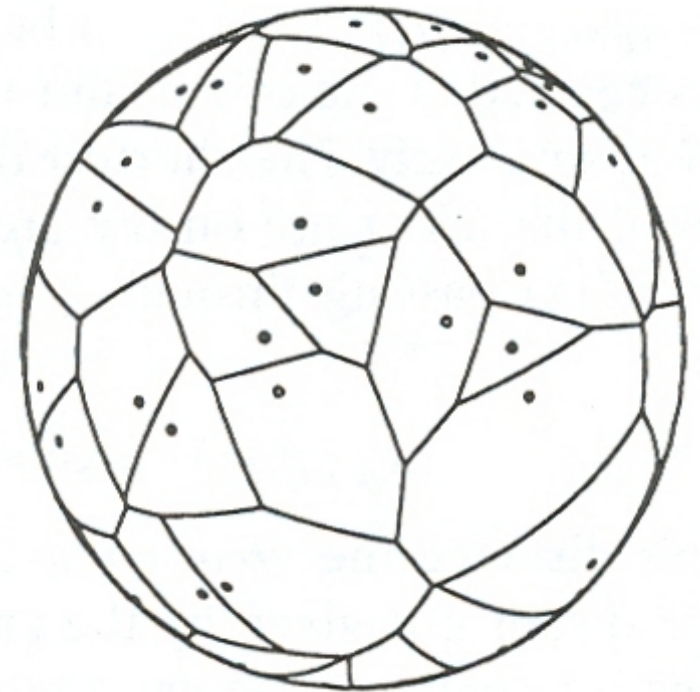
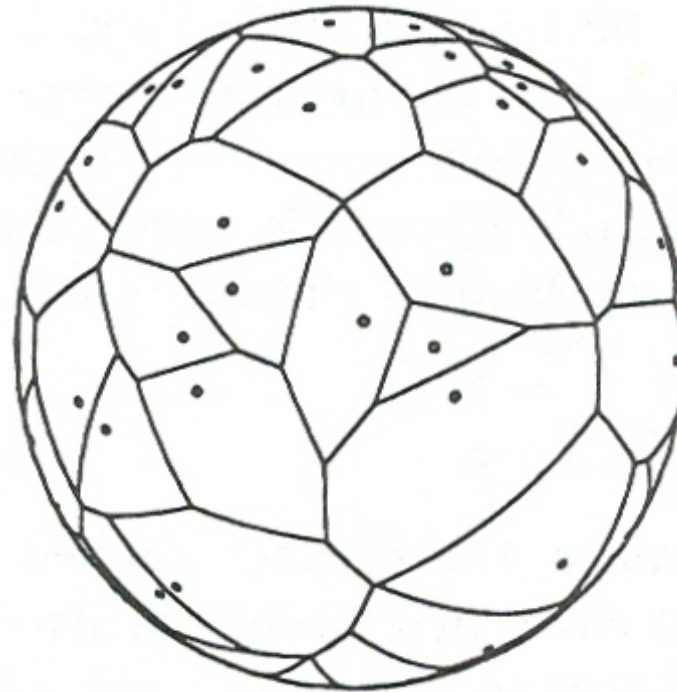
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APPLICATIONS

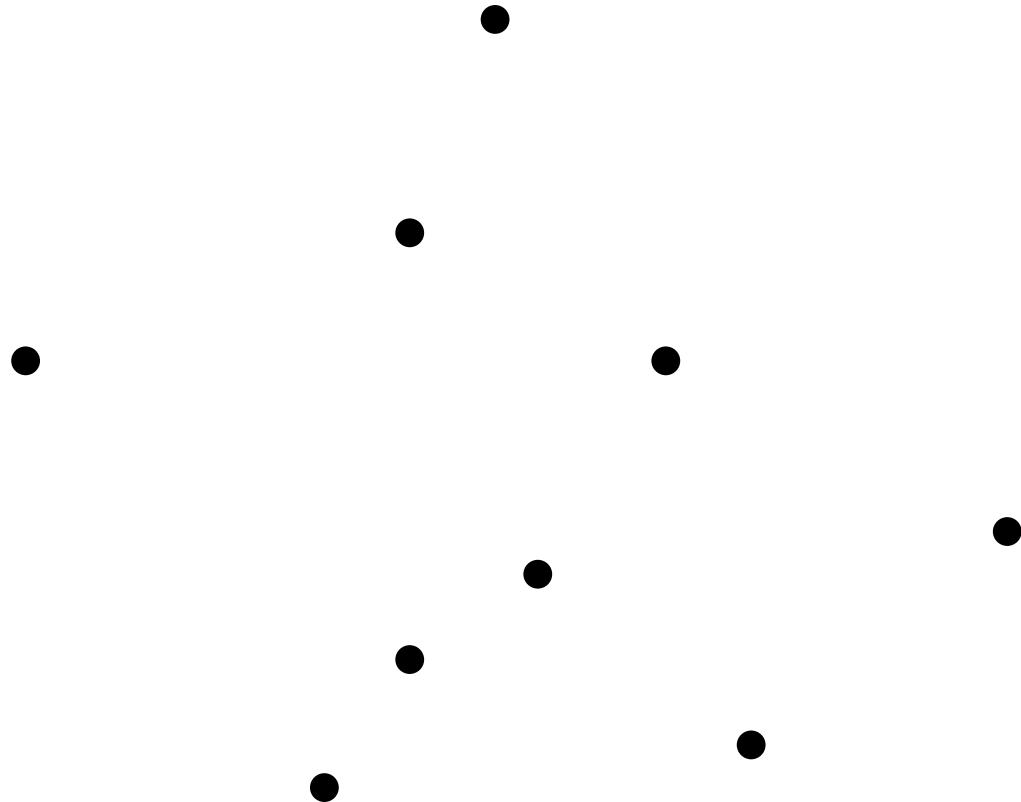
- Sphere: study of global Earth problems
- Cylinder: study of problems with cyclic periodicity in time

PROXIMITY

GENERALIZATIONS OF THE VORONOI DIAGRAM

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other
metrics

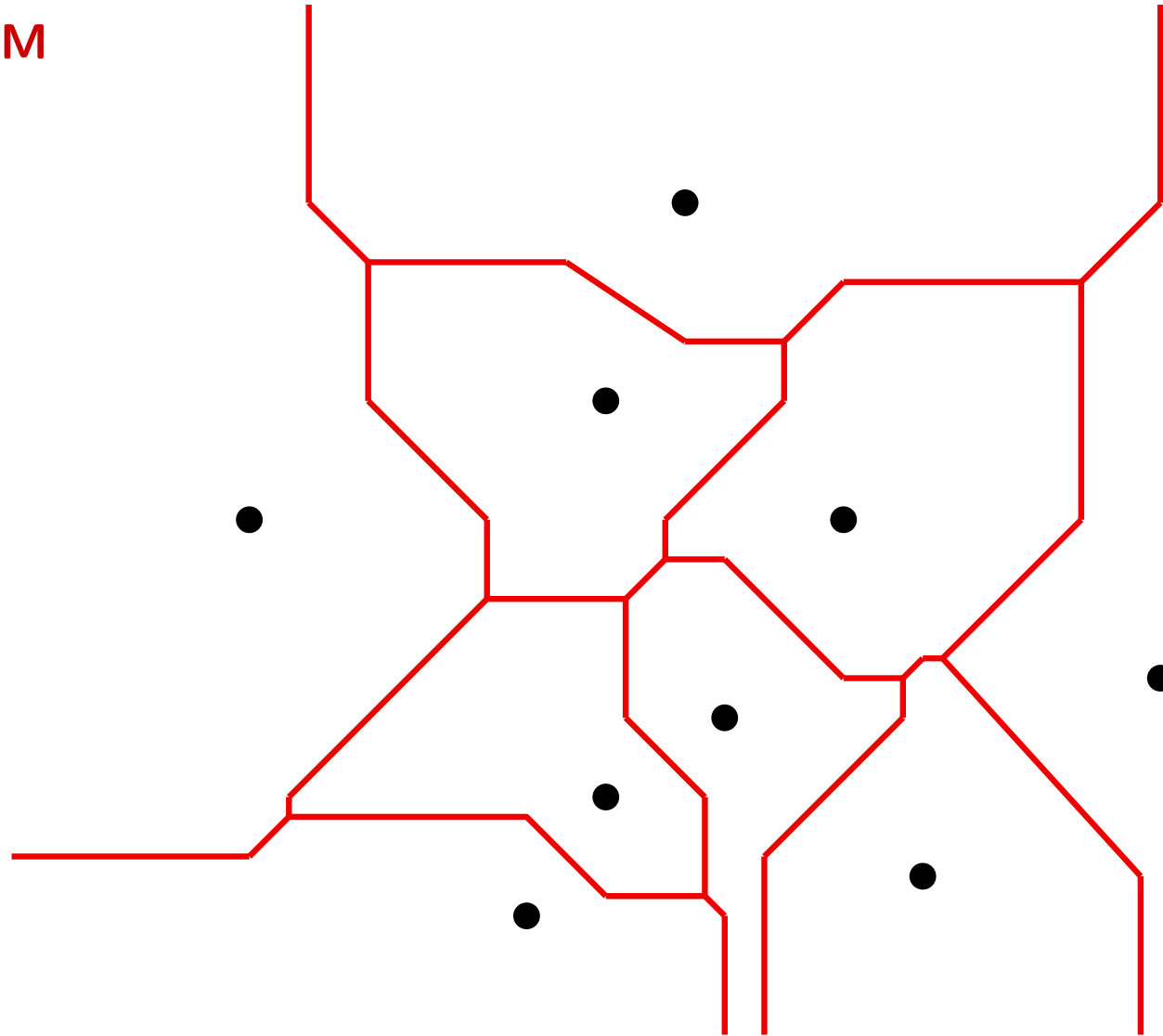


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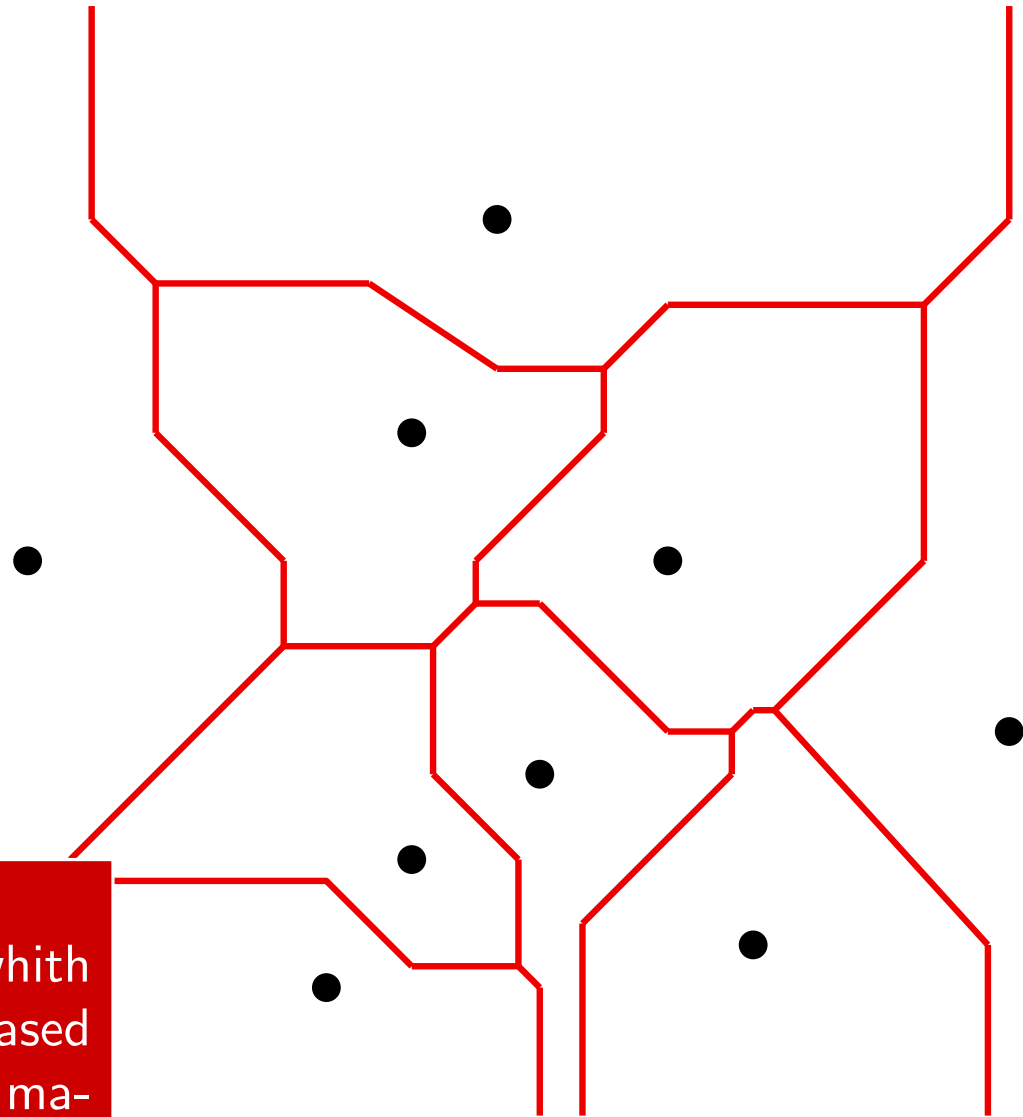
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other
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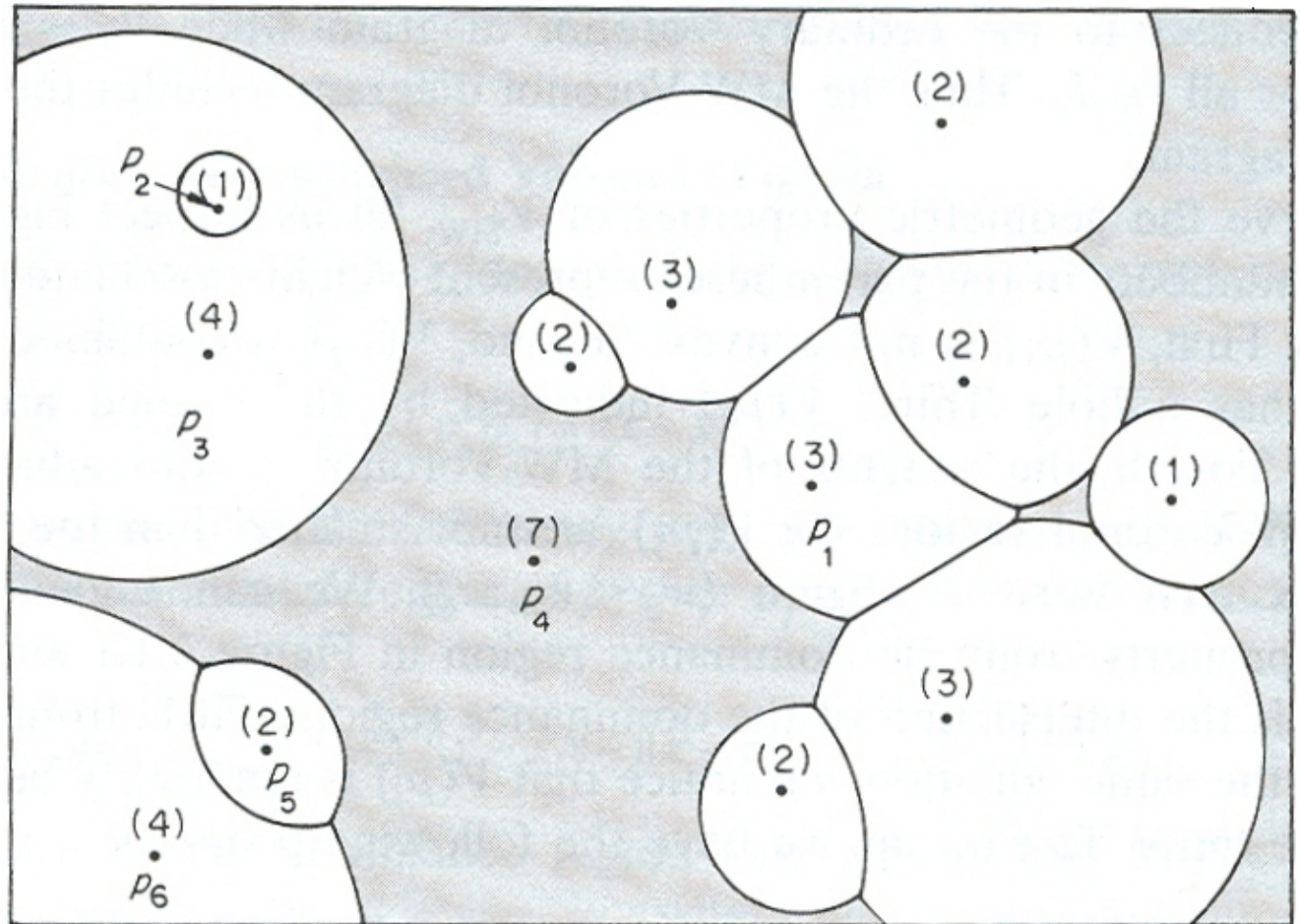
Massive storage system with a reading/writing head based on the Manhattan or the maximum metrics

PROXIMITY

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other
metrics

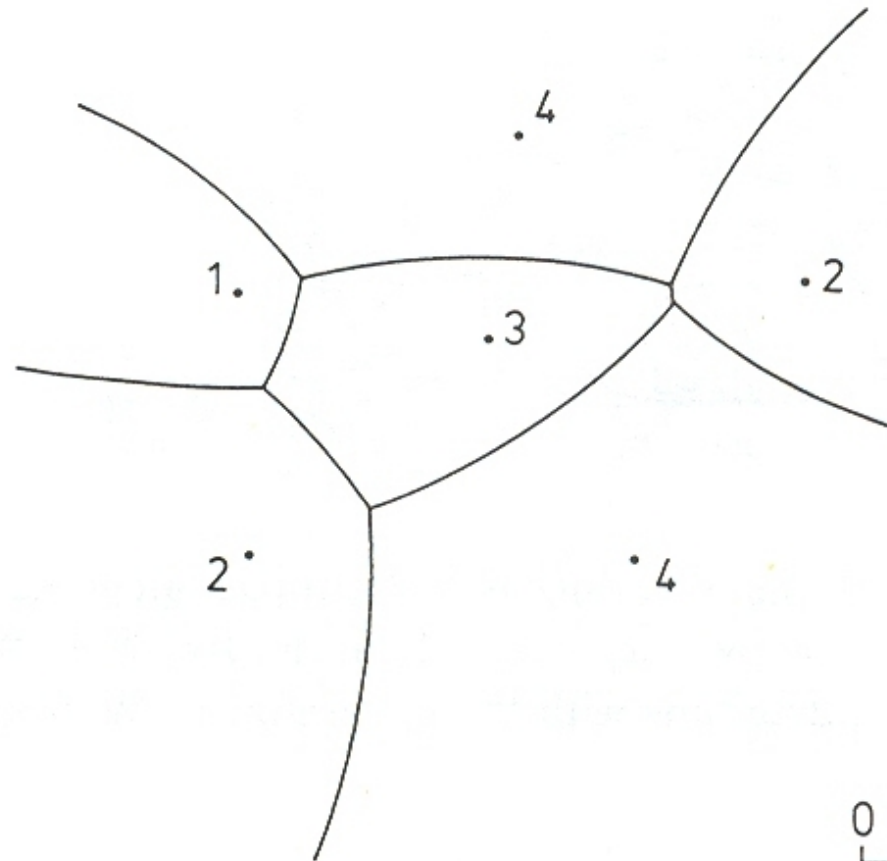


Numbers indicate multiplicative weights

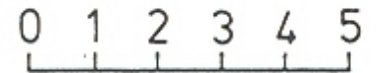
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other
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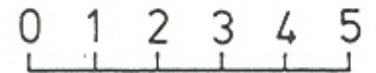
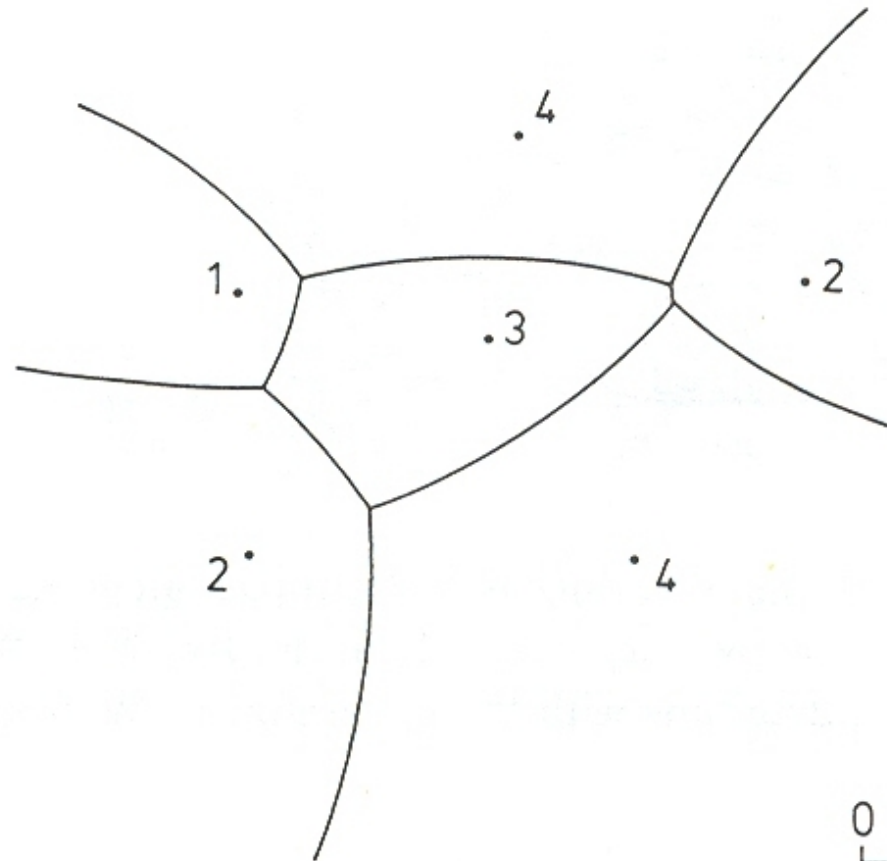


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other
metrics

APPLICATIONS

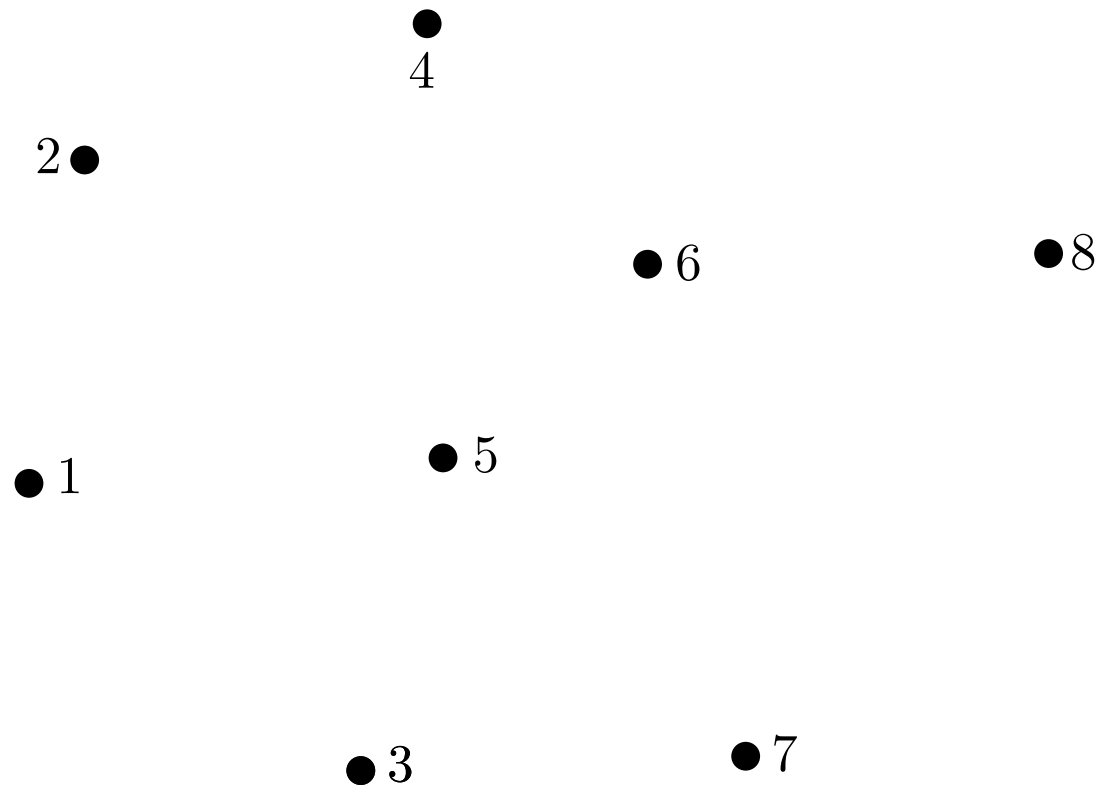
- Population of each city
- Amount of emissions of some polluting product
- Size of each atom in a crystalline structure
- Combination of transportation prices and costs

PROXIMITY

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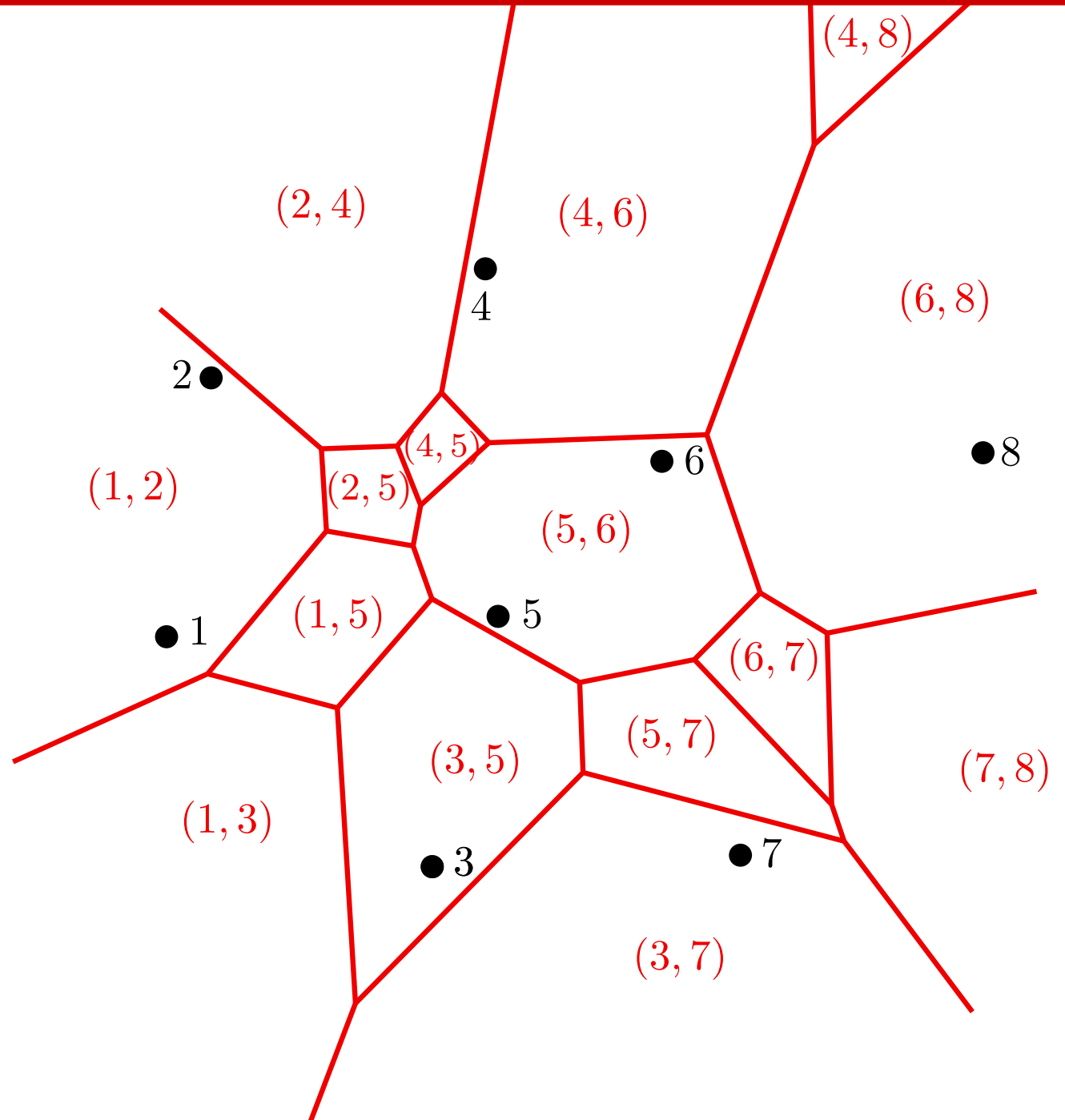
k



PROXIMITY

GENERALIZATIONS OF THE VORONOI DIAGRAM

The Voronoi diagram of a finite set P of **points** in the **plane** is a partition of the plane such that each region is the locus of the points that lie **closer** to ~~one~~ k element of P than to the remaining ones.



PROXIMITY

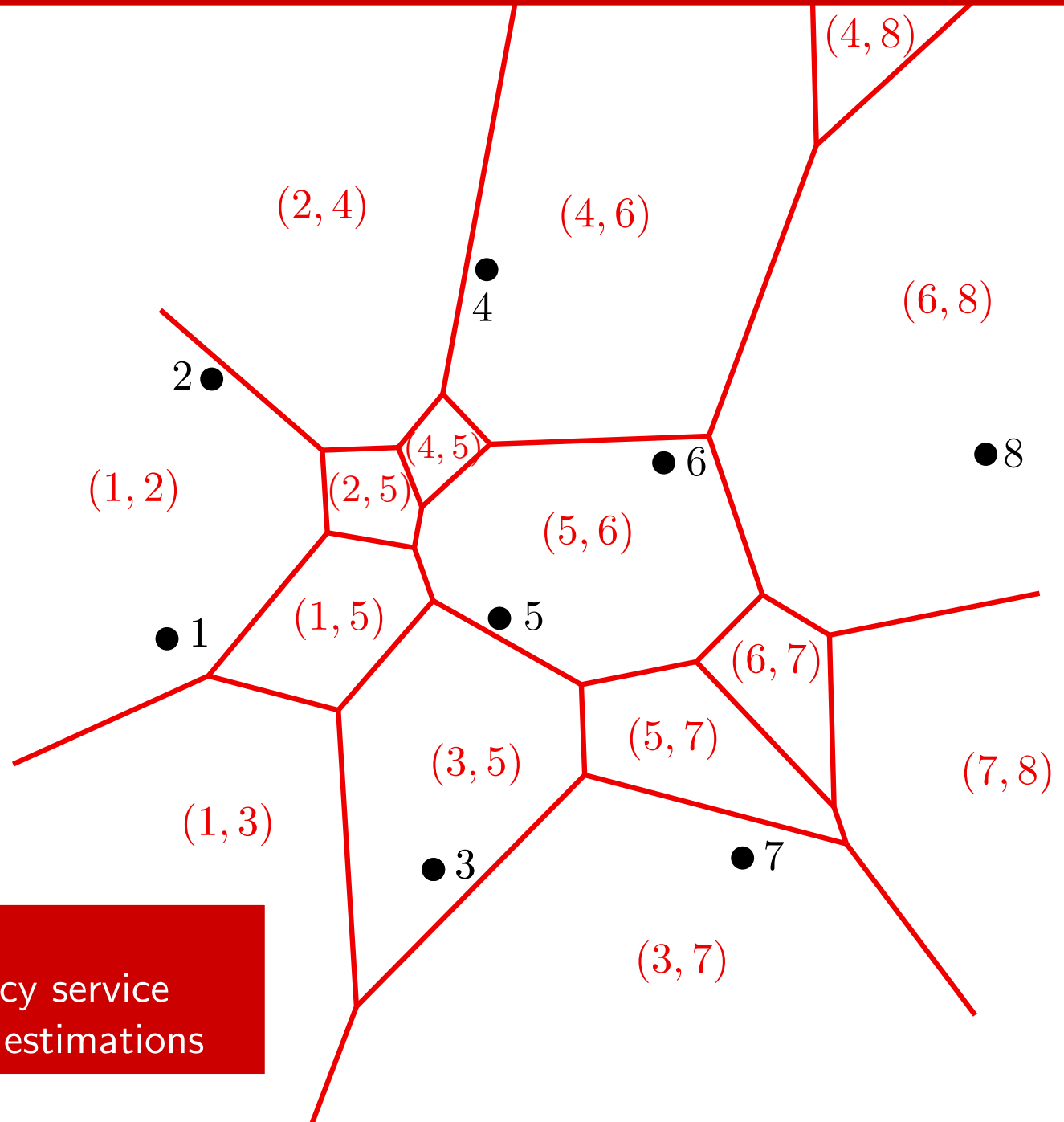
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k

APPLICATIONS

- Finding the k -th closest emergency service
- Spatial interpolation in statistics estimations



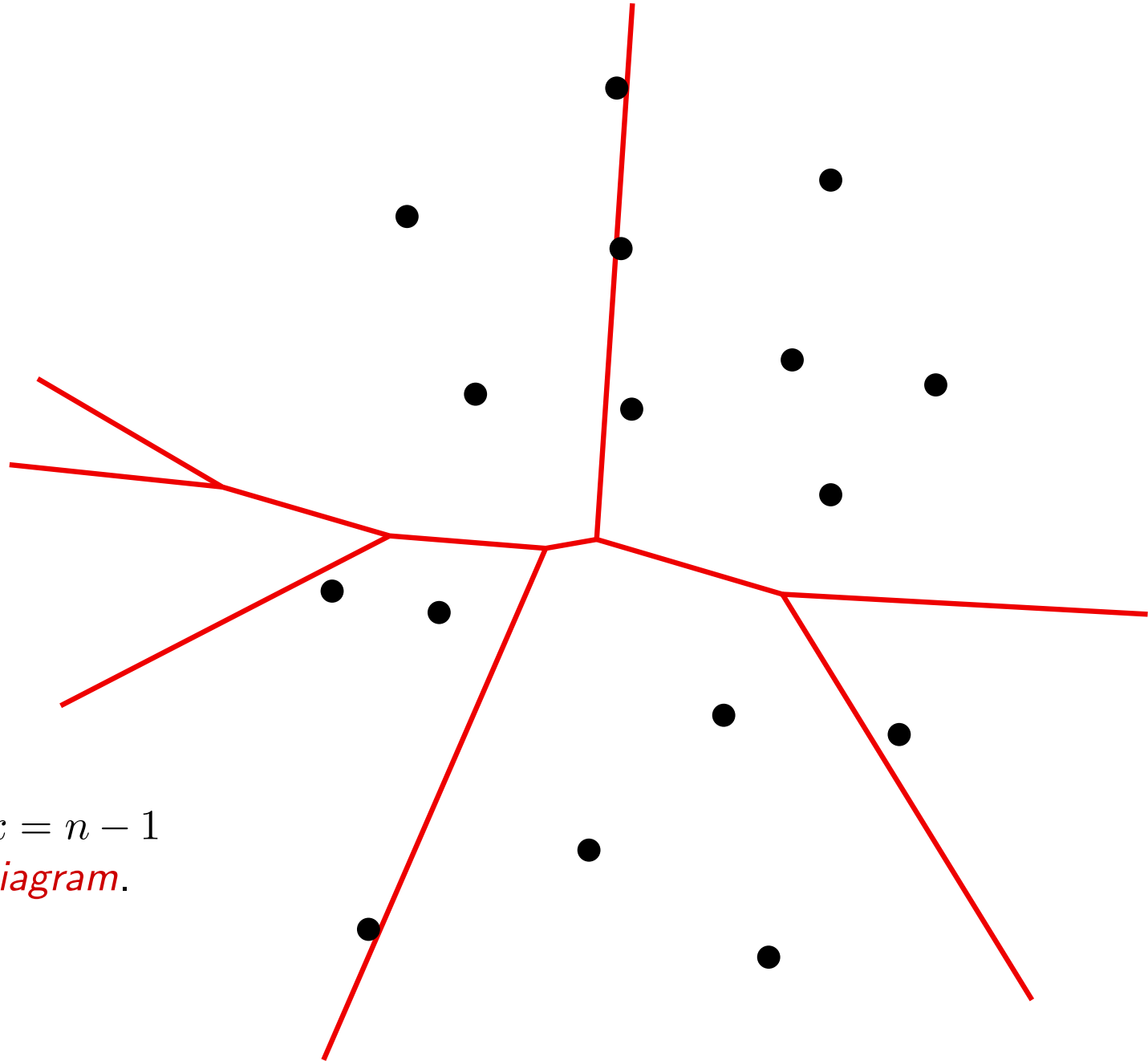
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k

The Voronoi diagram of order $k = n - 1$ is the *Furthest Point Voronoi diagram*.



PROXIMITY

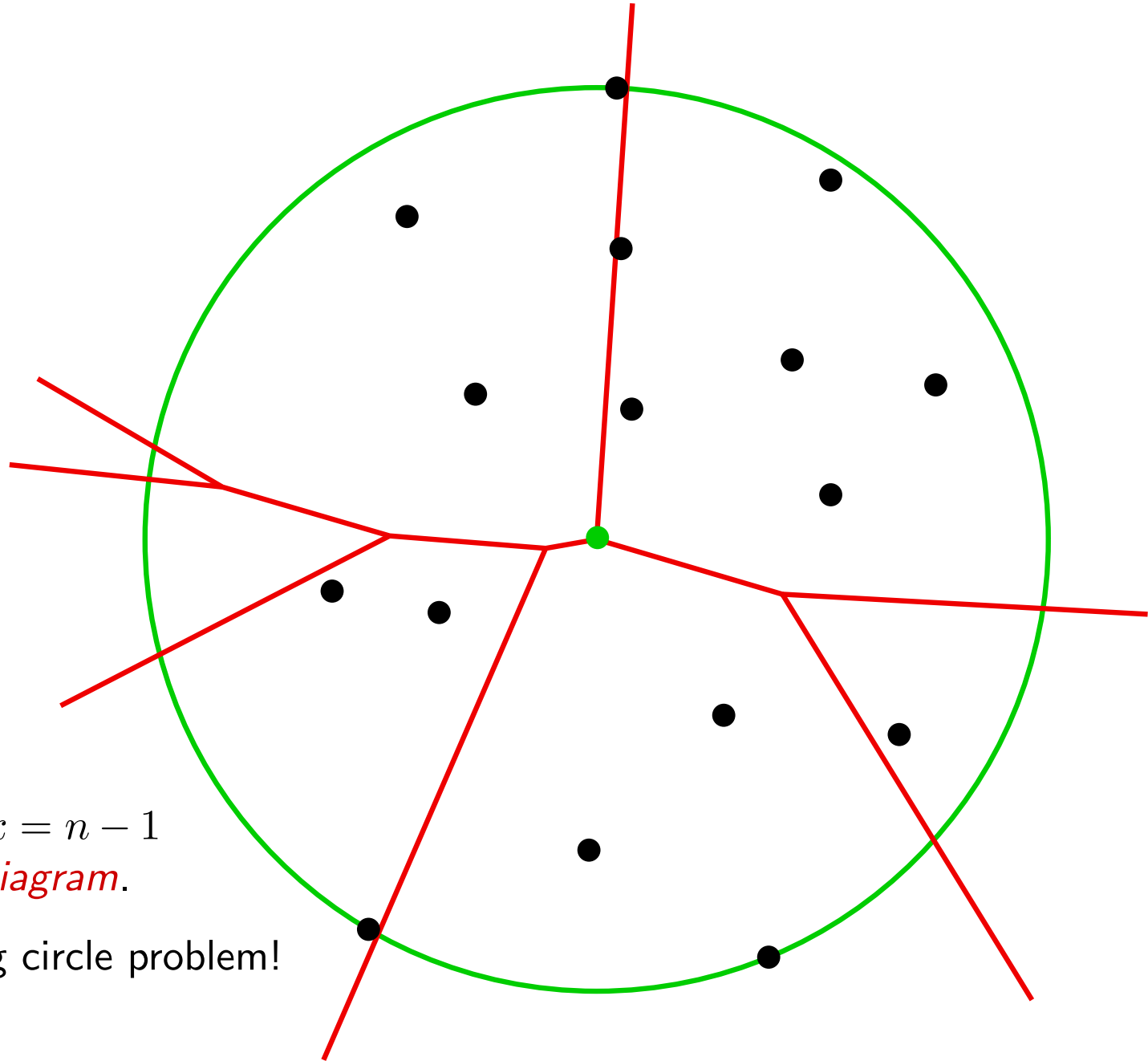
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It solves the minimum spanning circle problem!



PROXIMITY

VORONOI DIAGRAM AND INTERSECTION OF HALFSPACES IN 3D

Consider the point set P as being embedded in the plane $z = 0$.

Consider the paraboloid $z = x^2 + y^2$.

For each point $p_i \in P$ let p_i^* be its vertical projection of p_i onto the paraboloid, i.e.:

$$\text{if } p_i = (a_i, b_i, 0), \text{ then } p_i^* = (a_i, b_i, a_i^2 + b_i^2).$$

For each point p_i^* consider the plane which is tangent to the paraboloid at p_i^* .

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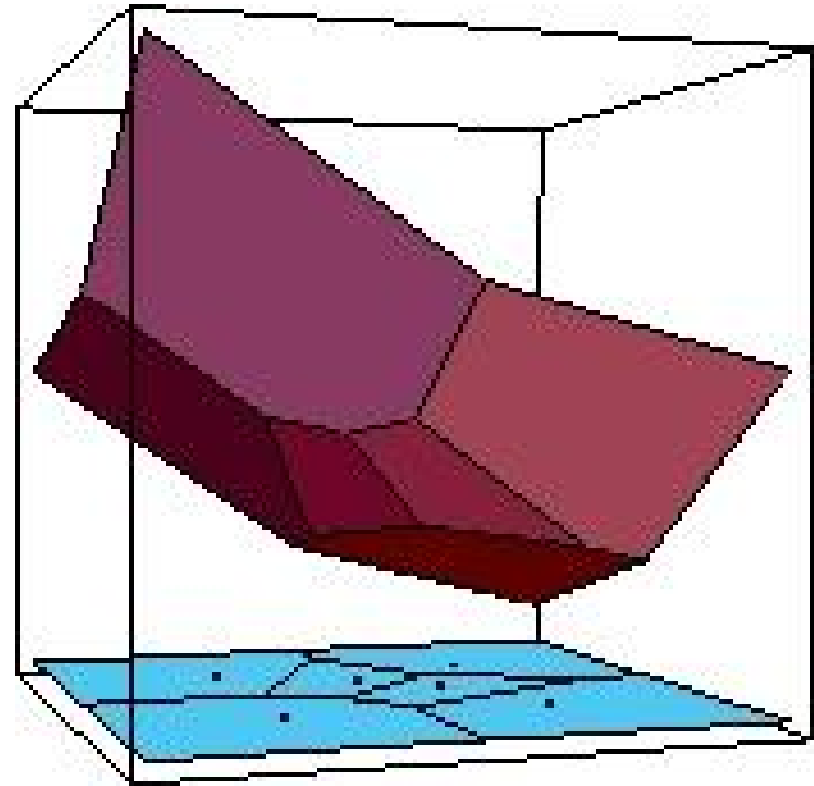
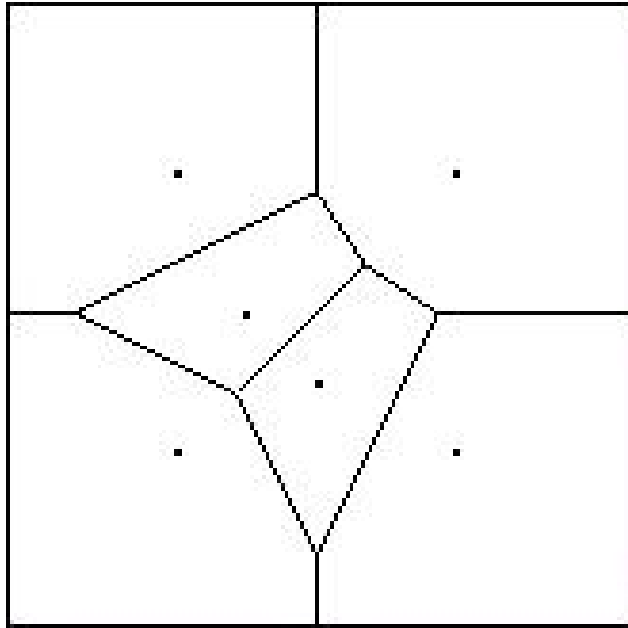
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The (farthest point) Voronoi diagram of P is the orthogonal projection onto the plane $z = 0$ of the polyhedral convex region obtained when intersecting the upper (lower) halfspaces defined by these planes.

PROXIMITY

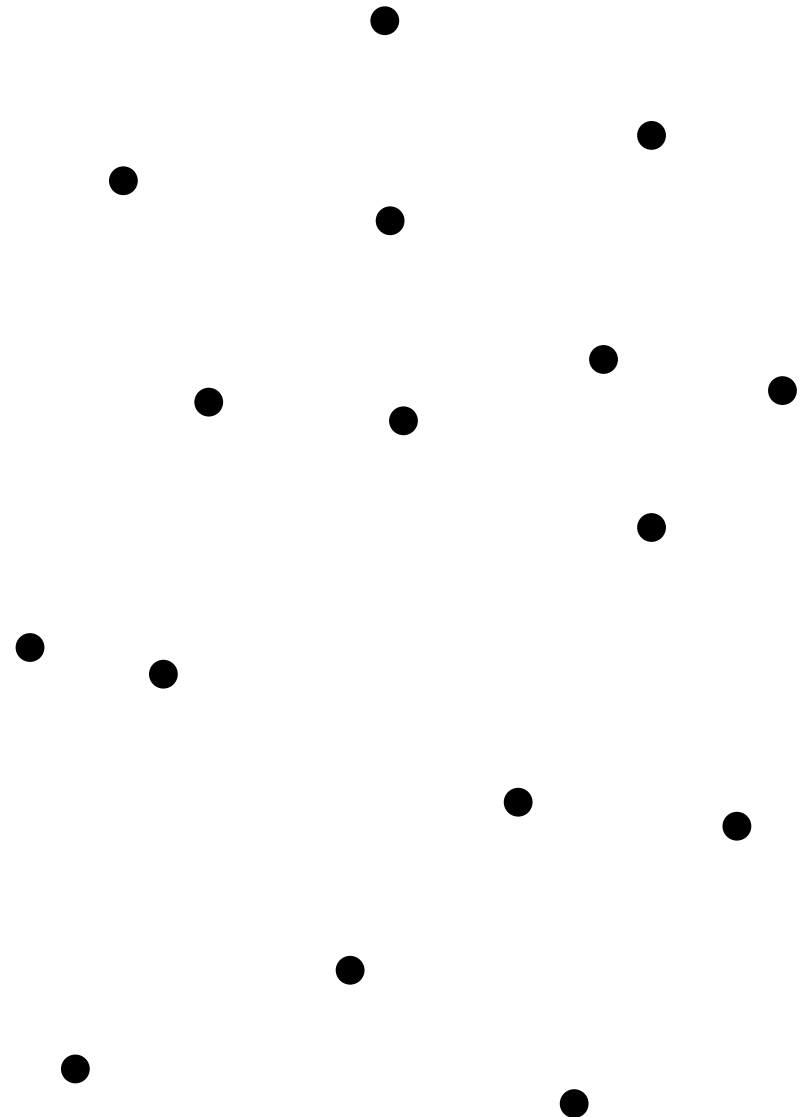


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PROXIMITY

THREE BIG QUESTIONS

1. How to **build** the Voronoi diagram of P ?
2. How to **store** the Voronoi diagram of P ?
3. How to **use** the Voronoi diagram to improve the solutions of the proximity problems?



PROXIMITY

TWO BOOKS WITH MUCH MORE INFORMATION

A. Okabe, B. Boots, K. Sugihara, S. N. Chiu

Spatial Tessellations

2nd ed., J. Wiley & Sons, 2000.

F. Aurenhammer, R. Klein, D.-T. Lee

Voronoi Diagrams and Delaunay Triangulations

World Scientific, 2013.